MATHEMATICS

INVERSE TRIGONOMETRY FUNCTION

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INVERSE TRIGONOMETRY FUNCTION

JEE EXERCISES

Exercise J-1

BASIC INTRODUCTION

1. If $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$, then $\sum_{i=1}^{20} x_i$ is equal to :

(A) 20 (B) 10 (C) 0 (D) None of these

यदि $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$, तो $\sum_{i=1}^{20} x_i$ का मान है :

(A) 20 (B) 10 (C) 0 (D) इनमें से कोई नहीं

Ans. A

Sol. $\sin^{-1} x_1 + \sin^{-2} x_2 + \sin^{-1} x_3 + \ldots\ldots\ldots + \sin^{-1} x_{20} = 10\pi$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$\frac{\pi}{2} \quad \frac{\pi}{2} \quad \frac{\pi}{2} \quad \frac{\pi}{2}$

$x_1 = 1 \quad x_2 = 1 \quad x_3 = 1 \quad x_{20} = 1$

$x_1 + x_2 + \ldots\ldots\ldots\ldots\ldots + x_{20} = 1 + 1 + 1 + \ldots\ldots\ldots + 1$

$= 20$

2. If $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$ and $f(1) = 2$, $f(x + y) = f(x)\, f(y)$ for all x, y; then $a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)} + b^{2f(2)} + c^{2f(3)}}$ is equal to :

यदि $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$ तथा $f(1) = 2$ एवं सभी x, y के लिए $f(x + y) = f(x)\, f(y)$ हो तो $a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)} + b^{2f(2)} + c^{2f(3)}}$ का मान है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$

$\downarrow \quad \downarrow \quad \downarrow$

$\pi \quad \pi \quad \pi$

$f(x + y) = f(x)\, f(y)$

$f(1) = 2$

$f(1 + 1) = f(1) \times f(1)$

$f(2) = (2)\,(2) = 4$

$f(2 + 1) = f(2)\, f(1)$

$f(3) = 4 \times 2 = 8$

$$\Rightarrow a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)} + b^{2f(2)} + c^{2f(3)}}$$

$$\Rightarrow (-1)^4 + (-1)^8 + (-1)^{16} + \frac{(-1-1-1)}{(-1)^4 + (-1)^8 + (-1)^{16}}$$

$$\Rightarrow 1+1+1+\frac{-3}{1+1+1}$$

$$= 3 - \frac{3}{3} = 3 - 1 = 2$$

DOMAIN AND RANGE OF ITFs

3. If $\sin^{-1}x + \tan^{-1}x = y\ (-1 < x < 1)$, then which is not possible :

यदि $\sin^{-1}x + \tan^{-1}x = y\ (-1 < x < 1)$, तो कौनसा सम्भव नहीं है :

(A) $y = \frac{3\pi}{2}$ (B) $y = 0$ (C) $y = \frac{\pi}{2}$ (D) $y = -\frac{\pi}{2}$

Ans. A

Sol. $\sin^{-1}x + \tan^{-1}x = y\ \ (-1 < x < 1)$

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) + \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) = y$$

$$\left(-\frac{3\pi}{4}, \frac{3\pi}{4}\right) = y$$

$$y = \frac{3\pi}{2}$$

EVALUATION

4. The value of $\sin^2\left(\cos^{-1}\frac{1}{2}\right)+\cos^2\left(\sin^{-1}\frac{1}{3}\right)$ is :

(A) $\frac{17}{36}$ (B) $\frac{59}{36}$ (C) $\frac{36}{59}$ (D) None of these

$\sin^2\left(\cos^{-1}\frac{1}{2}\right)+\cos^2\left(\sin^{-1}\frac{1}{3}\right)$ का मान है :

(A) $\frac{17}{36}$ (B) $\frac{59}{36}$ (C) $\frac{36}{59}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\sin^2\left(\cos^{-1}\frac{1}{2}\right)+\cos^2\left(\sin^{-1}\frac{1}{3}\right)$

$= \sin^2\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)+\cos^2\left(\cos^{-1}\left(\frac{\sqrt{8}}{3}\right)\right)$

$\Rightarrow\left(\frac{\sqrt{3}}{2}\right)^2+\left(\frac{\sqrt{8}}{3}\right)^2$

$=\frac{3}{4}+\frac{8}{9}$

$=\frac{27+32}{36}=\frac{59}{36}$

5. $\cot\left[\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{8}\right]=$

(A) 1 (B) -1 (C) $\sqrt{2}$ (D) $-\sqrt{2}$

Ans. A

Sol. $\cot\left[\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{8}\right]$

$$\cot\left[\tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{8}}{1-\frac{1}{16}}\right)+\tan^{-1}\frac{1}{5}\right]$$

$$\cot\left[\tan^{-1}\left(\frac{2}{3}\right)+\tan^{-1}\frac{1}{5}\right]$$

$$\cot\left[\tan^{-1}\left(\frac{\frac{2}{3}+\frac{1}{5}}{1-\frac{2}{15}}\right)\right]$$

$$\cot\left[\tan^{-1}\left(\frac{10+3}{13}\right)\right]$$

$\cot(\tan^{-1}(1))$

$= 1$

6. $\cot\left[\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)\right]$ is equal to :

$\cot\left[\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)\right]$ का मान है :

(A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) $\frac{7}{9}$ (D) $\frac{9}{7}$

Ans. B

Sol. $\cot\left[\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)\right]$

$$\cot\left[\tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{13}}{1-\frac{1}{91}}\right)\right]$$

$$\cot\left[\tan^{-1}\left(\frac{13+7}{91-1}\right)\right]$$

$$\cot\left[\tan^{-1}\left(\frac{20}{91}\right)\right]$$

$$\cot\left(\cot^{-1}\left(\frac{9}{2}\right)\right)$$

$$= \frac{9}{2}$$

PROPERTIES OF ITF

7. If $0 < x < 1$, then $\sqrt{1+x^2}\,[\{x\cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1]^{1/2} =$ **[JEE-2008]**

यदि $0 < x < 1$, तो $\sqrt{1+x^2}\,[\{x\cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1]^{1/2} =$ **[JEE-2008]**

(A) $\frac{x}{\sqrt{1+x^2}}$ (B) x (C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$

Ans. C

Sol. $\sqrt{1+x^2}\,[\{x\cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1]^{1/2}$

$$\sqrt{1+x^2}\left[\left\{x\frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}\right\}^2 - 1\right]^{1/2}$$

$$\sqrt{1+x^2}\left[\frac{(x^2+1)^2}{x^2+1} - 1\right]^{1/2}$$

$\sqrt{1+x^2}$ (x)

OTHER PROPERTIES OF ITF

8. If $a \le \tan^{-1}x + \cot^{-1}x + \sin^{-1}x \le b$. Then :

(A) $a = 0, b = \pi$ (B) $b = \frac{\pi}{2}$ (C) $a = \frac{\pi}{4}$ (D) None of these

यदि $a \le \tan^{-1}x + \cot^{-1}x + \sin^{-1}x \le b$. तब :

(A) $a = 0, b = \pi$ (B) $b = \frac{\pi}{2}$ (C) $a = \frac{\pi}{4}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $a \le \underbrace{\tan^{-1} x + \cot^{-1} x + \sin^{-1} x}_{y} \le b$

$x \in [-1, 1]$

$y = \left[\frac{-\pi}{4}, \frac{\pi}{4}\right] + \left[\frac{\pi}{4}, \frac{3\pi}{4}\right] + \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$y_{min} = \frac{-\pi}{4} + \frac{3\pi}{4} - \frac{\pi}{2} \Rightarrow a = 0$

$y_{max} = \frac{\pi}{4} + \frac{\pi}{4} + \frac{\pi}{2} = \pi \Rightarrow b = \pi$

9. $\sin^{-1}x + \sin^{-1}\frac{1}{x} + \cos^{-1}x + \cos^{-1}\frac{1}{x} =$

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) None of these

$\sin^{-1}x + \sin^{-1}\frac{1}{x} + \cos^{-1}x + \cos^{-1}\frac{1}{x} =$

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\sin^{-1}x + \sin^{-1}\frac{1}{x} + \cos^{-1}x + \cos^{-1}\frac{1}{x}$

$\sin^{-1} x + \cos^{-1} + \text{cosec}^{-1} x + \sec^{-1} x$

$= \frac{\pi}{2} + \frac{\pi}{2} = \pi$

10. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y =$

यदि $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, तो $\cos^{-1}x + \cos^{-1}y =$

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) π

Ans. B

Sol. $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$

$\frac{\pi}{2} - \cos^{-1} x + \frac{\pi}{2} - \cos^{-1} y = \frac{2\pi}{3}$

$\cos^{-1} x + \cos^{-1} y = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$

11. $\sin\left[2\cos^{-1}\left(-\frac{3}{5}\right)\right]$ is equal to :

$\sin\left[2\cos^{-1}\left(-\frac{3}{5}\right)\right]$ का मान है :

(A) $\frac{6}{25}$ (B) $\frac{24}{25}$ (C) $\frac{4}{5}$ (D) $-\frac{24}{25}$

Ans. D

Sol. $\sin\left[2\cos^{-1}\left(-\frac{3}{5}\right)\right]$

$\cos^{-1}\left(\frac{-3}{5}\right) = A$

$\pi + \cos^{-1}\left(\frac{3}{5}\right) = A$

$-\frac{3}{5} = \cos A$

$\sin 2A = 2 \sin A \cos A$

$= 2\left(\frac{4}{5}\right)\left(-\frac{3}{5}\right) = -\frac{24}{25}$

ADDITION OR SUBTRACTION OF TWO OR MORE ANGLES

12. If $\cos^{-1}\frac{3}{5} - \sin^{-1}\frac{4}{5} = \cos^{-1} x$,then x =

यदि $\cos^{-1}\frac{3}{5} - \sin^{-1}\frac{4}{5} = \cos^{-1}x$, तो x =

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Ans. B

Sol. $\cos^{-1}\frac{3}{5} - \sin^{-1}\frac{4}{5} = \cos^{-1}x$

$\cos^{-1}\left(\frac{3}{5}\right) - \cos^{-1}\left(\frac{3}{1}\right) = \cos^{-1}x$

$\cos^{-1}x = 0$

$x = 1$

13. If $\sin^{-1}\frac{1}{3} + \sin^{-1}\frac{2}{3} = \sin^{-1}x$, then x is equal to :

यदि $\sin^{-1}\frac{1}{3} + \sin^{-1}\frac{2}{3} = \sin^{-1}x$, तो x बराबर है :

(A) 0 (B) $\frac{\sqrt{5}-4\sqrt{2}}{9}$ (C) $\frac{\sqrt{5}+4\sqrt{2}}{9}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. $\sin^{-1}\frac{1}{3} + \sin^{-1}\frac{2}{3} = \sin^{-1}x$

$\sin A = \frac{1}{3}$ $\sin B = \frac{2}{3}$

$\cos A = \frac{\sqrt{8}}{3}$ $\cos B = \frac{\sqrt{5}}{3}$

$A + B = \sin^{-1}x$

$\sin(A+B) = x$

$x = \sin A\cos B + \cos A \sin B$

$= \frac{1}{3}\frac{\sqrt{5}}{3} + \frac{\sqrt{8}}{3}\frac{2}{3}$

$x = \frac{\sqrt{5}+4\sqrt{2}}{9}$

14. If $\theta = \cot^{-1}\sqrt{\cos x} - \tan^{-1}\sqrt{\cos x}$, then $\sin\theta =$

(A) $\tan\frac{1}{2}x$ (B) $\tan^2(x/2)$ (C) $\frac{1}{2}\tan^{-1}(x/2)$ (D) None of these

यदि $\theta = \cot^{-1}\sqrt{\cos x} - \tan^{-1}\sqrt{\cos x}$, तब $\sin\theta =$

(A) $\tan\frac{1}{2}x$ (B) $\tan^2(x/2)$ (C) $\frac{1}{2}\tan^{-1}(x/2)$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given that $\cot^{-1}(\sqrt{\cos\alpha}) - \tan^{-1}(\sqrt{\cos\alpha}) = x$ .....(i)

We know that, $\cot^{-1}(\sqrt{\cos\alpha}) + \tan^{-1}(\sqrt{\cos\alpha}) = \frac{\pi}{2}$ .....(ii)

On adding equations (i) and (ii), we get

$$2\cot^{-1}(\sqrt{\cos\alpha}) = \frac{\pi}{2} + x \quad \Rightarrow \quad \sqrt{\cos\alpha} = \cot\left(\frac{\pi}{4} + \frac{x}{2}\right) \Rightarrow \quad \sqrt{\cos\alpha} = \frac{\cot\frac{x}{2} - 1}{1 + \cot\frac{x}{2}}$$

$$\Rightarrow \quad \sqrt{\cos\alpha} = \frac{\cos\frac{x}{2} - \sin\frac{x}{2}}{\cos\frac{x}{2} + \sin\frac{x}{2}} \Rightarrow \quad \cos\alpha = \frac{1 - \sin x}{1 + \sin x} \quad \Rightarrow \quad \frac{1 - \tan^2\frac{\alpha}{2}}{1 + \tan^2\frac{\alpha}{2}} = \frac{1 - \sin x}{1 + \sin x}$$

Applying componendo and dividendo rule, we get $\sin x = \tan^2\left(\frac{\alpha}{2}\right)$

15. If a, b, c be positive real numbers and the value of

$\theta = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ then $\tan\theta$ is equal to :

(A) 0 (B) 1 (C) $\frac{a+b+c}{abc}$ (D) None of these

यदि a, b, c धनात्मक वास्तविक संख्याएँ है तथा $\theta = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ तो $\tan\theta$ बराबर है :

(A) 0 (B) 1 (C) $\frac{a+b+c}{abc}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\theta = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$

$\downarrow \qquad\qquad \downarrow \qquad\qquad \downarrow$

A B C

$\theta = A + B + C$

$\tan\theta = \tan(A + B + C)$

$$\Rightarrow \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan A - \tan B \tan C - \tan C \tan A}$$

$$\tan\theta = \frac{\sqrt{\frac{a(a+b+c)}{bc}} + \sqrt{\frac{b(a+b+c)}{ca}} + \sqrt{\frac{c(a+b+c)}{ab}} - \frac{(a+b+c)}{\sqrt{abc}}}{1 - \frac{a+b+c}{c} - \frac{a+b+c}{b} - \frac{a+b+c}{a}}$$

$\Rightarrow \tan\theta = 0$

ITF EQUATIONS

16. If $3\cos^{-1}(x^2 - 7x + 25/2) = \pi$, then x =

(A) Only 3 (B) Only 4 (C) 3 or 4 (D) None of these

यदि $3\cos^{-1}(x^2 - 7x + 25/2) = \pi$, तो x =

(A) केवल 3 (B) केवल 4 (C) 3 or 4 (D) इनमें से कोई नहीं

Ans. C

Sol. $3\cos^{-1}(x^2 - 7x + 25/2) = \pi$

$$\cos^{-1}\left(x^2 - 7x + \frac{25}{2}\right) = \frac{\pi}{3}$$

$$x^2 - 7x + \frac{25}{2} = \frac{1}{2}$$

$2x^2 - 14x + 25 = 1$

$2x^2 - 14x + 24 = 0$ $\qquad x = 3$

$x^2 - 7x + 12 = 0$ $\qquad \Rightarrow 4 - 21 + \frac{3}{2} = \frac{-x_1 + 25}{2} = \frac{1}{2}$

$x = 3, 4$

$x = 4$

17. If $\tan(x + y) = 33$ and $x = \tan^{-1} 3$, then y will be :

यदि $\tan(x + y) = 33$ तथा $x = \tan^{-1} 3$, तो y होगा :

(A) 0.3 (B) $\tan^{-1}(1.3)$ (C) $\tan^{-1}(0.3)$ (D) $\tan^{-1}\left(\frac{1}{18}\right)$

Ans. C

Sol. $\tan(x+y) = 33$ $\quad x = \tan^{-1} = 3$

$\tan x = 3$

$$\frac{\tan x + \tan y}{1 + \tan x \tan y} = 33$$

$$\frac{3+t}{1-3t} = 33$$

$3 + t = 33 - 99t$

$100 \tan y = 30$

$\tan y = 0.3$

$y = \tan^{-1}(0.3)$

18. If $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$, then x equals :

(A) –1 (B) 1 (C) 0 (D) None of these

यदि $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$, तो x बराबर है :

(A) –1 (B) 1 (C) 0 (D) इनमें से कोई नहीं

Ans. A

Sol. $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$

$$t^2 + \left(\frac{\pi}{2} - t\right)^2 = \frac{5\pi^2}{8}$$

$$t^2 + \frac{\pi^2}{4} + t^2 - \pi t = \frac{5\pi^2}{8}$$

$$2t^2 - \pi t - \frac{3\pi^2}{8} = 0$$

$16t^2 - 8\pi t - 3\pi^2 = 0$

$16t^2 - 12\pi t + 4\pi t - 3\pi^2 = 0$ $\quad \Rightarrow 4t(4t - 3\pi) + \pi(4t - 3\pi) = 0$

$(4\tan^{-1}x - 3\pi)(4\tan^{-1}x + \pi) = 0$

$\tan^{-1} x = \frac{3\pi}{4}$ $\tan^{-1} x = \frac{-\pi}{4}$

(rejected) $x = -1$

19. If $\tan^{-1}\frac{1}{a-1} = \tan^{-1}\frac{1}{x} + \tan^{-1}\frac{1}{a^2-x+1}$, then x is :

यदि $\tan^{-1}\frac{1}{a-1} = \tan^{-1}\frac{1}{x} + \tan^{-1}\frac{1}{a^2-x+1}$, तो x बराबर है :

(A) $\frac{a}{2}$ (B) a^3 (C) $a^2 - a + 1$ (D) $a^2 + a - 1$

Ans. C

Sol. $\tan^{-1}\frac{1}{a-1} = \tan^{-1}\frac{1}{x} + \tan^{-1}\frac{1}{a^2-x+1}$

$$\frac{1}{a-1} = \frac{\frac{1}{x}+\frac{1}{a^2-x+1}}{1-\frac{1}{x\left(a^2-x+1\right)}}$$

$$\frac{1}{a-1} = \frac{a^2-x+1+x}{x\left(a^2-x+1\right)-1}$$

$xa^2 - x^2 + x - 1 = (a - 1)(a^2 + 1)$

$xa^2 - x^2 + x - 1 = a^3 + a - a^2 - 1$

$x^2 - (a^2 + 1)x + a^3 - a^2 + a = 0$

$$x = \frac{\left(a^2+1\right) \pm \sqrt{a^4-4a^3+6a^2-4a+1}}{2}$$

$$= \frac{\left(a^2+1\right) \pm \sqrt{(a-1)^4}}{2} = \frac{\left(a^2+1\right) \pm \left(a^2-2a+1\right)}{2}$$

$$x = \frac{a^2+1+a^2-2a+1}{2}, \frac{a^2+1-a^2+2a-1}{2}$$

$x = a^2 - a + 1, a$

20. The trigonometric equation, $\sin^{-1} x = 2 \sin^{-1} a$ has a solution for :

(A) all real values of a (B) $|a| < \frac{1}{2}$ (C) $|a| \le \frac{1}{\sqrt{2}}$ (D) $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

त्रिकोणमितीय समीकरण $\sin^{-1} x = 2 \sin^{-1} a$ का एक हल सम्भव होगा यदि :

(A) a के सभी मान के लिए (B) $|a| < \frac{1}{2}$ (C) $|a| \le \frac{1}{\sqrt{2}}$ (D) $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

Ans. C

Sol. $-\frac{\pi}{2} \le 2\sin^{-1} a \le \frac{\pi}{2}$

$-\frac{\pi}{4} \le \sin^{-1} a \le \frac{\pi}{4}$

$-\frac{1}{\sqrt{2}} \le a \le \frac{1}{\sqrt{2}}$

$|a| \le \frac{1}{\sqrt{2}}$

21. If $\cos^{-1} x - \cos^{-1}\left(\frac{y}{2}\right) = \alpha$, then $4x^2 - 4xy\cos\alpha + y^2$ is equal to :

(A) 4 (B) $2\sin 2\alpha$ (C) $-4\sin^2\alpha$ (D) $4\sin^2\alpha$

यदि $\cos^{-1} x - \cos^{-1}\left(\frac{y}{2}\right) = \alpha$, तो $4x^2 - 4xy\cos\alpha + y^2$ बराबर होगा :

(A) 4 (B) $2\sin 2\alpha$ (C) $-4\sin^2\alpha$ (D) $4\sin^2\alpha$

Ans. D

Sol. $\sin^{-1} x = 2\sin^{-1} a$

$A - B = \alpha$

$\cos A \cos B + \sin A \sin B = \cos\alpha$

$xy + \sqrt{1+x^2}\sqrt{4-y^2} = \cos\alpha$

$(1 - x^2)(4 - y^2) = \cos^2\alpha + x^2y^2 - 4xy\cos\alpha$

$4 - y^2 - 4x^2 + x^2y^2 = \cos^2\alpha + x^2y - 4xy\cos\alpha$

$4 - 4\cos^2\alpha = 4x^2 - 4xy\cos\alpha + y^2$

$4(1-c^2)$

$\Rightarrow 4x^2 - 4xy\cos\alpha + y^2 = 4\sin^2\alpha$

SUMMATION OF SERIES

22. The sum of the infinite terms of the series $\cot^{-1}\left(1^2+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+\ldots..$ is equal to :

(A) $\tan^{-1}(1)$ (B) $\tan^{-1}(2)$ (C) $\tan^{-1}(3)$ (D) $\tan^{-1}(4)$

अनन्त श्रेणी $\cot^{-1}\left(1^2+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+\ldots..$ का योग होगा :

(A) $\tan^{-1}(1)$ (B) $\tan^{-1}(2)$ (C) $\tan^{-1}(3)$ (D) $\tan^{-1}(4)$

Ans. B

Sol. $T_n = \cot^{-1}\left(n^2+\frac{3}{4}\right) = \tan^{-1}\left(\frac{1}{n^2+(3/4)}\right) = \tan^{-1}\left(\frac{1}{1+n^2-(1/4)}\right)$

$$= \tan^{-1}\left(\frac{1}{1+\left(n-\frac{1}{2}\right)\left(n+\frac{1}{2}\right)}\right) = \tan^{-1}\left(\frac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)}\right)$$

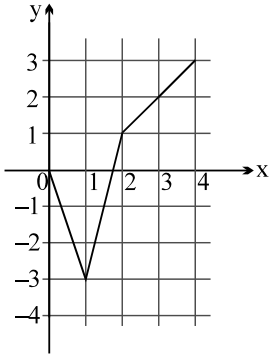
Exercise J-2

BASIC INTRODUCTION

1. Given is a partial graph of an even periodic function f whose period is 8. If [ ] denotes greatest integer function then find the value of the expression.

दिया गया ग्राफ एक सम व आवर्ती फलन f का है जिसका आवर्तकाल 8 है। [*] महत्तम पूर्णांक फलन को निरूपित करता है। निम्न का मान ज्ञात कीजिए।

$$f(-3) + 2\,|f(-1)| + \left[f\left(\frac{7}{8}\right)\right] + f(0) + \arccos\big(f(-2)\big) + f(-7) + f(20)$$



Ans. 5

Sol. Even function

Periodic with period = 8

$f(-x) = f(x)$

$$f(-3) + 2\,|f(-1)| + \left[f\left(\frac{7}{8}\right)\right] + f(0) + \cos^{-1}\big(f(-2)\big) + f(-7) + f(20)$$

$= 2 + 2(3) + (-3) + 0 + \cos^{-1}(1) + (-3) + 3$

$= 2 + 6 - 3 + 0$

$= 5$

DOMAIN AND RANGE BASED QUESTIONS

2. Find the domain of definition the following functions. (Read the symbols [ ] and { } as greatest integers and fractional part functions respectively.)

निम्नलिखित फलनों का प्रान्त ज्ञात कीजिये। ([*] व {*} महत्तम पूर्णांक फलन व भिन्नात्मक भाग फलन है)

(i) $f(x) = \arccos\dfrac{2x}{1+x}$

(ii) $f(x)=\sqrt{\cos(\sin x)}+\sin^{-1}\dfrac{1+x^2}{2x}$

(iii) $f(x)=\sin^{-1}\left(\dfrac{x-3}{2}\right)-\log_{10}(4-x)$

(iv) $f(x)=\sin^{-1}(2x+x^2)$

(v) $f(x)=\dfrac{\sqrt{1-\sin x}}{\log_5(1-4x^2)}+\cos^{-1}(1-\{x\})$

(vi) $f(x)=\sqrt{3-x}+\cos^{-1}\left(\dfrac{3-2x}{5}\right)+\log_6\left(2|x|-3\right)+\sin^{-1}\left(\log_2 x\right)$

(vii) $f(x)=\log_{10}(1-\log_7(x^2-5x+13))+\cos^{-1}\left(\dfrac{3}{2+\sin\frac{9\pi x}{2}}\right)$

(viii) $f(x)=e^{\sin^{-1}\left(\frac{x}{2}\right)}+\tan^{-1}\left[\dfrac{x}{2}-1\right]+\ell n\left(\sqrt{x-[x]}\right)$

Ans. (i) $-1/3\le x\le 1$ (ii) $\{1,-1\}$ (iii) $1\le x<4$ (iv) $[-(1+\sqrt{2}),(\sqrt{2},-1)]$
(v) $x\in(-1/2,1/2), x\neq 0$ (vi) $(3/2,2]$ (vii) $\{7/3, 25/9\}$
(viii) $(-2,2)-\{-1,0,1\}$ (ix) $\{x\,|\,x=2n\pi+\dfrac{\pi}{6}, n\in I\}$

Sol. (i) $f(x)=\cos^{-1}\left(\dfrac{2x}{1+x}\right)$

$$-1\le\frac{2x}{1+x}\le 1$$

$$\frac{2x}{x+1}+1\ge 0 \qquad \frac{2x}{x+1}-1\le 0$$

$$\frac{2x+x+1}{x+1}\ge 0 \qquad \frac{2x-x-1}{x+1}\le 0$$

$$x\in\left[-\frac{1}{3},1\right]$$

(ii) $f(x)=\sqrt{\cos(\sin x)}+\sin^{-1}\left(\dfrac{1+x^2}{2x}\right)$

$$-1\le\frac{1+x^2}{2x}\le 1$$

$\frac{1+x^2}{2x}+1\le 0$ $\frac{1+x^2}{2x}-1\le 0$

$\frac{x^2+2x+1}{2x}\ge 0$ $\frac{x^2-2x+1}{2x}\le 0$

$\frac{(x+1)^2}{2x}\ge 0$ $\frac{(x-1)^2}{2x}\le 0$

$x=-1,\ 1$

(iii) $f(x)=\sin^{-1}\left(\frac{x-3}{2}\right)-\log_{10}(4-x)$

$-1\le\frac{x-3}{2}\le 1$

$x\in[1,4]$

(iv) $f(x)=\sin^{-1}\left(2x+x^2\right)$

$-1\le x^2+2x\le 1$

$x^2+2x+1\ge 0$ $x^2+2x-1\le 0$

$x\in\left[-\left(1+\sqrt{2}\right),\sqrt{2}-1\right]$

(v) $f(x)=\frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)}+\cos^{-1}(1-\{x\})$

$1-4x^2>0$ $1-4x^2\ne 1\Rightarrow x\ne 0$

$(2x-1)(2x+1)<0$ $x\in 0$

$x\in\left(-\frac{1}{2},\frac{1}{2}\right)-\{0\}$

(vi) $f(x)=\sqrt{3-x}+\cos^{-1}\left(\frac{3-2x}{5}\right)+\log_6\left(2|x|-3\right)+\sin^{-1}\left(\log_2 x\right)$

$3-2\ge 0$ $-1\le\frac{3-2x}{5}\le 1$

$x\le 3$ $-5\le 3-2x$ $3-2x\le 5$

$x\le 4$ $x\ge -1$

$x\in[-1,4]$

$2|x|-3>0$

$x \in \left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{3}{2}, \infty\right)$

$-1 \le \log_2 x \le 1$

$\log_2 x \ge -1$ $\log_2 x \le 1$

$x \in \left[\frac{1}{2}, 2\right]$ f.a. $\Rightarrow x \in \left(\frac{3}{2}, 2\right]$

(vii) $(1 - \log_7 (x^2 - 5x + 13) > 0$

$\log_7 (x^2 - 5x + 13) < 1$ $x^2 - 5x + 13 > 0$

$x^2 - 5x + 13 < 7$ $x \in R$

$x^2 - 5x + 6 < 0$

$(x - 3)(x - 2) < 0$

$x \in (2, 3)$

$-1 \le \frac{3}{2 + \sin \frac{9\pi x}{2}} \le 1$

$x = \left(\frac{4n + 1}{9}\right)$

(viii) $f(x) = e^{\sin^{-1}\left(\frac{x}{2}\right)} + \tan^{-1}\left[\frac{x}{2} - 1\right] + \ell n\left(\sqrt{x - [x]}\right)$

$-1 \le \frac{x}{2} \le 1$ $x \in R$

$x \in [-2, 2]$

$x \in (-2, 2) - \{-1, 0, 1\}$

3. Identify the pair(s) of functions which are identical.

निम्न युग्मों में समरूप फलन ज्ञात कीजिये व ग्राफ बनाईये।

(a) $y = \tan(\cos^{-1} x); y = \frac{\sqrt{1 - x^2}}{x}$ (b) $y = \tan(\cot^{-1} x); y = \frac{1}{x}$

(c) $y = \sin(\text{arc} \tan x); y = \frac{x}{\sqrt{1 + x^2}}$ (d) $y = \cos(\text{arc} \tan x); y = \sin(\text{arc} \cot x)$

Ans. (a), (b), (c) and (d) all are identical

Sol. (a) $y = \tan(\cos^{-1} x)$

$= \tan\left(\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right) = \frac{\sqrt{1-x^2}}{x}$

(b) $y = \tan(\cot^{-1}x) = \tan\left(\tan^{-1}\right) = \frac{1}{x}$

(c) $y = \sin(\tan^{-1}x) = \sin\left(\sin^{-1}\frac{x}{\sqrt{1+x^2}}\right) = \frac{x}{\sqrt{1+x^2}}$

(d) $y = \cos(\text{arc}\tan x)$ $\qquad y = \sin\left(\cot^{-1}\frac{x}{1}\right)$

$= = \cos\left(\cos^{-1}\left(\frac{1}{1+x^2}\right)\right)$ $\qquad = \sin\left(\sin^{-1}\frac{1}{\sqrt{1+x^2}}\right)$

$= \frac{1}{\sqrt{1+x^2}}$ $\qquad = \frac{1}{\sqrt{1+x^2}}$

4. Find the domain and range of the following functions. (Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

निम्नलिखित फलनों का प्रान्त व परिसर ज्ञात कीजिए। ([*] व {*} महत्तम पूर्णांक फलन व अंशिक फलन निम्न है)

(i) $f(x) = \cot^{-1}(2x - x^2)$ $\qquad$ (ii) $f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$

(iii) $f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right)$ $\qquad$ (iv) $f(x) = \tan^{-1}\left(\log_{\frac{4}{5}}\left(5x^2 - 8x + 4\right)\right)$

Ans. (i) D : $x \in R$ R : $[\pi/4, \pi)$

(ii) D: $x \in \left(n\pi, n\pi + \frac{\pi}{2}\right) - \left\{x \middle| x = n\pi + \frac{\pi}{4}\right\}$ $n \in I$; R : $\left[\frac{\pi}{3}, \frac{2\pi}{3}\right] - \left[\frac{\pi}{2}\right]$

(iii) D : $x \in R$ $\qquad$ R : $\left[0, \frac{\pi}{2}\right)$

(iv) D : $x \in R$ $\qquad$ R : $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right]$

Sol. (i) $f(x) = \cot^{-1}(2x - x^2)$

$2x - x^2 \in R$

$-x^2 + 2x$

$$\left(-\infty, -\frac{D}{4a}\right]$$

$$\left(-\infty, +\frac{4}{4(+1)}\right] \Rightarrow (-\infty, 1]$$

$$\text{Rang} \Rightarrow \left[\frac{\pi}{4}, \pi\right)$$

(ii) $f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$

$$\left|\log_3 \tan x + \log_{\tan x} 3\right| \geq -1$$

$$\left|\log_3 \tan x + \frac{1}{\log_3 \tan x}\right| \geq 2$$

$\tan x > 0 \qquad \tan x \neq 1$

$$x \in \left(n\pi, n\pi + \frac{\pi}{2}\right) - \left\{n\pi + \frac{\pi}{4}\right\}$$

$$\text{Range } \sin^{-1}\left[(-\infty, -2] \cup [2, \infty)\right]$$

$$\left[\frac{\pi}{3}, \frac{2\pi}{3}\right] - \left[\frac{\pi}{2}\right]$$

(iii) $\qquad f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right)$

$$-1 \leq \frac{\sqrt{2x^2+1}}{x^2+1} \leq 1$$

$$\frac{\sqrt{2x^2+1}}{x^2+1} + 1 \geq 0 \qquad\qquad \frac{\sqrt{2x^2+1}}{x^2+1} - 1 \leq 0$$

$$\frac{\sqrt{2x^2+1} + x^2 + 1}{x^2+1} \geq 0 \qquad\qquad \frac{\sqrt{2x^2+1} + x^2 + 1}{x^2+1} \leq 0$$

$x \in R$ $\qquad \sqrt{2x^2+1} \le x^2+1$

$x^2 \ge 0$

$x \in R$

Domain $\Rightarrow x \in R$

Range $\Rightarrow \left[0, \frac{\pi}{2}\right)$

(iv) $\quad f(x) = \tan^{-1}\left(\log_{\frac{4}{5}}\left(5x^2 - 8x + 4\right)\right)$

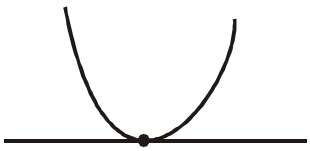
5. Let $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$ be a function defined $R \to \left(0, \frac{\pi}{2}\right]$ then find the complete set of real values of α for which $f(x)$ is onto.

$f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$ फलन $R \to \left(0, \frac{\pi}{2}\right]$ में परिभाषित है तो α के वह सभी मान ज्ञात कीजिए जिसके लिये $f(x)$ आच्छादक है।

Ans. $\frac{1 \pm \sqrt{17}}{2}$

Sol. $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$ $R \to \left(0, \frac{\pi}{2}\right]$

$[0, \infty)$



$D = 0$

$16 - 4(\alpha^2 - \alpha) = 0$

$\alpha^2 - \alpha - 4 = 0$

$\alpha = \frac{1 \pm \sqrt{17}}{2}$

6. If the range of the function $\cos^{-1}|1-x^2|$ is $\left[a, \frac{\pi}{b}\right]$ then find the value of b.

यदि फलन $\cos^{-1}|1-x^2|$ का परिसर $\left[a, \frac{\pi}{b}\right]$ है, तो b का मान ज्ञात कीजिए।

Ans. 2

Sol. $\cos^{-1}|1-x^2|$

$-1 \le |1-x^2| \le 1$

$0 \le |1-x^2| \le 1$

$\cos^{-1} 0 = \frac{\pi}{2} \quad \cos^{-1}(1) = 0$

$\left[0, \frac{\pi}{2}\right]$

EVALUATION

7. (a) Find the following

(a) निम्नलिखित के मान ज्ञात कीजिये।

(i) $\tan\left[\cos^{-1}\frac{1}{2} + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right]$

(ii) $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$

(iii) $\cos\left(\tan^{-1}\frac{3}{4}\right)$

(iv) $\tan\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$

(b) Find the following :

निम्नलिखित के मान ज्ञात कीजिये :

(i) $\sin\left[\frac{\pi}{2} - \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)\right]$

(ii) $\cos\left[\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \frac{\pi}{6}\right]$

(iii) $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

(iv) $\sin\left(\frac{1}{4}\text{arc}\sin\frac{\sqrt{63}}{8}\right)$

Ans. (a) (i) $\frac{1}{\sqrt{3}}$, (ii) $\frac{5\pi}{6}$, (iii) $\frac{4}{5}$, (iv) $\frac{17}{6}$ (b) (i) $\frac{1}{2}$, (ii) -1, (iii) $-\frac{\pi}{4}$, (iv) $\frac{\sqrt{2}}{4}$

Sol. (i) $\tan\left[\cos^{-1}\frac{1}{2}+\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right]$

$\tan\left[\tan^{-1}\sqrt{3}-\tan^{-1}\frac{1}{\sqrt{3}}\right]$

$\tan\left(\frac{\pi}{3}-\frac{\pi}{6}\right)$

$\tan\frac{\pi}{6}=\left(\frac{1}{\sqrt{3}}\right)$

(ii) $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$

$\pi-\frac{\pi}{6}\Rightarrow\frac{5\pi}{6}$

(iii) $\cos\left(\tan^{-1}\frac{3}{4}\right)=\cos\left(\cos^{-1}\frac{4}{5}\right)=\frac{4}{5}$

(iv) $\tan\left(\sin^{-1}\frac{3}{5}+\cot^{-1}\frac{3}{2}\right)$

$\tan\left[\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]$

$\tan\left[\tan^{-1}\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{1}{2}}\right]$

$=\tan\left[\tan^{-1}\left(\frac{9+8}{12\times\frac{1}{2}}\right)\right]=\frac{17}{6}$

(b) (i) $\sin\left[\frac{\pi}{2}-\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)\right]$

$$= \sin\left[\frac{\pi}{2}+\frac{\pi}{3}\right] = \cos\frac{\pi}{3} = \frac{1}{2}$$

(ii) $\cos\left[\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)+\frac{\pi}{6}\right]$

$$= \cos\left[\frac{5\pi}{6}+\frac{\pi}{6}\right] = \cos\pi = (1)$$

(iii) $\tan^{-1}\left(\tan\frac{3\pi}{4}\right) = -\frac{\pi}{4}$

(iv) $\sin\left(\frac{1}{4}\text{arc }\sin\frac{\sqrt{63}}{8}\right)$

$$\sin^{-1}\frac{\sqrt{63}}{8} = \theta \Rightarrow \sin\theta = \frac{\sqrt{63}}{8}$$

$\sin\frac{\theta}{4} = ?$

$$\sin\theta = \frac{\sqrt{63}}{8}$$

$$\cos\theta = \frac{1}{8}$$

$$\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}} = \frac{3}{4}$$

$$\sin\frac{\theta}{4} = \sqrt{\frac{1-\cos\frac{\theta}{2}}{2}} = \sqrt{\frac{1-\frac{3}{4}}{2}}$$

$$\Rightarrow \frac{1}{\sqrt{8}} = \frac{\sqrt{2}}{4}$$

PROPERTIES OF ITF

8. Show that : $\sin^{-1}\left(\sin\frac{33\pi}{7}\right)+\cos^{-1}\left(\cos\frac{46\pi}{7}\right)+\tan^{-1}\left(-\tan\frac{13\pi}{8}\right)+\cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)=\frac{13\pi}{7}$

प्रदर्शित कीजिए : $\sin^{-1}\left(\sin\frac{33\pi}{7}\right)+\cos^{-1}\left(\cos\frac{46\pi}{7}\right)+\tan^{-1}\left(-\tan\frac{13\pi}{8}\right)+\cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)=\frac{13\pi}{7}$

Ans.

Sol. $\sin^{-1}\left(\sin\frac{33\pi}{7}\right)+\cos^{-1}\left(\cos\frac{46\pi}{7}\right)+\tan^{-1}\left(-\tan\frac{13\pi}{8}\right)+\cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)$

LHS $5\pi-\frac{33\pi}{7}+\frac{46\pi}{7}-6\pi-\frac{3\pi}{8}+\pi-\left(\frac{19\pi}{8}-2\pi\right)$

$=\frac{13\pi}{7}$

9. Find the simplest value of:

मान ज्ञात कीजिये :

(a) $f(x)=\text{arc}\cos x+\text{arc}\cos\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right),\ x\in\left(\frac{1}{2},1\right)$

(b) $f(x)=\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right),\ x\in R-\{0\}$

Ans. **(a)** $\frac{\pi}{3}$; **(b)** $\frac{1}{2}\tan^{-1}x$

Sol. $y=\cos^{-1}x+\cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right),\ x\in\left(\frac{1}{2},1\right)$

$\cos^{-1}x=\theta \qquad \theta\in\left(0,\frac{\pi}{3}\right)$

$\cos\theta=x$

$\Rightarrow\theta+\cos^{-1}\left(\frac{\cos\theta}{2}+\frac{\sqrt{3}}{2}\sqrt{2}\theta\right)$

$\theta+\cos^{-1}\left(\cos\left(\theta-\frac{\pi}{3}\right)\right) \qquad 0<\theta<\frac{\pi}{3}$

$=\theta-\theta+\frac{\pi}{3}=\frac{\pi}{3}$

(b) $f(x)=\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) \qquad x\in R-\{0\}$

$\Rightarrow x=\tan\theta$

$$= \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$\Rightarrow \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right)$$

$$\Rightarrow \frac{\theta}{2} = \frac{\tan^{-1} x}{2}$$

10. Evaluate : $\left|\tan^{-1}\left(\frac{x-y}{1+xy}\right) + \tan^{-1}\left(\frac{y-z}{1+yz}\right) + \tan^{-1}\left(\frac{z-x}{1+zx}\right)\right|$

(i) when x, y, z all are of same sign.

(ii) When not all three are of the same sign and (one positive × one negative) < -1 and (one positive × one negative) > -1.

हल कीजिए : $\left|\tan^{-1}\left(\frac{x-y}{1+xy}\right) + \tan^{-1}\left(\frac{y-z}{1+yz}\right) + \tan^{-1}\left(\frac{z-x}{1+zx}\right)\right|$

(i) जब x, y, z सभी के समान चिन्ह हो।

(ii) जब सभी के समान समान चिन्ह नहीं हो (एक धनात्मक×एक ऋणात्मक) < -1 तथा (एक धनात्मक×एक ऋणात्मक) > -1

Ans. (i) 0 (ii) π

Sol. $\left|\tan^{-1}\left(\frac{x-y}{1+xy}\right) + \tan^{-1}\left(\frac{y-z}{1+yz}\right) + \tan^{-1}\left(\frac{z-x}{1+zx}\right)\right|$

(i) $|\tan^{-1}(x) - \tan^{-1} y + \tan^{-1} y - \tan^{-1} z + \tan^{-1} z - \tan^{-1} x|$

$= 0$

(ii) $x, y > 0 \qquad z < 0$

$x(z) < -1 \qquad y(z) > -1$

$z = -t$

$xt > 1 \qquad yt < 1$

$$\Rightarrow \left|\tan^{-1}\left(\frac{x-y}{1+xy}\right) + \tan^{-1}\left(\frac{y+t}{1-yt}\right) + \tan^{-1}\left(\frac{-t-x}{1-tx}\right)\right|$$

$$\left|\tan^{-1} x - \tan^{-1} y + \tan^{-1} y + \tan^{-1} t - \left(\pi + \tan^{-1}\left(\frac{t+x}{1-tx}\right)\right)\right|$$

$\Rightarrow |-\pi|$

$= \pi$

ADDITION OR SUBTRACTION OF TWO OR MORE ANGLES

11. Prove that : $\sin \cot^{-1} \tan \cos^{-1} x = \sin \operatorname{cosec}^{-1} \cot \tan^{-1} x = x$ where $x \in (0, 1]$

सिद्ध कीजिय : $\sin \cot^{-1} \tan \cos^{-1} x = \sin \operatorname{cosec}^{-1} \cot \tan^{-1} x = x$ where $x \in (0, 1]$

Ans.

Sol. $\sin\left(\cot^{-1}\left(\tan\left(\cos^{-1}\frac{x}{1}\right)\right)\right)$

$\Rightarrow \sin\left(\cot^{-1}\left(\tan\left(\tan^{-1}\frac{\sqrt{1-x^2}}{x}\right)\right)\right)$

$\Rightarrow \sin\left(\cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right)$

$\Rightarrow \sin\left(\sin^{-1}\left(\frac{x}{1}\right)\right)$

$= x$

$\sin \operatorname{cosec}^{-1}\left(\cot\left(\cot^{-1}\left(\frac{1}{x}\right)\right)\right)$

$\Rightarrow \sin\left(\operatorname{cosec}^{-1}\left(\cot^{-1}\left(\frac{1}{x}\right)\right)\right)$

$\Rightarrow \sin\left(\operatorname{cosec}^{-1}\left(\frac{1}{x}\right)\right)$

$\Rightarrow \sin(\sin^{-1}(x)) = x$

12. Prove the identities :

निम्न का सत्यापन कीजिये :

(a) $\sin^{-1}\cos(\sin^{-1}x)+\cos^{-1}\sin(\cos^{-1}x)=\frac{\pi}{2},\ |x|\le 1$

(b) $2\tan^{-1}(\operatorname{cosec}\tan^{-1}x-\tan\cot^{-1}x)=\tan^{-1}x \quad (x\neq 0)$

(c) $\tan(\tan^{-1}x+\tan^{-1}y+\tan^{-1}z)=\cot(\cot^{-1}x+\cot^{-1}y+\cot^{-1}z)$

Ans.

Sol. (a) $\sin^{-1}\cos(\sin^{-1}x)+\cos^{-1}\sin(\cos^{-1}x)=\frac{\pi}{2}$

LHS

$$\Rightarrow \sin^{-1}\left(\sqrt{1-x^2}\right)+\cos^{-1}\left(\sqrt{1-x^2}\right)$$

$$\Rightarrow \frac{\pi}{2}$$

(b) $2\tan^{-1}(\operatorname{cosec}\tan^{-1}x-\tan\cot^{-1}x)=\tan^{-1}x$

$$\text{LHS } 2\tan^{-1}\left(\operatorname{cosec}\left(\operatorname{cosec}^{-1}\frac{\sqrt{1+x^2}}{x}\right)-\tan\tan^{-1}\frac{1}{x}\right)$$

$$\Rightarrow 2\tan^{-1}\left(\frac{\sqrt{1+x^2}}{x}-\frac{1}{x}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{2\frac{\left(\sqrt{1+x^2}-1\right)}{x}}{1-\frac{\left(\sqrt{1+x^2}-1\right)^2}{x^2}}\right)=\tan^{-1}\left(\frac{\frac{2\left(\sqrt{1+x^2}-1\right)}{x}}{\frac{x^2-1-x^2-1+2\sqrt{1+x^2}}{x^2}}\right)\Rightarrow \tan^{-1}(x)$$

(c) $\tan(\tan^{-1}x+\tan^{-1}y+\tan^{-1}z)=\cot(\cot^{-1}x+\cot^{-1}y+\cot^{-1}z)$

$$\text{LHS} \Rightarrow \tan\left[\frac{\pi}{2}-\cot^{-1}x+\frac{\pi}{2}-\cot^{-1}y+\frac{\pi}{2}-\cot^{-1}z\right]$$

$$\tan\left[\frac{3\pi}{2}-\left(\cot^{-1}x+\cot^{-1}y+\cot^{-1}z\right)\right]$$

$\Rightarrow \cot\left(\cot^{-1} x + \cot^{-1} y + \cot^{-1} z\right)$

ADDITION OR SUBSTRACTION OF TWO OR MORE ANGLES

13. Let $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$, $\beta = \cos^{-1}\left(\frac{4}{5}\right)$ and $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$, find $(\alpha + \beta + \gamma)$ and hence prove that

यदि $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$, $\beta = \cos^{-1}\left(\frac{4}{5}\right)$ और $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$, $(\alpha + \beta + \gamma)$ ज्ञात कीजिए और सिद्ध कीजिए

(i) $\sum \cot\alpha = \prod \cot\alpha$, (ii) $\sum \tan\alpha \cdot \tan\beta = 1$

Ans. $\pi/2$

Sol. (i) $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$, $\beta = \cos^{-1}\left(\frac{4}{5}\right)$ and $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$

$$\alpha+\beta+\gamma = \sin^{-1}\left(\frac{36}{85}\right)+\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{8}{15}\right)$$

$$= \tan^{-1}\left(\frac{36}{77}\right)+\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{8}{15}\right)$$

$$= \tan^{-1}\left(\frac{36}{77}\right)+\tan^{-1}\left(\frac{\frac{3}{4}+\frac{8}{15}}{1-\frac{24}{60}}\right)$$

$$= \tan^{-1}\left(\frac{36}{77}\right)+\tan^{-1}\left(\frac{45+32}{60-24}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{36}{77}\right)+\tan^{-1}\left(\frac{77}{36}\right)$$

$$\Rightarrow \frac{\pi}{2}$$

(ii) $\cot\alpha + \cot\beta + \cot\gamma = \cot\alpha \cos\beta \cot\gamma$

$$\frac{1}{\tan\alpha}+\frac{1}{\tan\beta}+\frac{1}{\tan\gamma} = \frac{1}{\tan\alpha\tan\beta\tan\gamma}$$

$\tan\alpha \ \tan\beta + \tan\beta + \tan\gamma + \tan\gamma\tan\alpha = 1$

$$\alpha+\beta+\gamma = \frac{\pi}{2}$$

$\tan(\alpha+\beta+\gamma) : \tan\left(\frac{\pi}{2}\right)$

(ii) $\sum \tan\alpha \cdot \tan\beta = 1$

14. Prove that :

सिद्ध कीजिय :

(a) $2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25} = \pi$

(b) $\cos^{-1}\left(\frac{5}{13}\right) + \cos^{-1}\left(-\frac{7}{25}\right) + \sin^{-1}\frac{36}{325} = \pi$

(c) $\text{arc}\cos\sqrt{\frac{2}{3}} - \text{arc}\cos\frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$

Ans.

Sol. (a) $2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25} = \pi$

A B C

$2\cos^{-1}\left(\frac{3}{\sqrt{13}}\right) = A$

$\cos\frac{A}{2} = \frac{3}{\sqrt{13}}$

$\Rightarrow \tan\frac{A}{2} = \frac{2}{3}$

$\tan A = \frac{2\tan\frac{A}{2}}{1-\tan^{-1}\frac{A}{2}} = \frac{2\times\frac{2}{3}}{1-\frac{4}{9}} = \frac{\frac{4}{3}}{\frac{5}{9}}$

$\tan A = \frac{12}{5} \Rightarrow A = \tan^{-1}\left(\frac{12}{5}\right)$

$\cot^{-1}\left(\frac{16}{63}\right) = B$

$$\cot^{-1} B = \frac{16}{63}$$

$$\tan B = \frac{63}{16} \Rightarrow B = \tan^{-1}\left(\frac{63}{16}\right)$$

$$\frac{1}{2}\cos^{-1}\frac{7}{25} = C \Rightarrow \cos 2C = \frac{7}{25}$$

$$\frac{1-\tan^2 C}{1+\tan^2 C} = \frac{7}{25}$$

$\Rightarrow 25 - 25 \tan^2 C = 7 + 7 \tan^2 C$

$$\tan^2 C = \frac{9}{16} \Rightarrow \tan C = \frac{3}{4} \Rightarrow C = \tan^{-1}\left(\frac{3}{4}\right)$$

$\Rightarrow A + B + C$

$$= \tan^{-1}\left(\frac{12}{5}\right) + \tan^{-1}\left(\frac{63}{16}\right) + \tan^{-1}\left(\frac{3}{4}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{12}{5}\right) + \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \pi + \tan^{-1}\left(\frac{\frac{12}{5}+\frac{3}{4}}{1-\frac{12}{5}\times\frac{3}{4}}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \pi + \tan^{-1}\left(\frac{48+15}{20-36}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$\Rightarrow \pi + \tan^{-1}\left(\frac{-63}{16}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$\Rightarrow \pi$

(b) $$\cos^{-1}\left(\frac{5}{13}\right) + \cos^{-1}\left(-\frac{7}{25}\right) + \sin^{-1}\frac{36}{325} = \pi$$

$$\Rightarrow \tan^{-1}\left(\frac{12}{5}\right) + \pi - \tan^{-1}\left(\frac{24}{7}\right) + \tan^{-1}\left(\frac{36}{323}\right)$$

$$\Rightarrow \pi + \tan^{-1}\left(\frac{\frac{12}{5}-\frac{24}{7}}{1+\frac{12}{5}\times\frac{24}{7}}\right) + \tan^{-1}\left(\frac{36}{323}\right)$$

$$\Rightarrow \pi + \tan^{-1}\left(\frac{84-120}{35+288}\right) + \tan^{-1}\left(\frac{36}{323}\right)$$

$$\Rightarrow \pi + \tan^{-1}\left(-\frac{36}{323}\right) + \tan^{-1}\left(\frac{36}{323}\right)$$

$$\Rightarrow \pi$$

(c) $\cos\sqrt{\frac{2}{3}} - \text{arc}\cos\frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$

$$\text{LHS} \Rightarrow \tan^{-1}\frac{1}{\sqrt{2}} - \tan^{-1}\left(\frac{\sqrt{3}-\sqrt{2}}{\sqrt{6}+1}\right)$$

$$\Rightarrow \tan^{-1}\frac{1}{\sqrt{2}} - \left[\tan^{-1}\left(\sqrt{3}\right) - \tan^{-1}\sqrt{2}\right]$$

$$\Rightarrow \cot^{-1}\sqrt{2} - \tan^{-1}\sqrt{3} + \tan^{-1}\sqrt{2}$$

$$\Rightarrow \frac{\pi}{2} - \frac{\pi}{3}$$

$$= \frac{\pi}{6}$$

15. If α and β are the roots of the equation $x^2 + 5x - 49 = 0$ then find the value of $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$.

यदि α व β समीकरण $x^2 + 5x - 49 = 0$ के मूल है तो $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$ का मान ज्ञात कीजिए।

Ans. 10

Sol. $x^2 + 5x - 49 = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$\alpha + \beta = -5 \quad \alpha\beta = -49$

$\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$

$\cot A = \alpha \qquad \cot B = \beta$

$$\cot(A+B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$$

$$\Rightarrow \frac{\alpha\beta - 1}{\alpha + \beta} = \frac{-49-1}{-5} = \frac{-50}{-5} = 10$$

16. If $a > b > c > 0$ then find the value of : $\cot^{-1}\left(\frac{ab+1}{a-b}\right)+\cot^{-1}\left(\frac{bc+1}{b-c}\right)+\cot^{-1}\left(\frac{ca+1}{c-a}\right)$.

यदि $a > b > c > 0$ तो मान ज्ञात कीजिये : $\cot^{-1}\left(\frac{ab+1}{a-b}\right)+\cot^{-1}\left(\frac{bc+1}{b-c}\right)+\cot^{-1}\left(\frac{ca+1}{c-a}\right)$

Ans. π

Sol. $\cot^{-1}\left(\frac{ab+1}{a-b}\right)+\cot^{-1}\left(\frac{bc+1}{b-c}\right)+\cot^{-1}\left(\frac{ca+1}{c-a}\right)$

$\tan^{-1}\left(\frac{a-b}{1+ab}\right)+\tan^{-1}\left(\frac{b-c}{1+bc}\right)+\tan^{-1}\left(\frac{c-a}{1+ca}\right)$

$\Rightarrow \tan^{-1} a - \tan^{-1} b + \tan^{-1} b - \tan^{-1} c + \pi + \tan^{-1} c - \tan^{-1} a$

$= \pi$

ITF EQUATIONS

17. Prove that : $\tan^{-1} n + \cot^{-1} (n+1) = \tan^{-1} (n^2 + n + 1)$.

सिद्ध करो कि : $\tan^{-1} n + \cot^{-1} (n+1) = \tan^{-1} (n^2 + n + 1)$.

Ans.

Sol. $\tan^{-1} n + \cot^{-1} (n+1) = \tan^{-1} (n^2 + n + 1)$

LHS $\Rightarrow \tan^{-1}(n)+\tan^{-1}\left(\frac{1}{n+1}\right)$

$\Rightarrow \tan^{-1}\left(\frac{n+\frac{1}{n+1}}{1-\frac{n}{n+1}}\right)$

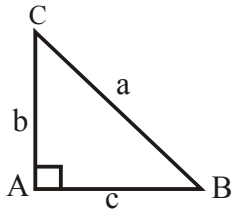
$\Rightarrow \tan^{-1}\left(\frac{n^2+n+1}{n+1-n}\right)=\tan^{-1}\left(n^2+n+1\right)$

18. In a triangle ABC if $\angle A = 90^\circ$, then prove that, $\tan^{-1}\frac{b}{c+a}+\tan^{-1}\frac{c}{a+b}=\frac{\pi}{4}$.

यदि त्रिभुज ABC में $\angle A = 90^\circ$ तब सिद्ध कीजिए कि $\tan^{-1}\frac{b}{c+a}+\tan^{-1}\frac{c}{a+b}=\frac{\pi}{4}$

Ans.

Sol.



$a^2 = b^2 + c^2$

$$\Rightarrow \tan^{-1}\left(\frac{b}{a+c}\right)+\tan^{-1}\left(\frac{c}{a+b}\right)=\frac{\pi}{4}$$

$$\text{LHS } \tan^{-1}\left(\frac{\frac{b}{a+c}+\frac{c}{a+b}}{1-\frac{bc}{(a+b)(a+c)}}\right)$$

$$\Rightarrow \tan^{-1}\left[\frac{ab+b^2+ac+c^2}{a^2+ac+ab+bc-bc}\right]=\tan^{-1}\left[\frac{ab+ac+a^2}{a^2+ac+ab}\right]$$

$$\Rightarrow \frac{\pi}{4}$$

19. Solve for x and y : $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$ and $\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$.

x व y के लिए समीकरण $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$ एवं $\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$ को हल कीजिए।

Ans. $x = \frac{1}{2}, y = 1$

Sol. $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$

$$\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$$

$$\frac{\pi}{2} - \sin^{-1}x - \frac{\pi}{2} + \sin^{-1}y = \frac{\pi}{3}$$

$$-\sin^{-1} x + \sin^{-1} y = \frac{\pi}{3}$$

$$2\sin^{-1} = \pi$$

$$\sin^{-1} = \frac{\pi}{2} \Rightarrow y = 1$$

$$\sin^{-1} x + \frac{\pi}{2} = \frac{2\pi}{3}$$

$$\sin^{-1} x = \frac{2\pi}{3} - \frac{\pi}{2} = \frac{\pi}{6}$$

$$x = \frac{1}{2}$$

20. Find all possible value of 'p' and 'q' for which $\cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$.

'p' व 'q' के सभी सम्भव मान ज्ञात करो जो कि समीकरण $\cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$

Ans. $0 \le p \le 1$ and $q = \frac{1}{2}$

Sol. $\cos^{-1}\sqrt{p} + \cos^{-1}\sqrt{1-p} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$

$$\tan^{-1}\frac{\sqrt{1-p}}{\sqrt{p}} + \tan^{-1}\frac{\sqrt{p}}{\sqrt{1-p}} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$$

$$\tan^{-1}\frac{\sqrt{1-p}}{\sqrt{p}} + \cot^{-1}\frac{\sqrt{1-p}}{\sqrt{p}} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$$

$$\frac{\pi}{2} + \cos^{-1}\sqrt{1-q} = \frac{3\pi}{4}$$

$$\cos^{-1}\sqrt{1-q} = \frac{\pi}{4}$$

$$\sqrt{1-q} = \frac{1}{\sqrt{2}}$$

$$1-q = \frac{1}{2}$$

$$q = \frac{1}{2}$$

$p \in [0, 1]$

21. Solve the following equations :

निम्नलिखित को हल कीजिये :

(a) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$

(b) $\tan^{-1}\frac{x-1}{x+1} + \tan^{-1}\frac{2x-1}{2x+1} = \tan^{-1}\frac{23}{36}$

(c) $\cos^{-1}\frac{x^2-1}{x^2+1} + \tan^{-1}\frac{2x}{x^2-1} = \frac{2\pi}{3}$

Ans. (a) $x = 0, \frac{1}{2}, -\frac{1}{2}$ (b) $x = \frac{4}{3}$ (c) $x = 2-\sqrt{3}$ or $\sqrt{3}$

Sol. (a) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$

A B C D

$A + B + C = D$

$\tan(A + B + C) = \tan(D)$

$$\frac{\tan A + \tan B + \tan C - \tan A + \tan B + \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = 3x$$

$$\frac{3x - x\left(x^2-1\right)}{1-x^2+x-x^2-x-x^2+1} = 3x$$

$$\frac{x\left(3-x^2+1\right)}{2-3x^2} = 3x$$

$\Rightarrow x = 0$

$4 - x^2 = 6 - 9x^2$

$8x^2 = 2$

$$\Rightarrow x^2 = \frac{1}{4}$$

$$\Rightarrow x = \pm\frac{1}{2}$$

(b) $\tan^{-1}\frac{x-1}{x+1} + \tan^{-1}\frac{2x-1}{2x+1} = \tan^{-1}\frac{23}{36}$

$A + B = C$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan C$$

$$\frac{\frac{x-1}{x+1} + \frac{2x-1}{2x+1}}{1 - \frac{x-1}{x+1}\times\frac{2x-1}{2x+1}} = \frac{23}{36}$$

$$\frac{2x^2 + x - 2x - 1 + 2x^2 + 2x - x - 1}{2x^2 + x + 2x + 1 - 2x^2 + x + 2x - 1} = \frac{23}{36}$$

$$\frac{4x^2 - 2}{6x} = \frac{23}{12}$$

$24x^2 - 12 = 23x$

$24x^2 - 23x - 12 = 0$

$24x^2 - 32x + 9x - 12 = 0$

$8x(3x - 4) + 3(3x - 4) = 0$

$$x = \frac{4}{3}$$

(c) $\cos^{-1}\frac{x^2 - 1}{x^2 + 1} + \tan^{-1}\frac{2x}{x^2 - 1} = \frac{2\pi}{3}$

$$x = \tan\theta \Rightarrow \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cos^{-1}\left(-\left(\frac{1 - \tan^{-1}\theta}{1 + \tan^{-1}\theta}\right)\right) + \tan^{-1}\left(-\frac{(2\tan\theta)}{1 - \tan^{-1}\theta}\right) = \frac{2\pi}{3}$$

$$\pi - \cos^{-1}(\cos 2\theta) - \tan^{-1}(\tan 2\theta) = \frac{2\pi}{3}$$

SUMMATION OF SERIES

22. $\tan^{-1}\frac{1}{x^2+x+1} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7} + \tan^{-1}\frac{1}{x^2+7x+13}$ to n terms, where $x > 0$.

$\tan^{-1}\frac{1}{x^2+x+1} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7} + \tan^{-1}\frac{1}{x^2+7x+13}$ n पदों तक , जहां $x > 0$

Ans. $\tan^{-1}(x+n) - \tan^{-1} x$

Sol. $\tan^{-1}\left(\frac{1}{x^2+x+1}\right) + \tan^{-1}\left(\frac{1}{x^2+3x+3}\right) + \tan\left(\frac{1}{x^2+5x+7}\right) + \tan^{-1}\left(\frac{1}{x^2+7x+13}\right)$

$1 + 3 + 5 + \ldots\ldots = n^2$

$S = 1 + 3 + 7 + 13 + \ldots\ldots\ldots\ldots T_n$

$S = \quad 1 + 3 + 7 + \ldots\ldots\ldots\ldots T_{n-1} + T_n$

$S - S = 1 + 2 + 4 + 6 + \ldots\ldots\ldots - T_n$

$$T_n = 1 + \frac{n-1}{2}\left[2\times 2 + (n-1)2\right]$$

$T_n = n^2 - n + 1$

$$T_n = \tan^{-1}\left[\frac{1}{x^2 + (2n-1)x + (n^2 - n + 1)}\right]$$

$$= \tan^{-1}\left[\frac{1}{1 + x^2 + nx + (n-1)x + n(x-1)}\right]$$

$$= \tan^{-1}\left[\frac{1}{1 + x(n+x) + (n-1)(n+x)}\right]$$

$$= \tan^{-1}\left[\frac{1}{1 + (n+x)(n+x-1)}\right]$$

$$= \tan^{-1}\left[\frac{(n+x) - (x+n-1)}{1 + (n+x)(n+x-1)}\right]$$

$T_n = \tan^{-1}(n+x) - \tan^{-1}(n+x-1)$

$T_1 = \tan^{-1}(1+x) - \tan^{-1}(x)$

$T_2 = \tan^{-1}(2+x) - \tan^{-1}(1+x)$

$T_3 = \tan^{-1}(3+x) - \tan^{-1}(2+x)$

$\vdots \quad \vdots \quad \vdots$

$T_n = \tan^{-1}(n+x) - \tan^{-1}(n+x-1)$

$S_n = \tan^{-1}(n+x) - \tan^{-1} x$

23. $\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\frac{\sqrt{2}-1}{\sqrt{6}} + \ldots\ldots + \sin^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} + \ldots\ldots\infty$

Ans. $\pi/2$

Sol. $\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\frac{\sqrt{2}-1}{\sqrt{6}} + \ldots\ldots + \sin^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} + \ldots\ldots\infty$

$T_n = \sin^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n}\sqrt{n+1}}\right)$

$n^2 + x - n - n + 1 - 2\sqrt{n(n-1)}$

$n^2 - n + 1 - 2\sqrt{n(n-1)}$

$1 + n(n-1) - 2\sqrt{n(n-1)}$

$\left(n^2+n\right) - \left(n + n - 1 - 2\sqrt{n(n-1)}\right)$

$n^2 + n - n - n + 1 + 2\sqrt{n(n-1)}$

$\sqrt{n^2 - n + 1 + 2\sqrt{n(n-1)}}$

$\left(n\sqrt{n-1}\right)^2 + 1 + n2\sqrt{n(n-1)}$

$\sqrt{n(n-1)} + 1$

$S_n = \tan^{-1}\sqrt{n} - 0$

$S_n = \tan^{-1} + \infty$

$= \frac{\pi}{2}$

Paragraph for Question Nos. 24 to 25

Consider the function $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ and $h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

माना फलन $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ तथा $h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

24. If $x \in (-1, 1)$, then the solution of the equation $f(x) + g(x) + h(x) = \frac{\pi}{2}$, is equal to :

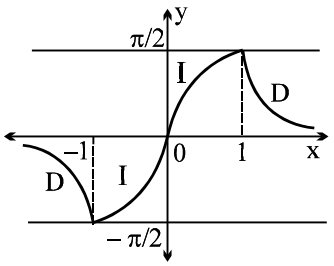
यदि $x \in (-1, 1)$ $x \in (-1, 1)$ है तो समीकरण $f(x) + g(x) + h(x) = \frac{\pi}{2}$ का हल होगा :

(A) $\sqrt{2} - 1$ (B) $2 - \sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2}$

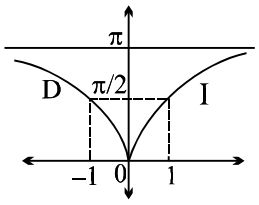
Ans. B

Sol. starting with $x = \tan\theta$

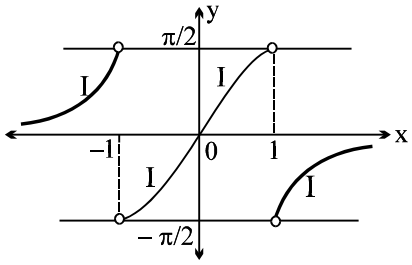
$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & -1 \le x \le 1 \\ \pi - 2\tan^{-1}x & \text{if} \quad x \ge 1 \\ -\pi - 2\tan^{-1}x & x \le -1 \end{cases}$$



$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & x \ge 0 \\ -2\tan^{-1}x & x < 0 \end{cases}$$



$$h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} \pi + 2\tan^{-1}x & x < -1 \\ 2\tan^{-1}x & -1 < x < 1 \\ 2\tan^{-1}x - \pi & x > 1 \end{cases}$$



in $(0, 1)$, $f(x) + g(x) + h(x) = \frac{\pi}{2}$

$2\tan^{-1}x + 2\tan^{-1}x + 2\tan^{-1}x = \frac{\pi}{2} \Rightarrow \tan^{-1}x = \frac{\pi}{12} \Rightarrow x = \tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}$ Ans.

in $(-1, 0)$, $2\tan^{-1}x - 2\tan^{-1}x + 2\tan^{-1}x = \frac{\pi}{2} \Rightarrow \tan^{-1}x = \frac{\pi}{4} \Rightarrow x = \tan\left(\frac{\pi}{4}\right) = 1$

25. $f(2) + g(2) + h(2)$ has the value equal to :

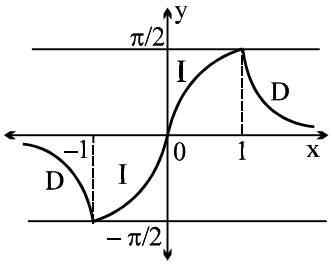
$f(2) + g(2) + h(2)$ का मान होगा :

(A) $\cot^{-1}\left(\frac{-4}{3}\right)$ (B) $\cot^{-1}\left(\frac{-3}{4}\right)$ (C) $\pi - \tan^{-1}\left(\frac{2}{3}\right)$ (D) $\cos^{-1}\left(\frac{-3}{4}\right)$

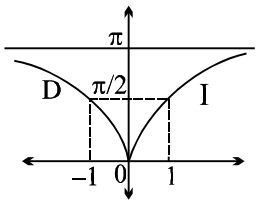
Ans. B

Sol. starting with $x = \tan\theta$

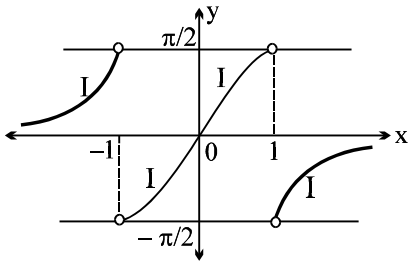
$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & -1 \le x \le 1 \\ \pi - 2\tan^{-1}x & \text{if} \quad x \ge 1 \\ -\pi - 2\tan^{-1}x & x \le -1 \end{cases}$$



$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & x \ge 0 \\ -2\tan^{-1}x & x < 0 \end{cases}$$



$$h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} \pi + 2\tan^{-1}x & x < -1 \\ 2\tan^{-1}x & -1 < x < 1 \\ 2\tan^{-1}x - \pi & x > 1 \end{cases}$$



Obvious from the graphs. $f(x) \to 2$; $g(x) \to 1$; $h(x) \to 0 \Rightarrow$ 3 critical points. Ans.

Paragraph for Question Nos. 26 to 28

Given that $\tan^{-1}\left(\frac{2x}{1-x^2}\right)=\begin{cases}2\tan^{-1}x & , \quad |x|<1\\ -\pi+2\tan^{-1}x & , \quad x>1\\ \pi+2\tan^{-1}x & , \quad x<-1\end{cases}$

$$\sin^{-1}\frac{2x}{1+x^2}=\begin{cases}2\tan^{-1}x & \text{if} \quad |x|\le 1\\ \pi-2\tan^{-1}x & \text{if} \quad x>1\\ -(\pi+2\tan^{-1}x) & \text{if} \quad x<-1\end{cases}$$

and $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$ for $-1\le x\le 1$

दिया गया है कि $\tan^{-1}\left(\frac{2x}{1-x^2}\right)=\begin{cases}2\tan^{-1}x & , \quad |x|<1\\ -\pi+2\tan^{-1}x & , \quad x>1\\ \pi+2\tan^{-1}x & , \quad x<-1\end{cases}$

$$\sin^{-1}\frac{2x}{1+x^2}=\begin{cases}2\tan^{-1}x & \text{यदि} \quad |x|\le 1\\ \pi-2\tan^{-1}x & \text{यदि} \quad x>1\\ -(\pi+2\tan^{-1}x) & \text{यदि} \quad x<-1\end{cases}$$

तथा $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$, जहाँ $-1\le x\le 1$

26. $\sin^{-1}\left(\frac{4x}{x^2+4}\right)+2\tan^{-1}\left(-\frac{x}{2}\right)$ is independent from x then :

$\sin^{-1}\left(\frac{4x}{x^2+4}\right)+2\tan^{-1}\left(-\frac{x}{2}\right)$, x से स्वतंत्र है, तो :

(A) $x\in[-3,4]$ (B) $x\in[-2,2]$ (C) $x\in[-1,1]$ (D) $x\in[1,\infty]$

Ans. B

Sol. $\sin^{-1}\left(\frac{4x}{x^2+4}\right)+2\tan^{-1}\left(-\frac{x}{2}\right)$

$$f(x)=\sin^{-1}\left(\frac{2\left(\frac{x}{2}\right)}{1+\left(\frac{x}{2}\right)^2}\right)-2\tan^{-1}\left(\frac{x}{2}\right)$$

$\frac{x}{2} = \tan\theta$

$= \sin^{-1}\left(\sin(2\theta)\right) - 2\tan^{-1}\left(\tan\theta\right)$

$\left|\frac{x}{2}\right| \le 1$

$-1 \le \frac{x}{2} = 1$

$-2 \le x \le 2$

27. If $\cos^{-1}\frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2\tan^{-1}3x$, then $x \in$

(A) $\left(\frac{1}{3}, \infty\right)$ (B) $(-1, \infty)$ (C) $(-\infty, -1)$ (D) None of these

यदि $\cos^{-1}\frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2\tan^{-1}3x$, तो $x \in$

(A) $\left(\frac{1}{3}, \infty\right)$ (B) $(-1, \infty)$ (C) $(-\infty, -1)$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\cos^{-1}\frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2\tan^{-1}3x$

$\cos^{-1}\left(\frac{3x+3x}{1+(3x)(3x)}\right) = -\frac{\pi}{2} + 2\tan^{-1}(3x)$

$\Rightarrow 3x = \tan\theta$ $\qquad |3x| \le 1 \Rightarrow |x| \le \frac{1}{3}$

$\cos^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$\Rightarrow \cos^{-1}\left(\sin 2\theta\right) = \frac{\pi}{2} - \sin^{-1}\left(\sin 2\theta\right)$

$\Rightarrow \frac{\pi}{2} - 2\theta$

$$\frac{\pi}{2} - 2\tan^{-1} 3x + \frac{\pi}{2} = 2\tan^{-1}(3x)$$

$4\tan^{-1}(3x) = \pi$

$$\tan^{-1}(3x) = \frac{\pi}{4}$$

$3x = 1$

$$x = \frac{1}{3}$$

where $3x > 1 \Rightarrow x > \frac{1}{3} \Rightarrow x \in \left(\frac{1}{3}, \infty\right)$ .....(2)

$\pi = \pi$

where $3x < 1 \Rightarrow x < \frac{-1}{3}\ x \in \left(-\infty \frac{-1}{3}\right)$ .....(3)

$\left(\pi + 2\tan^{-1} x\right) + 2\tan^{-1} x \neq \pi$

28. If $(x-1)(x^2+1) > 0$, then $\sin\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2} - \tan^{-1} x\right) =$

(A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) -1 (D) None of these

यदि $(x-1)(x^2+1) > 0$, तो $\sin\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2} - \tan^{-1} x\right) =$

(A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) -1 (D) इनमें से कोई नहीं

Ans. C

Sol. $(x-1)(x^2+1) > 0 \Rightarrow x > 1$

$$\sin\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2} - \tan^{-1} x\right)$$

$\tan^{-1} x = \theta$

$x = \tan\theta$

$$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$\sin^{-1}\left(\frac{2x}{1-x^2}\right)=\tan^{-1}(\tan 2\theta)\Rightarrow(2\theta-\pi);2\theta\in\left(\frac{\pi}{2},\pi\right)$$

$$\tan^{-1}(\tan\theta);\theta-0;\theta\in\left(\frac{\pi}{4},\frac{\pi}{2}\right)$$

$$\sin^{-1}\left[\frac{2\theta-\pi}{2}-\theta\right]$$

$$\sin^{-1}\left[\theta-\frac{\pi}{2}-\theta\right]$$

$$\Rightarrow -1$$

ONE OR MORE THAN ONE CORRECT TYPE

29. In the following functions defined from [-1, 1] to [-1, 1] the functions which are not bijective are :

अन्तराल [–1, 1] से [–1, 1] में परिभाषित कौनसे फलन एकैकी आच्छादक नहीं होंगे :

(A) $\sin(\sin^{-1}x)$ (B) $\frac{2}{\pi}\sin^{-1}(\sin x)$ (C) $(\text{sgn } x)\ln e^x$ (D) $x^3\,\text{sgn } x$

Ans. BCD

Sol. $f:[-1,1]\to[-1,1]$

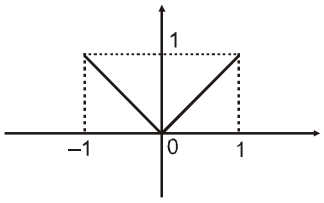
(A) $f(x)=\sin(\sin^{-1}x)=x,\quad x\in[-1,1]$ Bijective function एकैकी आच्छादक फलन $y\in[-1,1]$

(B) $f(x)=\frac{2}{\pi}\sin^{-1}(\sin x)=\frac{2x}{\pi},\quad x\in[-1,1]$

Not bijective एकैकी आच्छादक फलन नहीं है। $y\in\left[\frac{-2}{\pi},\frac{2}{\pi}\right]$

(C) $f(x)\,(\text{sgn } x)\,(\ell n\, e^x)$

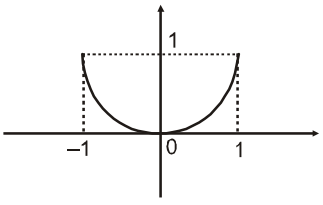
$$=\begin{cases}x & , \quad x\in(0,1]\\ 0 & , \quad x=0\\ -x & , \quad x\in[-1,0)\end{cases}$$



Not bijective एकैकी आच्छादक फलन नहीं है।

(D) $f(x)=x^3\,\text{sgn } x$

$$=\begin{cases}x^3 & , \quad x>0\\ 0 & , \quad x=0\\ -x^3 & , \quad x<0\end{cases}$$



Not bijective एकैकी आच्छादक फलन नहीं है।

30. If α, β, γ are three angles given by $\alpha = 2\tan^{-1}\left(\sqrt{2}-1\right)$, $\beta = 3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ and $\gamma = \cos^{-1}\frac{1}{3}$, then:

यदि α, β, γ तीन कोण निम्न प्रकार दिए गए $\alpha = 2\tan^{-1}\left(\sqrt{2}-1\right)$, $\beta = 3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ तथा $\gamma = \cos^{-1}\frac{1}{3}$, तब :

(A) $\alpha > \beta$ (B) $\beta > \gamma$ (C) $\alpha < \gamma$ (D) $\beta < \gamma$

Ans. BC

Sol. $\alpha = 2\tan^{-1}\left(\sqrt{2}-1\right) = 2 \times 15 = 30 = \frac{\pi}{6} = \frac{2\pi}{12}$

$$\beta = 3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$$

$$= 3\times\frac{\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$$

$$\gamma = \cos^{-1}\left(\frac{1}{3}\right)$$

$$\frac{\pi}{3} < \gamma < \frac{\pi}{2}$$

$$\frac{4\pi}{12} < \gamma < \frac{6\pi}{12}$$

31. Which of the following are identical functions? (where [x] denotes greatest integer less than or equal x, {x} denotes fractional part of x and sgn x denotes signum function of x respectively.)

निम्न में से कौन से फलन समान हैं? (जहाँ [x], महत्तम पूर्णांक को निरुपित करता है जो x अथवा x से छोटा है एवं {x}, x के भिन्नात्मक भाग को तथा sgn x सिग्नम फलन को दर्शाता है)

(A) $f(x) = \text{sgn}\left(|x|+1\right)$ (B) $g(x) = \sin^2(\ln x) + \cos^2(\ln x)$

(C) $h(x) = \frac{2}{\pi}\left(\sin^{-1}\{x\} + \cos^{-1}\{x\}\right)$ (D) $k(x) = \sec^2[\{x\}] - \tan^2\{[x]\}$

Ans. ACD

Sol. (A) $\text{sgn}\left(|x|+1\right) = 1 \ \forall\ x \in R$.

(B) $\sin^2(\ln x) + \cos^2(\ln x) = 1 \ \forall\ x \in R^+$

(C) $\frac{2}{\pi}\left(\sin^{-1}\{x\}+\cos^{-1}\{x\}\right)=1 \ \forall\ x \in R.$

(D) $\sec^2[\{x\}]-\tan^2\{[x]\} = \sec^2 0° - \tan^2 0° = 1 \ \forall\ x \in R.$

32. The value of a for which $ax + \sec^{-1}\sqrt{2x^2-x^4} + \text{cosec}^{-1}\sqrt{2x^2-x^4} = 0$:

a का मान ज्ञात कीजिये जिसके लिये $ax + \sec^{-1}\sqrt{2x^2-x^4} + \text{cosec}^{-1}\sqrt{2x^2-x^4} = 0$:

(A) $\frac{\pi}{2}$ (B) $-\frac{\pi}{2}$ (C) $\frac{2}{\pi}$ (D) $-\frac{2}{\pi}$

Ans. AB

Sol. for existance of $\sec^{-1}(\sqrt{2x^2-x^4})$ or $\text{cosec}^{-1}(\sqrt{2x^2-x^4})$

$2x^2 - x^4 \geq 1$

i.e. $(x^2-1)^2 \leq 0$ hence $x^2 = 1 \Rightarrow x = \pm 1$

if $x = 1,\ a + \frac{\pi}{2} = 0;$

if $x = -1,\ -a + \frac{\pi}{2} = 0 \Rightarrow a = \pm\frac{\pi}{2}$ Ans.

33. Let function f(x) be defined as $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$. Then which of the following is/are TRUE?

(A) f(x) is injective in its domain (B) f(x) is many-one in its domain

(C) Range of f is a singleton set (D) $\text{sgn}(f(x)) = 1$ where sgn x denotes signum function of x

माना फलन f(x) परिभाषित है $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$ तो निम्न में से कौनसा/कौनसे कथन सत्य है?

(A) f(x) स्वयं की प्रान्त में एकैकी है (B) f(x) स्वयं की प्रान्त में बहुएकैकी है

(C) f(x) का परिसर एकल समुच्चय है (D) $\text{sgn}(f(x)) = 1$ जहाँ sgn x, x का सिग्नम फलन है

Ans. AD

Sol. $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$

Domain of f(x) is $\{-1, 1\}$

$f(1) = \frac{\pi}{2}, f(-1) = \frac{3\pi}{2}$

So function f(x) is injective

$\text{sgn}(f(x)) = 1 \qquad (f(x) > 0)$

Range of $f(x) = \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$

34. Which of the following statement(s) is/are TRUE?

(A) Domain of $y = \cos^{-1}(e^x)$ is same as range of $y = -\sqrt{-x}$

(B) Number of elements common in the range of function $y = \tan^{-1}(\text{sgn } x)$ and $y = \cot^{-1}(\text{sgn } x)$ is only 1 (where sgn x denotes signum function of x)

(C) The function $y = \text{sgn}(\cot^{-1}x)$ and $y = 1$ are identical functions

(D) Number of integers in the solution set of $1 < \log_2(\tan^{-1}x) < 2$ is 4

निम्न में से कौन कौन से कथन सत्य हैं?

(A) $y = \cos^{-1}(e^x)$ का प्रान्त तथा $y = -\sqrt{-x}$ का परिसर समान है

(B) $y = \tan^{-1}(\text{sgn } x)$ और $y = \cot^{-1}(\text{sgn } x)$ के परिसर में उभयनिष्ठ अवयवों की संख्या 1 है

(C) $y = \text{sgn}(\cot^{-1}x)$ और $y = 1$ समान फलन है

(D) $1 < \log_2(\tan^{-1}x) < 2$ के हल समुच्चय में पूर्णांकों की संख्या 4 है

Ans. ABC

Sol. (A) For domain of $y = \cos^{-1}(e^x)$, $0 < e^x \le 1 \Rightarrow x \le 0$ or $x \in (-\infty, 0]$

For range of $y = -\sqrt{-x}$, $\quad -y = \sqrt{-x}$

$\sqrt{-x} \ge 0 \Rightarrow -y \ge 0 \Rightarrow y \in (-\infty, 0]$

$\therefore$ Domain of $y = \cos^{-1}(e^x)$ is identical with range of $y = -\sqrt{-x}$.

(B) $y = \tan^{-1}(\text{sgn } x) = \begin{cases} \tan^{-1}(-1), & \forall\, x < 0 \\ \tan^{-1}0, & \text{for } x = 0 \\ \tan^{-1}(1), & \forall\, x > 0 \end{cases} \Rightarrow$ Range of $y = \tan^{-1}(\text{sgn } x)$ is $\left\{\frac{-\pi}{4}, 0, \frac{\pi}{4}\right\}$

Also, $y = \cot^{-1}(\text{sgn } x) = \frac{\pi}{2} - \tan^{-1}(\text{sgn } x) \Rightarrow$ Range of $y = \cot^{-1}(\text{sgn } x)$ is $\left\{\frac{3\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4}\right\}$

$\therefore$ Number of elements common in the range of $y = \tan^{-1}(\text{sgn } x)$ and $y = \cot^{-1}\text{sgn } x$ is only one i.e $\frac{\pi}{4}$.

(C) $y = \text{sgn}(\cot^{-1}x) = 1 \quad \forall\, x \in R$

$\therefore$ $y = \text{sgn}(\cot^{-1}x)$ and $y = 1$ are identical functions.

(D) $1 < \log_2(\tan^{-1}x) < 2 \Rightarrow 2^1 < \tan^{-1}x < 2^2 \Rightarrow$ Not possible. $\left(\text{As } \frac{-\pi}{2} < \tan^{-1}x < \frac{\pi}{2} < 2\right)$

35. $2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$ is equal to :

$2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$ का मान होगा :

(A) $\cot^{-1} \frac{44}{117}$ (B) $\text{cosec}^{-1} \frac{125}{117}$ (C) $\tan^{-1} \frac{4}{117}$ (D) $\cos^{-1} \frac{44}{125}$

Ans. ABD

Sol. $2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$

$$2\tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{4}{3}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{2}{7}}{1-\frac{1}{49}}\right) + \tan^{-1}\left(\frac{4}{3}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{2\times 7}{48}\right) + \tan^{-1}\left(\frac{4}{3}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{7}{24}+\frac{4}{3}}{1-\frac{7}{24}\times\frac{4}{3}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{7+32}{24}}{\frac{18-7}{18}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{39}{4\times 11}\times 3\right)$$

$$\tan^{-1}\left(\frac{117}{44}\right)$$

36. If $\tan^2(\tan^{-1}x) = \text{cosec}^{-1}(\text{cosec}7) + \sec(\sec^{-1}8) + \cot^{-1}(\cot(-6))$, then $\tan^{-1}(\tan(18 + x^2))$:

(A) is always positive (B) is always negative (C) is irrational (D) is rational

यदि $\tan^2(\tan^{-1}x) = \text{cosec}^{-1}(\text{cosec}7) + \sec(\sec^{-1}8) + \cot^{-1}(\cot(-6))$, तो $\tan^{-1}(\tan(18 + x^2))$:

(A) सदैव धनात्मक होगा (B) सदैव ऋणात्मक होगा (C) अपरिमेय होगा (D) परिमेय होगा

Ans. BC

Sol. $\tan^2(\tan^{-1}x) = \text{cosec}^{-1}(\text{cosec}7) + \sec(\sec^{-1}8) + \cot^{-1}(\cot(-6))$

$x^2 = 7 - 2\pi + 8 - 2\pi - 6 - (-6)\,\pi$

$x^2 = 9 + 2\pi$

$\tan^{-1}\tan(18 + x^2) = \tan^{-1}(\tan(18 + 9 + 2\pi))$

$\Rightarrow \tan^{-1}(\tan 27)$

$\Rightarrow 27 - 9\pi$

37. Let $\tan^{-1}\left(\tan\frac{5\pi}{4}\right) = \alpha, \tan^{-1}\left(-\tan\frac{2\pi}{3}\right) = \beta$. Then :

माना $\tan^{-1}\left(\tan\frac{5\pi}{4}\right) = \alpha, \tan^{-1}\left(-\tan\frac{2\pi}{3}\right) = \beta$ है, तो :

(A) $\alpha > \beta$ (B) $4\alpha - 3\beta = 0$ (C) $\alpha + \beta = \frac{7\pi}{12}$ (D) $\alpha + \beta = \pi$

Ans. BC

Sol. $\tan^{-1}\tan\left(\frac{5\pi}{4}\right) = \alpha$

$\alpha = \tan^{-1}\tan\left(\frac{\pi}{4}\right)$

$\alpha = \frac{\pi}{4}$

$\beta = \tan^{-1}\left(-\tan\frac{2\pi}{3}\right)$

$\beta = -\tan^{-1}\left(\tan\left(\pi - \frac{\pi}{3}\right)\right)$

$\beta = +\tan^{-1}\left(\tan\frac{\pi}{3}\right)$

$\beta = +\frac{\pi}{3}$

38. $2\tan\left(\tan^{-1}(x)+\tan^{-1}(x^3)\right)$ where $x \in R-\{-1, 1\}$ is equal to :

$2\tan\left(\tan^{-1}(x)+\tan^{-1}(x^3)\right)$ जहां $x \in R-\{-1, 1\}$ बराबर होगा :

(A) $\frac{2x}{1-x^2}$ (B) $\tan(2\tan^{-1}x)$

(C) $\tan\left(\cot^{-1}(-x)-\cot^{-1}(x)\right)$ (D) $\tan(2\cot^{-1}x)$

Ans. ABC

Sol. $2\tan\left(\tan^{-1}(x)+\tan^{-1}(x^3)\right)$

$$\Rightarrow 2\tan\left(\tan^{-1}\left(\frac{x+x^3}{1-x^4}\right)\right)$$

$$=2\tan\left[\tan^{-1}\left(\frac{x\left(x^2+1\right)}{\left(1-x^2\right)\left(1+x^2\right)}\right)\right]$$

$$=2\tan\left[\tan^{-1}\left(\frac{x}{1-x^2}\right)\right]$$

$$=\frac{2x}{1-x^2}$$

(D) $\tan(2\cot^{-1}x)$ $\cot^{-1}x=\theta;\theta\in(0,\pi)$

$\cot\theta=x$

$$\tan\theta=\frac{1}{x}$$

$$\tan 2\theta=\frac{2\tan\theta}{1-\tan^{-1}\theta}=\frac{\frac{2}{x}}{1-\frac{1}{x^2}}=\frac{2x}{x^2-1}$$

(C) $\tan\left(\cot^{-1}(-x)-\cot^{-1}x\right)$

$\Rightarrow \tan\left(\pi-2\cot^{-1}x\right)$

$=-\tan(2\cot^{-1}x)$

$\Rightarrow \cot^{-1}x=\theta$

$=\tan 2\theta$

$$= \frac{-2\tan\theta}{1-\tan^2\theta} = \frac{-\frac{2}{x}}{1-\frac{1}{x^2}} = \frac{2x}{1-x^2}$$

(B) $\tan(2\tan^{-1}x)$

$\Rightarrow \tan^{-1} x = \theta$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2 x} = \frac{2x}{1-x^2}$$

39. For the equation $2x = \tan(2\tan^{-1}a) + 2\tan(\tan^{-1}a + \tan^{-1}a^3)$, which of the following is invalid?

समीकरण $2x = \tan(2\tan^{-1}a) + 2\tan(\tan^{-1}a + \tan^{-1}a^3)$ के लिए निम्न में से कौन सा असत्य है?

(A) $a^2x + 2a = x$ (B) $a^2 + 2ax + 1 = 0$ (C) $a \neq 0$ (D) $a \neq -1, 1$

Ans. BC

Sol. $2x = \tan(2\tan^{-1}a) + 2\tan(\tan^{-1}a + \tan^{-1}a^3)$

$$2x = \tan\left(\tan^{-1}\left(\frac{2a}{1-a^2}\right)\right) + 2\tan\left(\tan^{-1}\left(\frac{a+a^3}{1-a^4}\right)\right)$$

$$= \frac{2a}{1-a^2} + \frac{a}{1-a^2}$$

$$x = \frac{2a}{1-a^2}$$

$x - xa^2 = 2a$

$xa^2 + 2a = x$

$a \neq \pm 1$

40. If integral pair (x, y) satisfy the equation $\arctan\frac{1}{x} + \arctan\frac{1}{y} = \arctan\frac{1}{10}$, then sum of $x + y$ can be :

यदि पूर्णांक युग्म (x, y) समीकरण $\arctan\frac{1}{x} + \arctan\frac{1}{y} = \arctan\frac{1}{10}$ का संतुष्ट करते है तो $x + y$ का मान हो सकता/सकते है :

(A) 122 (B) 100 (C) 92 (D) – 92

Ans. A

Sol. $\tan^{-1}\left(\frac{1}{x}\right)+\tan^{-1}\left(\frac{1}{y}\right)=\tan^{-1}\left(\frac{1}{10}\right)$

$$=\tan^{-1}\left(\frac{\frac{1}{x}+\frac{1}{y}}{1-\frac{1}{xy}}\right)$$

$$\tan^{-1}\left(\frac{x+y}{2y-1}\right)=\tan^{-1}\left(\frac{1}{10}\right)$$

$$\frac{x+y}{xy-1}=\frac{1}{10}$$

$10x + 10y = xy - 1$

$10x - xy + 10y + 1 = 0$

$10x + 10y - xy + 1 = 0$

$x(10 + y) + 10y - y = -y$

$x(10 - y) + 10y - 10y - 100 = -101$

$x(10 - y) - 10y - 10y - 100 = -101$

$(x - 10) - 10(10 - y) = 101$

$(x - 10)(y - 10) = 101$

$x - 10 = 101$	$y - 10 = 1$	$x - 10 = 1$	$y - 10 = 101$
$x = 111$	$y = 11$	$x = 11$	$y = 111$
$x - 10 = -1,$	$y - 10 = -101$	$x - 10 = -101$	$y - 10 = -1$
$x = 9$	$y = -91$	$x = -91$	$y = 9$

MATCH THE COLUMN

41. Match the following column :

Column-I	**Column-II**
(A) Period of $\frac{1}{2}\left\{\frac{\lvert\sin x\rvert}{\cos x}+\frac{\lvert\cos x\rvert}{\sin x}\right\}$	(P) 2
(B) Range of $\sin^{-1}\sqrt{\log_{[x]}\frac{\lvert x\rvert}{x}}$	(Q) 2π

(C) Total number of solution $x^2 - 4 - [x] = 0$ (R) 1

(D) Period of $f(x) = \sin^2 x + \cos x$ (S) {0}

[where [ ] denotes G.I.F]

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $\frac{1}{2}\left\{\frac{\lvert\sin x\rvert}{\cos x}+\frac{\lvert\cos x\rvert}{\sin x}\right\}$ का आवर्तकाल होगा	(P) 2
(B) $\sin^{-1}\sqrt{\log_{[x]}\frac{\lvert x\rvert}{x}}$ का परिसर होगा	(Q) 2π
(C) $x^2 - 4 - [x] = 0$ के कुल हल होंगे	(R) 1
(D) $f(x) = \sin^2 x + \cos x$ का आवर्तकाल होगा	(S) {0}

[जहाँ [ ] महत्तम पूर्णांक फलन है]

Ans. (A) Q (B) S (C) P (D) Q

Sol. (A) $\frac{1}{2}\left\{\frac{\lvert\sin x\rvert}{\cos x}+\frac{\lvert\cos x\rvert}{\sin x}\right\}$

$P = 2\pi \qquad P = \pi$

LCM $\{2\pi,\ 2\pi\} = 2\pi$

(B) $\sin^{-1}\sqrt{\log_{[x]}\frac{\lvert x\rvert}{x}}$

$\frac{\lvert x\rvert}{x} > 0$ $[x] > 0,\ x > 0$

$x > 1,\ x > 2$

$\sin^{-1}\sqrt{\log_{[x]}(1)}$

$\sin^{-1}\sqrt{0} = 0$

(C) $x^2 - 4 - [x] = 0$

$x^2 = 4 - (x - \{x\}) = 6$

$x^2 - x - 4 + \{x\} = 0$

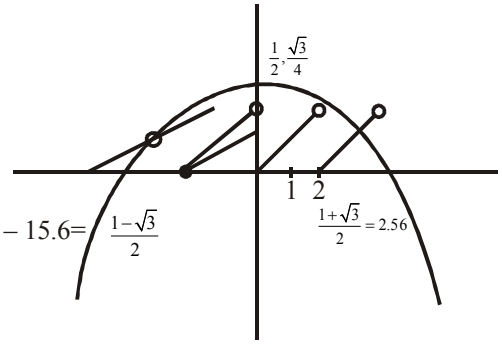
$\{x\} = -x^2 + x + 4$

$[0, 1)$

$-x^2 + x + 4 = 0$

$$x = \frac{-1 \pm \sqrt{1-16}}{-2}$$

$$x = \frac{1 \pm \sqrt{3}}{2}$$



(D) $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

$\pi \qquad 2\pi$

LCM $(\pi, 2\pi) = 2\pi$

42. Match the following column :

Column-I	**Column-II**
(A) $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$	(P) range is $[0, \pi]$
(B) $g(x) = \sin^{-1}\lvert x \rvert + 2\tan^{-1}\lvert x \rvert$	(Q) is increasing $\forall\ x \in (0, 1)$
(C) $h(x) = 2\sin^{-1}\left(\frac{2x}{1+x^2}\right),\ x \in [0, 1]$	(R) period is 2π
(D) $k(x) = \cot(\cot^{-1}x)$	(S) is decreasing $\forall\ x \in (0, 1)$

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$	(P) परिसर $[0, \pi]$ है
(B) $g(x) = \sin^{-1}\lvert x \rvert + 2\tan^{-1}\lvert x \rvert$	(Q) $\forall\ x \in (0, 1)$ के लिए वर्धमान
(C) $h(x) = 2\sin^{-1}\left(\frac{2x}{1+x^2}\right),\ x \in [0, 1]$	(R) आवृत काल 2π है
(D) $k(x) = \cot(\cot^{-1}x)$	(S) $\forall\ x \in (0, 1)$ के लिए ह्रासमान है

Ans. (A) PQR (B) PQ (C) PQ (D) Q

Sol. (A) $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

$2\pi \qquad 2\pi$

LCM $(\pi, 2\pi) = 2\pi$

$$f'(x) = \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{-1-\sin x}{\sqrt{1-\cos^2 x}}$$

$$= \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\sin x}{\sqrt{1-\cos^2 x}}$$

$f'(x) > 0$ in $x \in (0, 1)$

$f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

$\downarrow \qquad\qquad \downarrow$

$x \in \left[\frac{\pi}{2}, \frac{\pi}{2}\right] \qquad\qquad x \in [0, \pi]$

$x \in \left[0, \frac{\pi}{2}\right] \Rightarrow \text{Range} \Rightarrow \left[0, \frac{\pi}{2}\right] + \left[0, \frac{\pi}{2}\right] = [0, \pi]$

(B) $g(x) = \sin^{-1}|x| + 2\tan^{-1}|x|$

$|x| \leq 1$

$\text{Range} \Rightarrow \left[0, \frac{\pi}{2}\right] + 2\left[0, \frac{\pi}{4}\right]$

$\Rightarrow [0, \pi]$

$f(x) = 3x \Rightarrow$ increasing

(C) $h(x) = 2\sin^{-1}\left(\frac{2x}{1+x^2}\right),\ x \in [0, 1]$

$x = \tan\theta \qquad\qquad \theta \in \left[0, \frac{\pi}{4}\right]$

$= 2\sin^{-1}(\sin 2\theta) \qquad 2\theta \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow \qquad 2(2\theta)$

$= 4\theta$

$h(x) = \tan^{-1}(x)$ is increasing in $x \in (0, 1)$

$x \in [0, 1]$

$\text{Range} \Rightarrow [0, \pi]$

(D) $k(x) = \cot(\cot^{-1}x)$ increasing in $(0, 1)$

$\text{Range} = (0, \pi)$

43. If $S_n = \sum_{r=1}^{n} r!$ then for $n > 6$ $\left(\text{given } \sum_{r=1}^{6} r! = 873\right)$: (where [ ] denotes greatest integer function)

यदि $S_n = \sum_{r=1}^{n} r!$ सभी $n > 6$ के लिये $\left(\sum_{r=1}^{6} r! = 873 \right)$ (जहाँ [ ] महत्तम पूर्णांक फलन को दर्शाता है) सही स्तम्भों का मिलान कीजिए।

Column-I	**Column-II**
(A) $\sin^{-1}\left(\sin\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(P) $5 - 2\pi$
(B) $\cos^{-1}\left(\cos\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(Q) $2\pi - 5$
(C) $\tan^{-1}\left(\tan\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(R) $6 - 2\pi$
(D) $\cot^{-1}\left(\cot\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(S) $5 - \pi$
	(T) $\pi - 4$

Ans. (A) P (B) Q (C) P (D) S

Sol. (A) $S_n = \sum_{r=1}^{n} r!$

$$\sin^{-1}\left(\sin\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$$

$S_n = 1! + 2! + 3! + \ldots\ldots 6! + 7! + \ldots\ldots n!$

$S_n = 873 + 7! + 8! + \ldots\ldots + n!$

$S_n = 873 + 7k$

$$\left[\frac{S_n}{7}\right] = \left[\frac{873}{7} + \frac{7k}{7}\right]$$

$$\Rightarrow \left[\frac{873}{7}\right] + k$$

$\Rightarrow 124 + k$

$$\Rightarrow 7\left[\frac{S_n}{7}\right] = 7(124 + k)$$

$= 868 + 7k$

$$\Rightarrow S_n - 7\left[\frac{S_n}{7}\right] = 873 + 7k - 868 - 7k$$

$\Rightarrow 5$

(A) $\sin^{-1}(\sin 5) = 5 - \pi$

(B) $\cos^{-1}(\cos 5) = 2\pi - 5$

(C) $\tan^{-1}(\tan 5) = 5 - 2\pi$

(D) $\cot^{-1}(\cot 5) = 5 - \pi$

44. Let (x, y) be such that $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$.

Match the statements in Column I with statements in Column II and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in the ORS.

Column-I	**Column-II**
(A) If a = 1 and b = 0, then (x, y)	(P) lies on the circle $x^2 + y^2 = 1$
(B) If a = 1 and b = 1, then (x, y)	(Q) lies on $(x^2 - 1)(y^2 - 1) = 0$
(C) If a = 1 and b = 2, then (x, y)	(R) lies on $y = x$
(D) If a = 2 and b = 2, then (x, y)	(S) lies on $(4x^2 - 1)(y^2 - 1) = 0$

यदि (x, y) इस प्रकार है कि $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$ तो स्तम्भ–I का स्तम्भ–II से सही मिलान कीजिए व अपने उत्तरों को

Column-I	**Column-II**
(A) यदि a = 1 व b = 0, तो (x, y)	(P) वृत $x^2 + y^2 = 1$ पर स्थित है
(B) यदि a = 1 व b = 1, तो (x, y)	(Q) $(x^2 - 1)(y^2 - 1) = 0$ पर स्थित है
(C) यदि a = 1 व b = 2, तो (x, y)	(R) $y = x$ पर स्थित है
(D) यदि a = 2 व b = 2, तो (x, y)	(S) $(4x^2 - 1)(y^2 - 1) = 0$ पर स्थित है

Ans. (A) P (B) Q (C) P (D) S

Sol. $\sin^{-1}(ax) + \cos^{-1}y + \cos^{-1}(bxy) = \frac{\pi}{2}$

(A) $a = 1, \ b = 0 \Rightarrow \sin^{-1}(x) + \cos^{-1}(y) + \cos^{-1}(0) = \frac{\pi}{2} \Rightarrow \sin^{-1}x + \cos^{-1}y = 0$

$\Rightarrow \cos^{-1}y = -\sin^{-1}x \Rightarrow \cos^{-1}y = \cos^{-1}\sqrt{1 - x^2} \Rightarrow x^2 + y^2 = 1$

(B) $\sin^{-1}(x) + \cos^{-1}y + \cos^{-1}(xy) = \frac{\pi}{2} \Rightarrow \cos^{-1}(y) + \cos^{-1}(xy) = \cos^{-1}x.$

$\Rightarrow \cos^{-1}\left(xy^2 - \sqrt{(1 - y^2)(1 - x^2y^2)}\right) = \cos^{-1}x. \Rightarrow xy^2 - \sqrt{(1 - y^2)(1 - x^2y^2)} = x$

$\Rightarrow 1 - x^2 - y^2 + x^2y^2 = 0 \Rightarrow (1 - x^2)(1 - y^2) = 0$

(C) $\sin^{-1}(x) + \cos^{-1} y + \cos^{-1}(2xy) = \frac{\pi}{2} \Rightarrow \quad \cos^{-1}\left(2xy^2 - \sqrt{(1-y^2)(1-4x^2y^2)}\right) = \cos^{-1}x.$

$\Rightarrow \quad 2xy^2 - \sqrt{(1-y^2)(1-4x^2y^2)} = x \quad \Rightarrow \quad 2xy^2 - x = \sqrt{(1-y^2)(1-4x^2y^2)}$

$\Rightarrow \quad 4x^2y^4 + x^2 - 4x^2y^2 = 1 - y^2 - 4x^2y^2 + 4x^2y^4 \Rightarrow \quad x^2 + y^2 = 1.$

(D) $\sin^{-1}(2x) + \cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2} \Rightarrow \quad \cos^{-1}\left(2y^2x - \sqrt{(1-y^2)(1-4x^2y^2)}\right) = \cos^{-1}(2x)$

$\Rightarrow \quad 2y^2x - \sqrt{1-y^2-4x^2y^2+4x^2y^4} = 2x. \Rightarrow \quad 1 - 4x^2 - y^2 + 4x^2y^2 = 0$

$\Rightarrow \quad (1-4x^2)(1-y^2) = 0.$

Exercise J-M

1. The number of real roots of the equation $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{4}$ is : **[JEE(Main)July2021]**

समीकरण $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{4}$ के वास्तविक मूलों की संख्या है : **[JEE(Main)July2021]**

(A) 1 (B) 4 (C) 0 (D) 2

Ans. C

Sol. Check domain of LHS

$x(x+1)\geq 0 \qquad 0\leq x^2+x+1\leq 1$

$x(x+1)\geq 0 \qquad 0\leq x^2+x+1\leq 1$

$\Rightarrow x^2+x=0$

$x=0,-1$

Both values of x do not satisfy the given equation.

Ans = 0

2. The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to : **[JEE(Main)July2021]**

$\tan\left(2\tan^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{5}{13}\right)\right)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $\frac{151}{63}$ (B) $\frac{-291}{76}$ (C) $\frac{-181}{69}$ (D) $\frac{220}{21}$

Ans. D

Sol. $2\tan^{-1}\left(\frac{3}{5}\right)=\tan^{-1}\frac{\frac{3}{5}+\frac{3}{5}}{1-\frac{9}{25}}$

$=\tan^{-1}\left(\frac{15}{8}\right)$

$=\tan\left(\tan^{-1}\frac{15}{8}+\tan^{-1}\frac{5}{12}\right)$

$$= \tan\left(\tan^{-1}\frac{\frac{15}{8}+\frac{5}{12}}{1-\frac{75}{96}}\right)$$

$$= \frac{220}{21}$$

3. If the domain of the function $f(x) = \dfrac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\dfrac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$ is equal to :

[JEE(Main)July2021]

यदि फलन, $f(x) = \dfrac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\dfrac{2x-1}{2}\right)}}$ का प्रान्त, अन्तराल $(\alpha, \beta]$ है, तो $\alpha + \beta$ बराबर है :

[JEE(Main)July2021]

(A) 3/2 (B) 1/2 (C) 2 (D) 1

Ans. A

Sol. here

$0 \le x^2 - x + 1 \le 1$

Which is

$x^2 - x + 1 \ge 0$ and $x^2 - x + 1 \le 1$

$x \in R$ ....(1) $x^2 - x \le 0$

$x \in [0, 1]$ ....(2)

by (1) and (2)

$x \in [0, 1]$ ....(3)

Also

$$0 < \sin^{-1}\left(\frac{2x-1}{2}\right) \le \frac{\pi}{2}$$

$$0 < \frac{2x-1}{2} \le 1$$

$$0 < 2x - 1 \le 2$$

$$1 < 2x \le 3$$

$$\frac{1}{2} < x \le \frac{3}{2} \quad ....(4)$$

hence final answer is intersection of (3) and (4)

$x \in \left(\frac{1}{2}, 1\right]$

$\alpha = \frac{1}{2}$

$\beta = 1$

$\alpha + \beta = 3/2$

4. The domain of the function $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is : **[JEE(Main)Aug.2021]**

फलन $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ का प्रांत है : **[JEE(Main)Aug.2021]**

(A) $x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$ (B) $\left[-\frac{1}{2}, 0\right) \cup [0, \infty)$ (C) $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$ (D) $\left(-\frac{1}{2}, \infty\right) - \{0\}$

Ans. A

Sol. $\frac{1+x}{x} \le -1$ or $\frac{1+x}{x} \ge 1$

$\frac{1+2x}{x} \le 0$ or $\frac{1}{x} \ge 1$

$x \in \left[\frac{-1}{2} 0\right)$ or $x \in (0, \infty)$

$x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$

5. If $\sum_{r=1}^{50} \tan^{-1} \frac{1}{2r^2} = p$, then the value of tan p is : **[JEE(Main)Aug.2021]**

यदि $\sum_{r=1}^{50} \tan^{-1} \frac{1}{2r^2} = p$ है, तो tan p का मान है : **[JEE(Main)Aug.2021]**

(A) $\frac{50}{51}$ (B) 100 (C) $\frac{101}{102}$ (D) $\frac{51}{50}$

Ans. A

Sol. $\because \tan^{-1}\frac{1}{2r^2} = \tan^{-1}\left(\frac{2}{1+\left(4r^2-1\right)}\right)$

$= \tan^{-1}\left(\frac{(2r+1)-(2r-1)}{1+(2r+1)(2r-1)}\right)$

$= \tan^{-1}(2r+1) - \tan^{-1}(2r-1)$

$\therefore \sum_{r=1}^{50} \tan^{-1}\left(\frac{1}{2r^2}\right) = \sum_{r=1}^{50}\left(\tan^{-1}(2r+1) - \tan^{-1}(2r-1)\right)$

$\therefore p = \tan^{-1}(101) - \tan^{-1}1$

$= \tan^{-1}\left(\frac{101-1}{1+101\cdot 1}\right)$

$\therefore \tan p = \frac{50}{51}$

6. If $\left(\sin^{-1}x\right)^2 - \left(\cos^{-1}x\right)^2 = a; 0 < x < 1, a \neq 0,$ then the value of $2x^2 - 1$ is : **[JEE(Main)Aug.2021]**

यदि $\left(\sin^{-1}x\right)^2 - \left(\cos^{-1}x\right)^2 = a; 0 < x < 1, a \neq 0$ है, तो $2x^2 - 1$ का मान है : **[JEE(Main)Aug.2021]**

(A) $\sin\left(\frac{4a}{\pi}\right)$ (B) $\cos\left(\frac{2a}{\pi}\right)$ (C) $\cos\left(\frac{4a}{\pi}\right)$ (D) $\sin\left(\frac{2a}{\pi}\right)$

Ans. D

Sol. $(\sin^{-1}x + \cos^{-1}x)(\sin^{-1}x - \cos^{-1}x) = a$

$\Rightarrow \frac{\pi}{2}\left(\frac{\pi}{2} - 2\cos^{-1}x\right) = a$

$\Rightarrow 2\cos^{-1}x = \frac{\pi}{2} - \frac{2a}{\pi}$

$\Rightarrow \cos^{-1}(2x^2-1) = \frac{\pi}{2} - \frac{2a}{\pi}$

$\Rightarrow 2x^2 - 1 = \cos\left(\frac{\pi}{2} - \frac{2a}{\pi}\right)$

$\Rightarrow 2x^2 - 1 = \sin\left(\frac{2a}{\pi}\right)$

7. Let M and m respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $\left[0, \frac{\pi}{2}\right]$.

Then the value of tan(M–m) is equal to : **[JEE(Main)Aug.2021]**

माना $\left[0, \frac{\pi}{2}\right]$ में फलन $f(x) = \tan^{-1}(\sin x + \cos x)$ के अधिकतम तथा न्यूनतम मान क्रमशः M तथा m हैं तो tan(M–m) का मान बराबर है : **[JEE(Main)Aug.2021]**

(A) $3+2\sqrt{2}$ (B) $2-\sqrt{3}$ (C) $3-2\sqrt{2}$ (D) $2+\sqrt{3}$

Ans. C

Sol. $y = \tan^{-1}\left(\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\right)$

$y = \tan^{-1}\left[\sqrt{2}\left[\frac{1}{\sqrt{2}}, 1\right]\right]$

$y = \tan^{-1}\left[1, \sqrt{2}\right]$

$y = \left[\frac{\pi}{4}, \tan^{-1}\sqrt{2}\right]$

Now $m = \frac{\pi}{4}$ $\qquad M = \tan^{-1}\sqrt{2} = \lambda \Rightarrow \tan\lambda = \sqrt{2}$

$= \tan(M-m)$

$= \tan\left(\lambda - \frac{\pi}{4}\right)$

$= \frac{\tan\lambda - 1}{\tan\lambda + 1}$

$= \frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$

$= \left(\sqrt{2}-1\right)^2$

$= 2+1-2\sqrt{2}$

$= 3-2\sqrt{2}$

8. The domain of the function $f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$ is : **[JEE(Main)Aug.2021]**

फलन $f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$ का प्रांत है : **[JEE(Main)Aug.2021]**

(A) $\left[0,\frac{1}{4}\right]$ (B) $\left[\frac{1}{4},\frac{1}{2}\right]\cup\{0\}$ (C) $\left[0,\frac{1}{2}\right]$ (D) $[-2,0]\cup\left[\frac{1}{4},\frac{1}{2}\right]$

Ans. B

Sol. $f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$

$-1\le\frac{3x^2+x-1}{x^2-2x+1}\le 1 \qquad -1\le\frac{x-1}{x+1}\le 1$

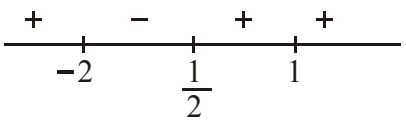
$\frac{3x^2+x-1}{x^2-2x+1}\ge -1$

$\frac{4x^2-x}{(x-1)^2}\ge 0\Rightarrow\frac{x(4x-1)}{(x-1)^2}\ge 0$

$x\in(-\infty,0]\cup\left[\frac{1}{4},1\right)\cup(1_1+\infty)$

$\frac{3x^2+x-1}{x^2-2x+1}-1\le 0$

$\frac{(x+2)(2x-1)}{(x-1)^2}\le 0$

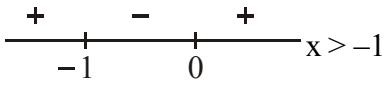


$x\in\left[-2,\frac{1}{2}\right]$

$x\in[-2,0]\cup\left[\frac{1}{4},\frac{1}{2}\right]$

$\frac{x-1}{x+1}\ge -1 \qquad \frac{x-1}{x+1}\le 1$

$\frac{x}{x+1} \geq 0$ $\qquad$ $\frac{1}{x+1} \geq 0$



$x \in (-\infty, -1) \cup [0, +\infty)$

$x \in [0, +\infty)$

$x \in \left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$

9. $\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ is equal to : (The inverse trigonometric functions take the principal values) **[JEE(Main)Aug.2021]**

$\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ बराबर है : (प्रतिलोम त्रिकोणमितीय फलन मुख्य मान लेते हैं ।) **[JEE(Main)Aug.2021]**

(A) $3\pi + 1$ (B) $4\pi - 11$ (C) $4\pi - 9$ (D) $3\pi - 11$

Ans. B

Sol. $\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin 6) - \tan^{-1}(\tan 12)$

$= \cos^{-1}(\cos 5) + \sin^{-1}(\sin 6) - \tan^{-1}(\tan 12)$

$= (2\pi - 5) + 6 - 2\pi - (12 - 4\pi)$

$= 2\pi - 5 + 6 - 2\pi - 12 + 4\pi$

$= 4\pi - 11$

10. The set of all values of k for which $\left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3 = k\pi^3, x \in R$ has a solution in the interval : **[JEE(Main)June2022]**

k के सभी मानों, जिनके लिए $\left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3 = k\pi^3, x \in R$ है, का हल कौनसे अंतराल में है : **[JEE(Main)June2022]**

(A) $\left[\frac{1}{32}, \frac{7}{8}\right)$ (B) $\left(\frac{1}{24}, \frac{13}{16}\right)$ (C) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (D) $\left[\frac{1}{32}, \frac{9}{8}\right)$

Ans. A

Sol. Note : Language is incomplete

$\left(\tan^{-1}(x)\right)^3 + \left(\cot^{-1}(x)\right)^3 = k\pi^3$ has a solution

$\Rightarrow k\pi^3$ will lie in range of LHS

$y = (\tan^{-1}(x))^3 + (\cot^{-1}(x))^3$

put $\tan^{-1}(x) = \alpha$, $\cot^{-1}(x) = \beta$

$$\alpha+\beta=\frac{\pi}{2}$$

$$\alpha\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$$

$y=\alpha^3+\beta^3$

$=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$$y=\frac{\pi^3}{8}-\frac{3\pi}{2}\alpha\left(\frac{\pi}{2}-\alpha\right),\ \alpha\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$$

$$y)_{min}=\frac{\pi^3}{32} \qquad \text{when } \alpha=\frac{\pi}{4}$$

$$y)_{max}\to\frac{7\pi^3}{8} \qquad \text{when } \alpha\to-\frac{\pi}{2}$$

$$y\in\left[\frac{\pi^3}{32},\frac{7\pi^3}{8}\right)$$

$$\Rightarrow k\in\left[\frac{1}{32},\frac{7}{8}\right)$$

11. The domain of the function $f(x)=\dfrac{\cos^{-1}\left(\dfrac{x^2-5x+6}{x^2-9}\right)}{\log_e\left(x^2-3x+2\right)}$ is : **[JEE(Main)June2022]**

फलन $f(x)=\dfrac{\cos^{-1}\left(\dfrac{x^2-5x+6}{x^2-9}\right)}{\log_e\left(x^2-3x+2\right)}$ का प्रांत है : **[JEE(Main)June2022]**

(A) $(-\infty,1)\cup(2,\infty)$ (B) $(2,\infty)$

(C) $\left[-\frac{1}{2},1\right)\cup(2,\infty)$ (D) $\left[-\frac{1}{2},1\right)\cup(2,\infty)-\left\{\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2}\right\}$

Ans. Bonus

Sol. $$-1\le\frac{x^2-5x+6}{x^2-9}\le1$$

$x^2-3x+2>0$

$x^2-3x+2\neq1$

$$\Rightarrow x\in\left[-\frac{1}{2},1\right)\cup(2,\infty)-\left\{\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2},3\right\}$$

Bonus

12. Let $x*y = x^2 + y^3$ and $(x*1)*1 = x*(1*1)$. Then a value of $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ is :

[JEE(Main)June2022]

माना $x*y = x^2 + y^3$ तथा $(x*1)*1 = x*(1*1)$ हैं। तो $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ का एक मान है :

[JEE(Main)June2022]

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. given $x * y = x^2 + y^3$

Now

$(x * 1) * 1 = x * (1 * 1)$

Using the given condition

$(x^2 + 1^3) * 1 = x * (1^2 + 1^3)$

$(x^2 + 1) * 1 = x * (2)$

$((x^2 + 1)^2 + 1) = x^2 + 8$

$x^4 + 2x^2 + 2 = x^2 + 8$

$x^4 + x^2 - 6 = 0$

$(x^2 + 3)(x^2 - 2) = 0$

$x^2 = 2$ is possible answer

Now

$$2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$$

$$2\sin^{-1}\left(\frac{4}{8}\right) = \frac{\pi}{3}$$

13. The value of $\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to : **[JEE(Main)June2022]**

$\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $-\frac{\pi}{4}$ (B) $-\frac{\pi}{8}$ (C) $-\frac{5\pi}{12}$ (D) $-\frac{4\pi}{9}$

Ans. B

Sol. $\tan^{-1}\left(\dfrac{\cos\left(4\pi-\dfrac{\pi}{4}\right)-1}{\sin\dfrac{\pi}{4}}\right)$

$-\tan^{-1}\left(\dfrac{1-\cos\dfrac{\pi}{4}}{\sin\dfrac{\pi}{4}}\right)$

$-\tan^{-1}\left(\tan\dfrac{\pi}{8}\right)=-\dfrac{\pi}{8}$

14. If the inverse trignometric functions take principal values, then $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)+\dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)$ is equal to : **[JEE(Main)June2022]**

यदि प्रतिलोम त्रिकोणमितीय फलन मुख्यमान लेते हैं, तो $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)+\dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)$ बराबर है : **[JEE(Main)June2022]**

(A) 0 (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{\pi}{6}$

Ans. C

Sol. $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)+\dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)=0$

$\Rightarrow \tan\theta=\dfrac{4}{3}$

$\cos^{-1}\left(\dfrac{3}{10}\cos\theta+\dfrac{2}{5}\sin\theta\right)$

$\tan\theta=\dfrac{4}{3}=\dfrac{P}{B}$

$H=\sqrt{25}=25$

$\cos\theta=\dfrac{3}{5}$ $\sin\theta=\dfrac{4}{5}$

$\cos^{-1}\left(\dfrac{3}{10}\times\dfrac{3}{5}+\dfrac{2}{5}\times\dfrac{4}{5}\right)=\cos^{-1}\left(\dfrac{9}{50}+\dfrac{8}{25}\right)$

$$\cos^{-1}\left(\frac{25}{50}\right)=\cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}$$

15. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to : **[JEE(Main)June2022]**

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{11\pi}{12}$ (B) $\frac{17\pi}{12}$ (C) $\frac{31\pi}{12}$ (D) $-\frac{3\pi}{4}$

Ans. A

Sol. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

$\sin^{-1}\left(\sin\frac{\pi}{3}\right)+\cos^{-1}\left(\cos\left(\pi+\frac{\pi}{6}\right)\right)+\tan^{-1}\left(\tan\left(\pi-\frac{\pi}{4}\right)\right)$

$\sin^{-1}\left(\sin\frac{\pi}{3}\right)+\cos^{-1}\left(-\cos\frac{\pi}{6}\right)+\tan^{-1}\left(-\tan\frac{\pi}{4}\right)$

$\frac{\pi}{3}+\pi-\frac{\pi}{6}-\frac{\pi}{4}$

$=11\frac{\pi}{12}$

16. The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is : **[JEE(Main)June2022]**

$\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ का मान है : **[JEE(Main)June2022]**

(A) $\frac{26}{25}$ (B) $\frac{25}{26}$ (C) $\frac{50}{51}$ (D) $\frac{52}{51}$

Ans. A

Sol. $\cot\sum\tan^{-1}\frac{(n+1)-n}{1+n(n+1)}$

$\cot\sum_{n=1}^{50}\tan^{-1}(n+1)-\tan^{-1}n$

$\cot\left(\tan^{-1}51-\tan^{-1}1\right)$

$\cot\tan^{-1}\frac{50}{52}=\frac{52}{50}=\frac{26}{25}$

17. The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is : **[JEE(Main)June2022]**

फलन $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ का प्रांत है : **[JEE(Main)June2022]**

(A) $R-\left\{-\frac{1}{2},\frac{1}{2}\right\}$ (B) $(-\infty,-1]\cup[1,\infty)\cup\{0\}$

(C) $\left(-\infty,\frac{-1}{2}\right)\cup\left(\frac{1}{2},\infty\right)\cup\{0\}$ (D) $\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$

Ans. D

Sol. For domain $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$

$$-1\le\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}-\le1$$

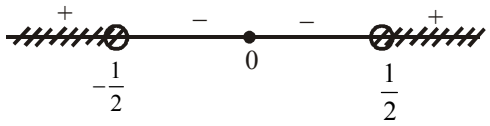
$$\Rightarrow\ -\frac{\pi}{2}\le\sin^{-1}\left(\frac{1}{4x^2-1}\right)\le\frac{\pi}{2}$$

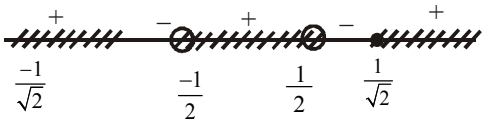
$$\Rightarrow\ -1\le\frac{1}{4x^2-1}\le1$$

$$\Rightarrow\ \frac{1}{4x^2-1}\le1\ \text{and}\ -1\le\frac{1}{4x^2-1}$$

$$\Rightarrow\ 0\le\frac{4x^2-1-1}{4x^2-1}\ \text{and}\ \ 0\le\frac{x^2}{(2x-1)(2x+1)}$$

$$\Rightarrow 0\le\frac{\left(\sqrt{2}x-1\right)\left(\sqrt{2}x+1\right)}{(2x-1)(2x+1)}\ \text{and}\ \le\frac{x^2}{(2x-1)(2x+1)}$$





$x \in \left(-\infty, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

18. $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$ is equal to _____.

[JEE(Main)June2022]

$50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$ का मान बराबर है _____

[JEE(Main)June2022]

Ans. 29

Sol. $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$

$= 50\tan\left(\tan^{-1}\left(\frac{1}{2}\right)+2\tan^{-1}\left(\frac{1}{2}\right)+2\tan^{-1}(2)\right)+4\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)$

$= 50\tan\left(\pi+\tan^{-1}\left(\frac{1}{2}\right)\right)+4$

$= 25 + 4$

$= 29$

19. $\tan\left(2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{\sqrt{5}}{2}+2\tan^{-1}\frac{1}{8}\right)$ is equal to : **[JEE(Main)July2022]**

$\tan\left(2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{\sqrt{5}}{2}+2\tan^{-1}\frac{1}{8}\right)$ बराबर है : **[JEE(Main)July2022]**

(A) 1 (B) 2 (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Ans. B

Sol. $\tan\left(2\left(\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{8}\right)+\tan^{-1}\left(\frac{1}{2}\right)\right)$

$= \tan\left[2\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{2}\right)\right]$

$= 2$

20. If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$, then a value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is : **[JEE(Main)July2022]**

यदि $0 < x < \frac{1}{\sqrt{2}}$ तथा $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$ हैं, तो $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ का एक समान है : **[JEE(Main)July2022]**

(A) $4\sqrt{\left(1-x^2\right)}\left(1-2x^2\right)$ (B) $4x\sqrt{\left(1-x^2\right)}\left(1-2x^2\right)$

(C) $2x\sqrt{\left(1-x^2\right)}\left(1-4x^2\right)$ (D) $4\sqrt{\left(1-x^2\right)}\left(1-4x^2\right)$

Ans. B

Sol. $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta} = k$

$\sin^{-1} x = k\alpha$

$\cos^{-1} x = k\beta$

$k = \frac{\pi}{2(\alpha+\beta)}$ .....(i)

$\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right) = \sin\left(4\sin^{-1} x\right)$

$= 2\sin\left(2\sin^{-1} x\right)\cos\left(2\sin^{-1} x\right)$

$= 4x\sqrt{1-x^2}\left(1-2x^2\right)$

21. Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is : **[JEE(Main)July2022]**

प्रतिलोम त्रिकोणमितीय फलन के केवल मुख्य मान लेते हुए, फलन $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ का प्रांत है: **[JEE(Main)July2022]**

(A) $\left(-\infty, \frac{1}{4}\right]$ (B) $\left[-\frac{1}{4}, \infty\right)$ (C) $\left(-\frac{1}{3}, \infty\right)$ (D) $\left(-\infty, \frac{1}{3}\right]$

Ans. B

Sol. $\left|\frac{x^2+4x+2}{x^2+3}\right| \le 1$

$\Leftrightarrow \left(x^2-4x+2\right)^2 \le \left(x^2+3\right)^2$

$$\Leftrightarrow \left(x^2-4x+2\right)^2-\left(x^2+3\right)^2 \le 0$$

$$\Leftrightarrow \left(2x^2-4x+5\right)\left(-4x-1\right) \le 0$$

$$\Leftrightarrow -4x-1 \le 0 \rightarrow x \ge -\frac{1}{4}$$

22. Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to : **[JEE(Main)July2022]**

प्रतिलोम त्रिकोणमितीय फलन के मुख्य मान लेते हुए समीकरण $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ के सभी हलों का योग है : **[JEE(Main)July2022]**

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Ans. A

Sol. $\cos^{-1} x = 2\sin^{-1} x = \cos^{-1} 2x$

$$\cos^{-1} x - 2\left(\frac{\pi}{2} - \cos^{-1} x\right) = \cos^{-1} 2x$$

$$\cos^{-1} x - \pi + 2\cos^{-1} x = \cos^{-1} 2x$$

$$3\cos^2 x = \pi + \cos^{-1} 2x \quad \text{....(1)}$$

$$\cos\left(3\cos^{-1} x\right) = \cos\left(\pi + \cos^{-1} 2x\right)$$

$$4x^3 - 3x = -2x$$

$$4x^3 = x \Rightarrow x = 0, \pm\frac{1}{2}$$

All satisfy the original equation

$$\text{sum} = -\frac{1}{2} \text{ to } +\frac{1}{2} = 0$$

23. The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is : **[JEE(Main)July2022]**

फलन $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ का प्रांत है : **[JEE(Main)July2022]**

(A) $[1, \infty)$ (B) $[-1, 2]$ (C) $[-1, \infty)$ (D) $(-\infty, 2]$

Ans. C

Sol. $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ Domain

$\frac{x^2-3x+2}{x^2+2x+7} \geq -1$ and $\frac{x^2-3x+2}{x^2+2x+7} \leq 1$

$2x^2 - x + 9 \geq 0$ and $5x \geq -5 \Rightarrow x \geq -1$

$x \in R$

Hence Domain $x \in [-1, \infty)$

24. Let $x = \sin(2\tan^{-1}\alpha)$ and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S = \{\alpha \in R : y^2 = 1 - x\}$, then $\sum_{\alpha \in S} 16\alpha^3$ is equal to _____.

[JEE(Main)July2022]

माना $x = \sin(2\tan^{-1}\alpha)$ तथा $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$ हैं। यदि $S = \{\alpha \in R : y^2 = 1 - x\}$ है, तो $\sum_{\alpha \in S} 16\alpha^3$ बराबर है _____

[JEE(Main)July2022]

Ans. 130

Sol. $\because x = \sin\left(2\tan^{-1}\alpha\right) = \frac{2\alpha}{1+\alpha^2}$ ...(i)

and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right) = \sin\left(\sin^{-1}\frac{1}{\sqrt{5}}\right) = \frac{1}{\sqrt{5}}$

Now, $y2 = 1 - x$

$\frac{1}{5} = 1 - \frac{2\alpha}{1+\alpha^2}$

$\Rightarrow 1 + \alpha^2 = 5 + 5\alpha^2 - 10\alpha$

$\Rightarrow 2\alpha^2 - 5\alpha + 2 = 0$

$\therefore \alpha = 2, \frac{1}{2}$

$\therefore \sum_{\alpha \in S} 16\alpha^3 = 16 \times 2^3 + 16 \times \frac{1}{2^3}$

$= 130$

25. For $k \in R$, let the solutions of the equation $\cos(\sin^{-1}(x \cot(\tan^{-1}(\cos(\sin^{-1} x))))) = k$, $0 < |x| < \frac{1}{\sqrt{2}}$ be α and β, where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2 - bx - 5 = 0$ are $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$, then $\frac{b}{k^2}$ is equal to _____. **[JEE(Main)July2022]**

माना $k \in R$ के लिए समीकरण $\cos(\sin^{-1}(x \cot(\tan^{-1}(\cos(\sin^{-1} x))))) = k$, $0 < |x| < \frac{1}{\sqrt{2}}$ के हल α तथा β हैं, जहाँ प्रतिलोम त्रिकोणमितीय फलन केवल मुख्य मान लेते हैं। यदि समीकरण $x^2 - bx - 5 = 0$ के हल $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ तथा $\frac{\alpha}{\beta}$ है, तो $\frac{b}{k^2}$ बराबर है _____ **[JEE(Main)July2022]**

Ans. 12

Sol. $\cos\left(\sin^{-1} x\right) = \cos\left(\cos^{-1}\sqrt{1-x^2}\right) = \sqrt{1-x^2}$

$$\cot\left(\tan^{-1} 1-x^2\right) = \cot\cot^{-1}\left(\sqrt{\frac{1}{\sqrt{1-x^2}}}\right) = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \cos\left(\sin^{-1}\left(\frac{x}{1-x^2}\right)\right) = \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}} = k$$

$$\Rightarrow 1-2x^2 = k^2\left(1-x^2\right)$$

$$\Rightarrow \left(k^2-2\right)x^2 = k^2-1$$

$$x^2 = \frac{k^2-1}{k^2-2}$$

$$\alpha = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \alpha = \frac{k^2-1}{k^2-2}$$

$$\beta = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \beta^2 = \frac{k^2-1}{k^2-2}$$

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2} = 2\left(\frac{k^2-2}{k^2-1}\right) \& \frac{\alpha}{\beta} = -1$$

$$\text{Sum of roots} = \frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{\alpha}{\beta} = b$$

$$\Rightarrow \frac{2\left(k^2-2\right)}{k^2-1}-1 = b \quad .....(1)$$

$$\text{Products of roots} = \left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}\right)\frac{\alpha}{\beta} = -5$$

$$\Rightarrow \frac{2\left(k^2-2\right)}{k^2-1}(-1)=-5$$

$$\Rightarrow 2k^2-4=5k^2-5$$

$$\Rightarrow 3k^2=1 \Rightarrow k^2=\frac{1}{3} \text{ .....Putin (1)}$$

$$\Rightarrow b=\frac{2\left(k^2-2\right)}{k^2-1}-1=5-1=4$$

$$\frac{b}{k^2}=\frac{4}{\frac{1}{3}}=12$$

Exercise J - A

1. The value of $\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$ is : **[JEE(Adv.)2013]**

$\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$ का मान है : **[JEE(Adv.)2013]**

(A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

Ans. B

Sol. $\cot\sum_{n=1}^{23}\cot^{-1}(1+2+4+6+.....+2n) \Rightarrow \cot\Sigma\cot^{-1}(1+n(n+1))$

$\cot\Sigma\tan^{-1}\frac{(n+1)-n}{1+n(n+1)} \Rightarrow \cot\sum_{n=1}^{23}\left(\tan^{-1}(n+1)-\tan^{-1}n\right)$

$\cot(\tan^{-1}24-\tan^{-1}1) \Rightarrow \cot\left(\tan^{-1}\frac{24-1}{1+24}\right) \Rightarrow \cot\left(\cot^{-1}\frac{25}{23}\right)=\frac{25}{23}$

2. Match Column I with Column II and select the correct answer using the code given below the columns : **[JEE(Adv.)2013]**

Column- I		**Column - II**	
p.	$\left(\frac{1}{y^2}\left(\frac{\cos\left(\tan^{-1}y\right)+y\sin\left(\tan^{-1}y\right)}{\cot\left(\sin^{-1}y\right)+\tan\left(\sin^{-1}y\right)}\right)^2+y^4\right)^{1/2}$ takes value	1.	$\frac{1}{2}\sqrt{\frac{5}{3}}$
Q.	If $\cos x+\cos y+\cos z=0=\sin x+\sin y+\sin z$ then possible value of $\cos\frac{x-y}{2}$ is	2.	$\sqrt{2}$
R.	If $\cos\left(\frac{\pi}{4}-x\right)\cos 2x+\sin x\sin 2x\sec x=\cos x\sin 2x\sec x+\cos\left(\frac{\pi}{4}+x\right)\cos 2x$ then possible value of sec x is	3.	$\frac{1}{2}$
S.	If $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6})), x\neq 0$ then possible value of x is	4.	1

कॉलम I को कॉलम II से सुमेलित कीजिए तथा सूचियों के नीचे दिए गए कोड का प्रयोग करके सही उतर चुनिये :

[JEE(Adv.)2013]

	Column- I		**Column - II**
p.	$\left(\frac{1}{y^2}\left(\frac{\cos\left(\tan^{-1}y\right)+y\sin\left(\tan^{-1}y\right)}{\cot\left(\sin^{-1}y\right)+\tan\left(\sin^{-1}y\right)}\right)^2+y^4\right)^{1/2}$ का मान है	1.	$\frac{1}{2}\sqrt{\frac{5}{3}}$
Q.	यदि $\cos x+\cos y+\cos z=0=\sin x+\sin y+\sin z$ तब $\cos\frac{x-y}{2}$ का सम्भावित मान है	2.	$\sqrt{2}$
R.	यदि $\cos\left(\frac{\pi}{4}-x\right)\cos 2x+\sin x\sin 2x\sec x=\cos x\sin 2x\sec x+\cos\left(\frac{\pi}{4}+x\right)\cos 2x$ तब sec x का सम्भावित मान है	3.	$\frac{1}{2}$
S.	यदि $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6})), x\neq 0$ तब x का सम्भावित मान है	4.	1

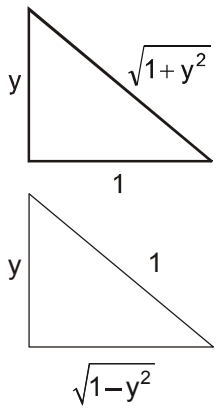
(A) P - 4; Q - 3; R - 1; S - 2 (B) P - 4; Q - 3; R - 2; S - 1

(C) P - 3; Q - 4; R - 2; S - 1 (D) P - 3; Q - 4; R - 1; S - 2

Ans. B

Sol. (P) $\left(\frac{1}{y^2}\left(\frac{\cos(\tan^{-1}y)+y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y)+\tan(\sin^{-1}y)}\right)^2+y^4\right)^{1/2}$

$$=\left[\frac{1}{y^2}\left[\frac{\left(\frac{1}{\sqrt{1+y^2}}+\frac{y.y}{\sqrt{1+y^2}}\right)}{\left(\frac{\sqrt{1-y^2}}{y}+\frac{y}{\sqrt{1-y^2}}\right)}\right]^2+y^4\right]^{1/2}$$



$$=\left(\frac{1}{y^2}.y^2(1-y^4)+y^4\right)^{1/2}=1$$

Ans. 4

(Q) $\cos x+\cos y=-\cos z$

$\sin x+\sin y=-\sin z$ square and add वर्ग करके जोडने पर

$2+2\cos(x-y)=1$

$\Rightarrow \quad \cos(x-y)=-1/2$

$\Rightarrow \quad 2\cos^2\left(\frac{x-y}{2}\right)-1=-1/2, \quad \Rightarrow \cos\left(\frac{x-y}{2}\right)=1/2$

Ans. 3

(R) $\cos 2x\left(\cos\left(\frac{\pi}{4}-x\right)-\cos\left(\frac{\pi}{4}+x\right)\right)+2\sin^2 x=2\sin x\cos x$

$\cos 2x\,(\sqrt{2}\sin x)+2\sin^2 x=2\sin x\cos x \Rightarrow \sqrt{2}\sin x[\cos 2x+\sqrt{2}\sin x-\sqrt{2}\cos x]=0$

या तो Either $\sin x=0$ OR या $\cos^2 x-\sin^2 x=\sqrt{2}(\cos x-\sin x)$

$\boxed{\sec x=1}$ OR या $\cos x=\sin x \Rightarrow \boxed{\sec x=\sqrt{2}}$

Ans. 2 OR या 4

(S) $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6}))$



$\frac{x}{\sqrt{1-x^2}}=\frac{x\sqrt{6}}{\sqrt{1+6x^2}} \Rightarrow 1+6x^2=6-6x^2 \Rightarrow 12x^2=5$

$x=\sqrt{\frac{5}{12}}=\frac{1}{2}\sqrt{\frac{5}{3}}$

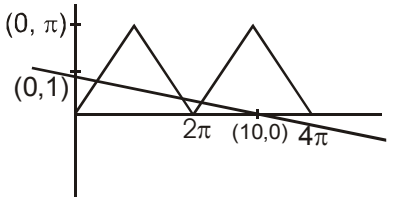
3. Let $f:[0,4\pi]\to[0,\pi]$ be defined by $f(x)=\cos^{-1}(\cos x)$. The number of points $x\in[0,4\pi]$ satisfying the equation $f(x)=\frac{10-x}{10}$ is: **[JEE(Adv.)2014]**

माना फलन $f:[0,4\pi]\to[0,\pi]$ इस प्रकार की $f(x)=\cos^{-1}(\cos x)$ है। अन्तराल $x\in[0,4\pi]$ में स्थित बिन्दुओं की संख्या जिसके लिए समीकरण $f(x)=\frac{10-x}{10}$ सन्तुष्ट होगा : **[JEE(Adv.)2014]**

Ans. 3

Sol. $f(x)=\cos^{-1}(\cos x)\in[0,4\pi]$ & $f(x)=\frac{10-x}{10}=1-\frac{x}{10}$

so, 3 solution. इसलिए 3 हल



4. If $\alpha=3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta=3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct option (s) is (are) : **[JEE(Adv.)2015]**

यदि $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ और $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, जहाँ प्रतिलोम त्रिकोणमितीय फलन (inverse trigonometric functions) केवल मुख्य मान (principal values) ही लेते हैं, तब सही कथन है (हैं) : **[JEE(Adv.)2015]**

(A) $\cos\beta > 0$ (B) $\sin\beta < 0$ (C) $\cos(\alpha+\beta) > 0$ (D) $\cos\alpha < 0$

Ans. BCD

Sol. $\alpha = 3\sin^{-1}\frac{6}{11} > 3\sin^{-1}\frac{6}{12}$ and $\beta = 3\cos^{-1}\frac{4}{9} > 3\cos^{-1}\frac{4}{8}$

$\Rightarrow \quad \alpha > \frac{\pi}{2}$ & $\beta > \pi$

$\Rightarrow \quad \alpha + \beta > \frac{3\pi}{2}$

5. The number of real solutions of the equation $\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty}(-x)^i\right)$ lying in the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$ is _____. (Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.) **[JEE(Adv.)2018]**

समीकरण $\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty}(-x)^i\right)$ के उन वास्तविक हलों की संख्या जो अन्तराल $\left(-\frac{1}{2}, \frac{1}{2}\right)$ में विद्यमान हैं, है _____. (यहाँ प्रतिलोम त्रिकोणमितीय फलन $\sin^{-1}x$ और $\cos^{-1}x$ क्रमश : $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ व $[0, \pi]$ में मान धारण करते हैं ।) **[JEE(Adv.)2018]**

Ans. 2

Sol. $\lim_{n\to\infty}\sin^{-1}\left(\sum_{i=1}^{n} x^{i+1} - x\sum_{i=1}^{n}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \lim_{n\to\infty}\cos^{-1}\left(\sum_{i=1}^{n}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{n}(-x)^i\right)$

$$\left(\frac{x^2}{1-x} - x\frac{\frac{x}{2}}{1-\frac{x}{2}}\right) = \frac{x}{1+x} + \frac{\left(-\frac{x}{2}\right)}{1+\frac{x}{2}}$$

$$\frac{x^2}{1-x}-\frac{x^2}{2-x}=\frac{x}{1+x}-\frac{x}{2+x}$$

$$\frac{x^2}{1-x}-\frac{x}{1+x}=\frac{x^2}{2-x}-\frac{x}{2+x}$$

$$\frac{x(1+x)-(1-x)}{1-x^2}=\frac{2x+x^2-2+x}{4-x^2} \text{ or } x=0$$

$$\frac{x^2+2x-1}{1-x^2}=\frac{x^2+3x-2}{4-x^2}$$

$\Rightarrow \quad x^3+2x^2+5x-2=0$

Let $f(x)=x^3+2x^2+5x-2$

$f'(x)>0$

$f(0)=-2$ and $f(1/2)=9/8$ so one root in $\left(0,\frac{1}{2}\right)$

$\Rightarrow$ 2 roots

6. For any positive integer n, define $f_n:(0,\infty)\to R$ as $f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$ for all $x\in(0,\infty)$.

(Here, the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$). Then, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) $\sum_{j=1}^{5}\tan^2\left(f_j(0)\right)=55$

(B) $\sum_{j=1}^{10}\left(1+f_j'(0)\right)\sec^2\left(f_j(0)\right)=10$

(C) For any fixed positive integer n, $\lim_{x\to\infty}\tan\left(f_n(x)\right)=\frac{1}{n}$

(D) For any fixed positive integer n, $\lim_{x\to\infty}\sec^2\left(f_n(x)\right)=1$

किसी भी धनात्मक पूर्णांक n के लिये, $f_n:(0,\infty)\to R$, $f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$ सभी $x\in(0,\infty)$ के लिये,

($\tan^{-1}x$, $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ में मान धारण करता है।) तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) $\sum_{j=1}^{5}\tan^2\left(f_j(0)\right)=55$

(B) $\sum_{j=1}^{10}\left(1+f_j'(0)\right)\sec^2\left(f_j(0)\right)=10$

(C) किसी भी नियत धनात्मक पूर्णांक n के लिये, $\lim_{x\to\infty}\tan\left(f_n(x)\right)=\frac{1}{n}$

(D) किसी भी नियत धनात्मक पूर्णांक n के लिये, $\lim_{x\to\infty}\sec^2\left(f_n(x)\right)=1$

Ans. D

Sol. $f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$

$f_n(x)=\tan^{-1}(x+n)-\tan^{-1}(x) \quad \Rightarrow \quad f_n'(x)=\frac{1}{1+(x+n)^2}-\frac{1}{1+x^2}$

$f_n(0)=\tan^{-1}(n) \quad \Rightarrow \quad \tan^2\left(\tan^{-1}n\right)=n^2$

(A) $\sum_{j=1}^{5}\tan^2\left(f_j(0)\right)=\sum_{j=1}^{5}j^2=\frac{5.6.11}{6}=55$

(B) $f_n'(0)=\frac{1}{1+n^2}-1 \quad \Rightarrow \quad 1+f_n(0)=\frac{1}{1+n^2}$

$\sec^2\left(f_n(0)\right)=\sec^2\left(\tan^{-1}(n)\right)=1+n^2$

Hence $\left(1+f_n'(0)\right)\cdot\sec^2\left(f_n(0)\right)=\left(\frac{1}{1+n^2}\right)\left(1+n^2\right)=1$

So $\sum_{i=1}^{10}\left(1+f_{ii}(0)\right)\sec^2\left(f_i(0)\right)=\sum_{i=1}^{10}1=10$

$\lim_{x\to\infty}f_n(x)=\lim_{x\to\infty}\tan^{-1}\left(\frac{n}{1+x(n+x)}\right)=0$

$\lim_{x\to\infty}\tan\left(f_n(x)\right)=0$ and $\lim_{x\to\infty}\sec^2\left(f_n(x)\right)=1$

7. Let $E_1=\left\{x\in\mathbb{R}:x\neq1 \text{ and } \frac{x}{x-1}>0\right\}$ and $E_2=\left\{x\in E_1:\sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)\text{ is a real number}\right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.) Let $f:E_1\to\mathbb{R}$ be the function

defined by $f(x)=\log_e\left(\frac{x}{x-1}\right)$ and $g:E_2\to\mathbb{R}$ be the function defined by $g(x)=\sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$

Column-I	**Column-II**
P. The range of f is	1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
Q. The range of g contains	2. $(0, 1)$
R. The domain of f contains	3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
S. The domain of g is	4. $(-\infty, 0) \cup (0, \infty)$
	5. $\left(-\infty, \frac{e}{e-1}\right]$
	6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{1-e}\right]$

The correct option is : **[JEE(Adv.)2018]**

माना कि $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ और } \frac{x}{x-1} > 0\right\}$ और $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ एक वास्तविक संख्या है।}\right\}$.

(यहाँ प्रतिलोम त्रिकोणमितीय फलन (inverse trigonometric function) $\sin^{-1}x$, $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ में मान धारण करता है।) माना कि फलन

$f : E \to \mathbb{R}$, $f(x) = \log_e\left(\frac{x}{x-1}\right)$ के द्वारा परिभाषित है और फलन $g : E_2 \to \mathbb{R}$, $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$ के द्वारा परिभाषित है।

Column-I	**Column-II**
P. f का परिसर है	1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
Q. g के परिसर में समाहित है	2. $(0, 1)$
R. f के प्रान्त में समाहित है	3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
S. g का प्रानत है	4. $(-\infty, 0) \cup (0, \infty)$
	5. $\left(-\infty, \frac{e}{e-1}\right]$
	6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{1-e}\right]$

दिए हुए विकल्पों में से सही विकल्प है : **[JEE(Adv.)2018]**

(A) P - 4; Q - 2; R - 1; S - 1 (B) P - 3; Q - 3; R - 6; S - 5

(C) P - 4; Q - 2; R - 1; S - 6 (D) P - 4; Q - 3; R - 6; S - 5

Ans. A

Sol. $E_1 \dfrac{x}{x-1} > 0 \quad \Rightarrow \quad x \in (-\infty, 0) \cup (1, \infty)$

$E_2 : -1 \le \ell n\left(\dfrac{x}{x-1}\right) \le 1 \quad \Rightarrow \quad \dfrac{1}{e} \le \dfrac{x}{x-1} \le e \quad \Rightarrow \quad \dfrac{1}{e} \le 1 + \dfrac{1}{x-1} \le e$

$\dfrac{1}{e} - 1 \le \dfrac{1}{x-1} \le e - 1 \quad \Rightarrow \quad (x-1) \in \left(-\infty, \dfrac{e}{1-e}\right] \cup \left[\dfrac{1}{1-e}, \infty\right)$

$x \in \left(-\infty, \dfrac{1}{e-1}\right] \cup \left[\dfrac{e}{e-1}, \infty\right)$

Now $\dfrac{x}{x-1} \in (0, \infty) - \{1\} \; \forall \in E_1 \quad \Rightarrow \quad \ell n\left(\dfrac{x}{x-1}\right) \in (-\infty, \infty) - \{0\}$

$\sin^{-1}\left(\ell n\left(\dfrac{x}{x-1}\right)\right) \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right] - \{0\}$

8. For non negative integers n let $f(n) = \dfrac{\sum_{k=0}^{n} \sin\left(\dfrac{k+1}{n+2}\pi\right) \sin\left(\dfrac{k+2}{n+2}\pi\right)}{\sum_{k=0}^{n} \sin^2\left(\dfrac{k+1}{n+2}\pi\right)}$ Assuming $\cos^{-1} x$ takes values in $[0,\pi]$

which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) $f(4) = \dfrac{\sqrt{3}}{2}$

(B) If $\alpha = \tan(\cos^{-1} f(6))$ then $\alpha^2 + 2\alpha - 1 = 0$

(C) $\lim\limits_{n \to \infty} f(n) = \dfrac{1}{2}$

(D) $\sin(7\cos^{-1} f(5)) = 0$

अऋणात्मक पूर्णांकों (non negative integers) n के लिए माना कि $f(n) = \dfrac{\sum_{k=0}^{n} \sin\left(\dfrac{k+1}{n+2}\pi\right) \sin\left(\dfrac{k+2}{n+2}\pi\right)}{\sum_{k=0}^{n} \sin^2\left(\dfrac{k+1}{n+2}\pi\right)}$ माना कि $\cos^{-1} x$

का मान $[0,\pi]$ में है, तब निम्न में से कौन सा (से) विकल्प सही है (हैं) : **[JEE(Adv.)2019]**

(A) $f(4) = \dfrac{\sqrt{3}}{2}$

(B) यदि $\alpha = \tan(\cos^{-1} f(6))$ then $\alpha^2 + 2\alpha - 1 = 0$

(C) $\lim\limits_{n \to \infty} f(n) = \dfrac{1}{2}$

(D) $\sin(7\cos^{-1} f(5)) = 0$

Ans. ABD

Sol. $$f(n)=\frac{\sum_{k=0}^{n}2\sin\left((k+1)\frac{\pi}{n+2}\right)\sin\left((k+2)\frac{\pi}{n+2}\right)}{\sum_{k=0}^{n}2\sin^2\left((k+1)\frac{\pi}{n+2}\right)}$$

$$\frac{\sum_{k=0}^{n}\left(\cos\frac{\pi}{n+2}-\cos\left((2k+3)\frac{\pi}{n+2}\right)\right)}{\sum_{k=0}^{n}1-\cos\left((2k+2)\frac{\pi}{n+2}\right)}$$

$$f(n)=\frac{(n+1)\cos\left(\frac{\pi}{n+2}\right)-\sum_{k=0}^{n}\cos\left((2k+3)\frac{\pi}{n+2}\right)}{(n+1)-\sum_{k=0}^{n}\cos\left((2k+2)\frac{\pi}{n+2}\right)} \quad .....(1)$$

$$\sum_{k=0}^{n}\cos\left((2k+3)\frac{\pi}{n+2}\right)=\cos\left(\frac{3\pi}{n+2}\right)+\cos\frac{5\pi}{n+2}+\cos\frac{7\pi}{n+2}+.....+\cos(2n+3)\frac{\pi}{n+2}$$

$$=\frac{\sin\left(\frac{(n+1)\frac{2\pi}{n+2}}{2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}.\cos\left(\frac{\frac{3\pi}{n+2}+(2n+3)\frac{\pi}{n+2}}{2}\right)$$

$$=\frac{\sin\left((n+1)\frac{\pi}{n+2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}.\cos\left((n+3)\frac{\pi}{n+2}\right)\ [\sin(\pi-\theta)=\sin\theta]$$

$$=\cos\left((n+3)\frac{\pi}{n+2}\right)=\cos\left(\pi+\frac{\pi}{n+2}\right)=-\cos\left(\frac{\pi}{n+2}\right)$$

$$\sum_{k=0}^{h}\cos\left((2k+2)\frac{\pi}{n+2}\right)=\cos\left(\frac{2\pi}{n+2}\right)+\cos\frac{4\pi}{n+2}+.....+\cos\left((2n+2)\frac{\pi}{n+2}\right)$$

$$=\frac{\sin\left((n+1)\frac{\pi}{n+2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}\cos\left((n+2)\frac{\pi}{n+2}\right)$$

$= \cos\pi = -1$

$$f(n) = \frac{(n+1)\cos\left(\frac{\pi}{n+2}\right) - \left(-\cos\frac{\pi}{n+2}\right)}{n+1-(-1)}$$

$$f(n) = \cos\frac{(n+2)\cos\left(\frac{\pi}{n+2}\right)}{n+2}$$

$$f(n) = \cos\left(\frac{\pi}{n+2}\right)$$

Option: (1) $f(4) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

(2) $f(6) = \cos^{-1}(f(6) = \frac{\pi}{8}$

$\alpha = \tan\frac{\pi}{8}$

$\alpha^2 + 2\alpha - 1 = 0$

$2\alpha = 1 - \alpha^2$

$$\frac{2\alpha}{1-\alpha^2} = 1$$

$$\tan\left(\frac{\pi}{8}\times 2\right) = 1$$

$\tan 45° = 1$ true

(3) $n \to \infty$

$f(n) \to 1$

(4) $f(5) = \cos\left(\frac{\pi}{7}\right)$

$\cos^{-1}(f(5)) = \frac{\pi}{7}$

$$\sin\left(7\times\frac{\pi}{7}\right) = \sin\pi = 0$$

9. The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4},\frac{3\pi}{4}\right]$ equals :

[JEE(Adv.)2019]

अन्तराल $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ में $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ का मान बराबर है :

[JEE(Adv.)2019]

Ans. 0

Sol. $$\frac{1}{4}\sum_{k=0}^{10}\frac{1}{\cos\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\cos\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)}$$

$$= \frac{1}{4}\sum_{k=0}^{10}\frac{\sin\left(\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)-\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\right)}{\cos\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\cos\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)}$$

$$= \frac{1}{4}\sum_{k=0}^{10}\left(\tan\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)-\tan\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\right)$$

$$= \frac{1}{4}\left(\tan\left(\frac{11\pi}{2}+\frac{7\pi}{12}\right)-\tan\frac{7\pi}{12}\right)$$

$$= \frac{1}{4}\left(\tan\left(\frac{\pi}{12}\right)-\tan\frac{7\pi}{12}\right)$$

$$= \frac{1}{4}\left(\tan\frac{\pi}{12}+\cot\frac{\pi}{12}\right)$$

$$= \frac{1}{4}2\operatorname{cosec}\left(2\times\frac{\pi}{12}\right)$$

$$= \frac{1}{4}\times 2\operatorname{cosec}\left(\frac{\pi}{6}\right)$$

$$= \frac{1}{4}\times 2\times 2 = 1$$

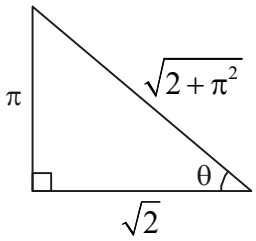
$\sec^{-1}(1) = 0$

10. Considering only the principal values of the inverse trigonometric functions, the value of $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$ is ____. **[JEE(Adv.)2022]**

प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मानों पर विचार करते हुए, $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$ का मान ज्ञात कीजिए। **[JEE(Adv.)2022]**

Ans. $\frac{3\pi}{4}$

Sol. $\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}=\tan^{-1}\frac{\pi}{\sqrt{2}}$



$$\sin^{-1}\left(\frac{2\sqrt{2}\pi}{2+\pi^2}\right)=\sin^{-1}\left(\frac{2\times\frac{\pi}{\sqrt{2}}}{1+\left(\frac{\pi}{\sqrt{2}}\right)^2}\right)$$

$$=\pi-2\tan^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

$$\left(\text{ As, }\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\pi-2\tan^{-1}x, x\geq 1\right)$$

$$\text{and }\tan^{-1}\frac{\sqrt{2}}{\pi}=\cot^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

$$\therefore \text{Expression}=\frac{3}{2}\left(\tan^{-1}\frac{\pi}{\sqrt{2}}\right)+\frac{1}{4}\left(\pi-2\tan^{-1}\frac{\pi}{\sqrt{2}}\right)+\cot^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

$$=\left(\frac{3}{2}-\frac{2}{4}\right)\tan^{-1}\frac{\pi}{\sqrt{2}}+\frac{\pi}{4}+\cot^{-1}\frac{\pi}{\sqrt{2}}$$

$$=\left(\tan^{-1}\frac{\pi}{\sqrt{2}}+\cot^{-1}\frac{\pi}{\sqrt{2}}\right)+\frac{\pi}{4}$$

$$=\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}$$

$= 2.35$ or 2.36

11. Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in R$. Then the number of real solutions of the equation $\sqrt{1+\cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to_____.

[JEE(Adv.)2023]

माना कि $x \in R$ के लिए $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ है। तब समुच्चय $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ में समीकरण $\sqrt{1+\cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ के वास्तविक हलों की संख्या है _____ **[JEE(Adv.)2023]**

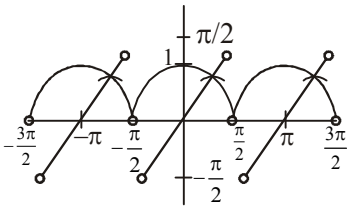
Ans. 3

Sol. $\sqrt{1+\cos 2x} = \sqrt{2}\tan^{-1}(\tan x)$

$\sqrt{2\cos^2 x} = \sqrt{2}\tan^{-1}(\tan x)$

$\sqrt{2}|\cos x| = \sqrt{2}\tan^{-1}(\tan x)$

$y = |\cos x|$ $\qquad$ $y = \tan^{-1}(\tan x)$



3 Solution

12. For any $y \in R$, let $\cot^{-1}(y) \in (0, \pi)$ and $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the sum of all the solutions of the equation $\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \cot^{-1}\left(\frac{9-y^2}{6y}\right) = \frac{2\pi}{3}$ for $0 < |y| < 3$, is equal to : **[JEE(Adv.)2023]**

किसी $y \in R$, के लिए माना कि $\cot^{-1}(y) \in (0, \pi)$ एवं $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ है। तब समीकरण

$\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \cot^{-1}\left(\frac{9-y^2}{6y}\right) = \frac{2\pi}{3}$, जहाँ $0 < |y| < 3$ है, के सभी हलों का योगफल है : **[JEE(Adv.)2023]**

(A) $2\sqrt{3} - 3$ (B) $3 - 2\sqrt{3}$ (C) $4\sqrt{3} - 6$ (D) $6 - 4\sqrt{3}$

Ans. C

Sol. Case-I : $y \in (-3, 0)$

$$\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \pi + \tan^{-1}\left(\frac{6y}{9-y^2}\right) = \frac{2\pi}{3}$$

$$2\tan^{-1}\left(\frac{6y}{9-y^2}\right) = -\frac{\pi}{3}$$

$y^2 - 6\sqrt{3}y - 9 = 0 \Rightarrow y = 3\sqrt{3} - 6 (\because y \in (-3, 0))$

Case-I : $y \in (0,3)$

$2\tan^{-1}\left(\frac{6y}{9-y^2}\right) = \frac{2\pi}{3} \Rightarrow \sqrt{3}y^2 + 6y - 9\sqrt{3} = 0$

$y = \sqrt{3}$ or $y = -3\sqrt{3}$ (rejected)

$\text{sum} = \sqrt{3} + 3\sqrt{3} - 6 = 4\sqrt{3} - 6$

13. Considering only the principal values of the inverse trigonometric functions, the value of $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ is : **[JEE(Adv.)2024]**

प्रतिलोम त्रिकोणमितीय फलनों (inverse trigonometric functions) के केवल मुख्य मानों (principal values) को ध्यान में रखते हुए, $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ का मान है : **[JEE(Adv.)2024]**

(A) $\frac{7}{24}$ (B) $\frac{-7}{24}$ (C) $\frac{-5}{24}$ (D) $\frac{5}{24}$

Ans. B

Sol. $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$

$\sin^{-1}\left(\frac{3}{5}\right) = \theta$ $\qquad$ $2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \phi$

$\sin\theta = \frac{3}{5}$ $\qquad$ $\cos\frac{\phi}{2} = \frac{2}{\sqrt{5}}$

$\tan\theta = \frac{3}{4}$ $\qquad$ $\tan\frac{\phi}{2} = \frac{1}{2}$

$\tan\phi = \frac{2\tan\frac{\phi}{2}}{1 - \tan^2\frac{\phi}{2}}$

$$\tan\phi = \frac{2\left(\frac{1}{2}\right)}{1-\frac{1}{4}}$$

$$\tan\phi = \frac{4}{3}$$

$\tan(\phi - \theta)$

$$\frac{\tan\theta - \tan\phi}{1+\tan\theta\tan\phi}$$

$$\frac{\frac{3}{4}-\frac{4}{3}}{1+1} = \frac{9-16}{24}$$

$$= -\frac{7}{24}$$

14. The total number of real solutions of the equation $\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$ is

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\tan^{-1}x$ assume values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, respectively)

[JEE(Adv.)2025]

समीकरण (equation) $\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$ के कुल वास्तविक हलों की संख्या (total number of real solutions) है

(यहाँ प्रतिलोम त्रिकोणमितीय फलनों, (inverse trigonometric functions) $\sin^{-1}x$ तथा $\tan^{-1}x$ के मान क्रमशः $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ और $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ में हैं।)

[JEE(Adv.)2025]

(A) 1 (B) 2 (C) 3 (D) 5

Ans. C

Sol. $\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}\left(\frac{2\left(\frac{\tan\theta}{3}\right)}{1+\left(\frac{\tan\theta}{3}\right)^2}\right)$

Let $\frac{\tan\theta}{3} = \tan\alpha$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}(\sin 2\alpha)$

Case-I : $2\alpha \in \left(-\pi, -\frac{\pi}{2}\right)$

$\alpha \in \left(-\frac{\pi}{2}, -\frac{\pi}{4}\right)$

$\tan\alpha \in (-\infty, -1)$

$\frac{\tan\theta}{3} \in (-\infty, -1)$

$\tan\theta \in (-\infty, -3)$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}(-\pi - 2\alpha)$

$\theta - \frac{\pi}{2} = \tan^{-1}(2\tan\theta) + \alpha$

$\tan\left(\theta - \frac{\pi}{2}\right) = \frac{2\tan\theta + \tan\alpha}{1 - 2\tan\theta\cdot\tan\alpha}$

$-\frac{1}{\tan\theta} = \frac{2\tan\theta + \frac{\tan\theta}{3}}{1 - \frac{2\tan^2\theta}{3}}$

$\frac{-1}{\tan\theta} = \frac{7\tan\theta}{3 - 2\tan^2\theta}$

$5\tan^2\theta = -3$

No solution.

Case-II : $2\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\alpha \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

$\tan\theta \in [-3, 3]$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}(2\alpha)$

$\tan\theta = \frac{2\tan\theta - \tan\alpha}{1 + 2\tan\theta\tan\alpha}$

$\tan\theta = \frac{5\tan\theta}{3 + 2\tan^2\theta} \qquad \left(\tan\alpha = \frac{\tan\theta}{3}\right)$

$\tan\theta = 0$ or $\tan^2\theta = 1$

$\tan\theta = \pm 1$

3 solutions

Case-III : $2\alpha \in \left(\frac{\pi}{2}, \pi\right)$

$\alpha \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\tan\theta \in (3, \infty)$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}(\pi - 2\alpha)$

$\frac{\pi}{2} + \theta = \tan^{-1}(2\tan\theta) + \alpha$

$\tan\left(\frac{\pi}{2} + \theta\right) = \frac{2\tan\theta + \tan\alpha}{1 - 2\tan\theta\tan\alpha}$

$\frac{-1}{\tan\theta} = \frac{7\tan\theta}{3 - 2\tan^2\theta} \qquad \left(\tan\alpha = \frac{\tan\theta}{3}\right)$

$5\tan^2\theta = -3$

No solution

Total 3 solutions.

Exercise C - I

1. The principal value of $\sin^{-1}\left[3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]\right]$ is equal to :

$\sin^{-1}\left[3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]\right]$ का मुख्य मान होगा :

(A) $-\frac{2\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. $\sin^{-1}\left[3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]\right]$

$\Rightarrow \sin^{-1}\left[3\cos^4\alpha+3\sin^4\alpha-2\cos^6\alpha-2\sin^6\alpha\right]$

$\Rightarrow \sin^{-1}\left[3\left(1-2s^2c^2\right)-2\left(1-3s^2c^2\right)\right]$

$\Rightarrow \sin^{-1}\left[3-6s^2c^2-2+6s^2c^2\right]$

$=\sin^{-1}(1)=\frac{\pi}{2}$

2. If $\sin\left(\cot^{-1}(x+1)\right)=\cos(\tan^{-1}x)$, then $x=$

यदि $\sin\left(\cot^{-1}(x+1)\right)=\cos(\tan^{-1}x)$, तब $x=$

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 0 (D) $\frac{9}{4}$

Ans. A

Sol. $\sin\left(\cot^{-1}(x+1)\right)=\cos(\tan^{-1}\frac{x}{1})$

$\sin\left(\sin^{-1}\left(\frac{1}{\sqrt{x^2+2x+2}}\right)\right)=\cos\left(\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)$

$$\Rightarrow \frac{1}{\sqrt{x^2+2x+2}} = \frac{1}{\sqrt{1+x^2}}$$

$1 + x^2 = x^2 + 2x + 2$

$$x = \frac{-1}{2}$$

3. Sum of series $\cot^{-1}\left(\frac{5}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{9}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{15}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{23}{\sqrt{3}}\right)+.....$ upto infinite terms is equal to :

श्रेणी $\cot^{-1}\left(\frac{5}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{9}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{15}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{23}{\sqrt{3}}\right)+.....$ के अनन्त पदों का योग होगा :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Ans. B

Sol. $\tan^{-1}\left(\frac{\sqrt{3}}{5}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{9}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{15}\right)+\tan\left(\frac{\sqrt{3}}{13}\right)+.....\infty$

$$T_r = \tan^{-1}\left(\frac{\sqrt{3}}{r^2+r+3}\right)$$

$$T_r = \tan^{-1}\left(\frac{\frac{1}{\sqrt{3}}}{1+\frac{r}{\sqrt{3}}\frac{(r+1)}{\sqrt{3}}}\right)$$

$$T_r = \tan^{-1}\left(\frac{\frac{r+1}{\sqrt{3}}-\frac{r}{3}}{1+\frac{r}{\sqrt{3}}\frac{(r+1)}{\sqrt{3}}}\right)$$

$$T_r = \tan^{-1}\left(\frac{r+1}{\sqrt{3}}\right)-\tan^{-1}\left(\frac{r}{\sqrt{3}}\right)$$

$$T_1 = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$T_2 = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

$\vdots \qquad \vdots \qquad\qquad \vdots$

$$T_n = \tan^{-1}\left(\frac{n+1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{n}{\sqrt{3}}\right)$$

$$S_n = \tan^{-1}\left(\frac{n+1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$S_\infty = \tan^{-1}(\infty) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{\pi}{2} - \frac{\pi}{6}$$

$$S_\infty = \frac{\pi}{3}$$

4. If $2\tan^{-1}\left(\frac{1+x}{1-x}\right) = 5\tan^{-1} x$, then value of x is :

यदि $2\tan^{-1}\left(\frac{1+x}{1-x}\right) = 5\tan^{-1} x$, तो x का मान होगा :

(A) $\tan\left(\frac{1}{\sqrt{3}}\right)$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{\pi}{4}$ (D) $\sqrt{3}$

Ans. B

Sol. $2\tan^{-1}\left(\frac{1+x}{1-x}\right) = 5\tan^{-1} x$

$$2\tan^{-1}\left(\tan\left(\frac{\pi}{4} + \tan^{-1} x\right)\right) = 5\tan^{-1} x$$

$\tan^{-1} x = \theta$

$x = \tan\theta$

$$2\left(\frac{\pi}{4}+\theta\right)=5\theta$$

$$\frac{\pi}{2}+2\theta=5\theta$$

$$3\theta=\frac{\pi}{2}$$

$$\theta=\frac{\pi}{6}$$

$$x=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}$$

5. If $x \in (0,1)$, then solution of inequality $\sin(2\sin^{-1}x) > \sqrt{2}\,x$ is given by :

यदि $x \in (0,1)$ हो, तो असमिका $\sin(2\sin^{-1}x) > \sqrt{2}\,x$ का हल होगा :

(A) $x \in \left(\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}\right)$ (B) $x \in \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ (C) (0, 1) (D) $\left(0, \frac{1}{\sqrt{2}}\right)$

Ans. D

Sol. $\sin(2\sin^{-1}x) > \sqrt{2}\,x$

$x \in (0, 1)$ .....(1)

$\sin^{-1}x = \theta$

$\sin\theta = x$

$\sin 2\theta > \sqrt{2}\sin\theta$

$\sin\theta\left(2\cos\theta - \sqrt{2}\right) > 0$

$\underset{x>0}{x}\left(\sqrt{1-x^2} - \frac{1}{\sqrt{2}}\right) > 0$

$\sqrt{1-x^2} - \frac{1}{\sqrt{2}} > 0$

$1 - x^2 > \frac{1}{2}$

$x^2 < \frac{1}{2}$

$|x| < \frac{1}{\sqrt{2}}$

$x \in \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ .....(2)

from (1) & (2)

$x \in \left(0, \frac{1}{\sqrt{2}}\right)$

6. If $\sin^{-1}(x^2) = \cos^{-1} x$, then $x^4 + x^2$ is :

यदि $\sin^{-1}(x^2) = \cos^{-1} x$ हो, तो $x^4 + x^2$ होगा :

(A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{2}{5}$

Ans. B

Sol. $\sin^{-1}(x^2) = \cos^{-1} x$

$\sin^{-1} x^2 = \sin^{-1}\sqrt{1-x^2}$

$x^2 = \sqrt{1-x^2}$

$x^4 = 1 - x^2$

$x^4 + x^2 = 1$

ONE OR MORE THAN ONE CORRECT TYPE

7. Which of the following functions represent identical graphs in x-y plane $\forall\ x \in (2, 3)$?

निम्न में से कौनसे फलन समरूप है $\forall\ x \in (2, 3)$?

(A) $y = \cos^{-1}\sqrt{3-x}$ (B) $y = \sin^{-1}\sqrt{x-2}$

(C) $y = \cot^{-1}\left(\sqrt{\frac{3-x}{x-2}}\right)$ (D) $y = \frac{1}{2}\sin^{-1}\left(2\sqrt{(3-x)(x-2)}\right)$

Ans. ABC

Sol. Let $\sin^{-1}\sqrt{x-2} = \alpha \quad \left(\text{where } \alpha \in \left(0, \frac{\pi}{2}\right)\right)$ .....(i)

then $(x-2) = \sin^2\alpha \quad \Rightarrow 3 - x = 1 - (x-2) = 1 - \sin^2\alpha = \cos^2\alpha$

$\therefore\ \alpha = \cos^{-1}\sqrt{3-x}$ .....(ii)

Also $\cot^2\alpha = \dfrac{\cos^2\alpha}{\sin^2\alpha} = \dfrac{3-x}{x-2}$ $\therefore\ \alpha = \cot^{-1}\left(\sqrt{\dfrac{3-x}{x-2}}\right)$ .....(iii)

Also, $\sin 2\alpha = 2\sin\alpha\cos\alpha = 2\sqrt{x-2}\sqrt{3-x}$

$$\therefore \quad 2\alpha = \begin{cases} \sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } 0<\alpha<\dfrac{\pi}{4} \\ \pi-\sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } \dfrac{\pi}{4}\le\alpha<\dfrac{\pi}{2} \end{cases}$$

$$\Rightarrow \alpha = \begin{cases} \dfrac{1}{2}\sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } 0<\alpha<\dfrac{\pi}{4} \\ \dfrac{\pi}{2}-\dfrac{1}{2}\sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } \dfrac{\pi}{4}\le\alpha<\dfrac{\pi}{2} \end{cases} \text{.....(iv)}$$

From (i), (ii), (iii) and (iv) it is clear that

$y = \cos^{-1}\sqrt{3-x}$, $y = \sin^{-1}\sqrt{x-2}$ and $y = \cot^{-1}\sqrt{\dfrac{3-x}{x-2}}$

have identical graphs in the range $\left(0, \dfrac{\pi}{2}\right)$ but $y = \dfrac{1}{2}\sin^{-1}\left(2\sqrt{(3-x)(x-2)}\right)$ has range $\left(0, \dfrac{\pi}{4}\right)$

8. Let x_1, x_2, x_3, x_4 be four non zero numbers satisfying the equation $\tan^{-1}\dfrac{a}{x}+\tan^{-1}\dfrac{b}{x}+\tan^{-1}\dfrac{c}{x}+\tan^{-1}\dfrac{d}{x}=\dfrac{\pi}{2}$ then which of the following relation(s) hold good?

यदि x_1, x_2, x_3, x_4 चार अशून्य संख्याऐं समीकरण $\tan^{-1}\dfrac{a}{x}+\tan^{-1}\dfrac{b}{x}+\tan^{-1}\dfrac{c}{x}+\tan^{-1}\dfrac{d}{x}=\dfrac{\pi}{2}$ को सन्तुष्ट करते हैं, तब निम्न में से कौन कौन से सम्बन्ध सत्य होंगे?

(A) $\sum_{i=1}^{4} x_i = a + b + c + d$

(B) $\sum_{i=1}^{4} \dfrac{1}{x_i} = 0$

(C) $\prod_{i=1}^{4} x_i = abcd$

(D) $(x_1 + x_2 + x_3)(x_2 + x_3 + x_4)(x_3 + x_4 + x_1)(x_4 + x_1 + x_2) = abcd$

Ans. BCD

Sol. Let $\tan^{-1}\frac{a}{x}=\alpha \quad \Rightarrow \quad \tan\alpha=\frac{a}{x}$ etc.

$\alpha+\beta+\gamma+\delta=\frac{\pi}{2}$

$\tan(\alpha+\beta+\gamma+\delta)=\tan\frac{\pi}{2}$

$\frac{S_1-S_3}{1-S_2+S_4}=\infty \quad \Rightarrow \quad 1-S_2+S_4=0 \Rightarrow \quad S_4-S_2+1=0$

Now, $S_4=\tan\alpha\cdot\tan\beta\cdot\tan\gamma\cdot\tan\delta=\frac{abcd}{x^4}$

$S_2=\sum\tan\alpha\cdot\tan\beta=\frac{\sum ab}{x^2}$

$\therefore \quad \frac{abcd}{x^4}-\frac{\sum ab}{x^2}+1=0$

$x^4-\sum ab\,x^2+abcd=0$ $\begin{matrix}\nearrow x_1\\ \rightarrow x_2\\ \rightarrow x_3\\ \searrow x_4\end{matrix}$

$\therefore \quad x_1+x_2+x_3+x_4=0 \quad ....(1)$

$\sum x_1x_2x_3=0 \quad ....(2)$

$\sum x_1x_2x_3=\underbrace{x_1x_2x_3x_4}_{\text{non zero}}\left[\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}\right]=0 \Rightarrow \quad (B)$

$x_1x_2x_3x_4=abcd \Rightarrow \quad (C)$

9. If $x^2+2x+n>10+\sin^{-1}(\sin 9)+\tan^{-1}(\tan 9)$ for all real x, then the possible value of n can be :

यदि $x^2+2x+n>10+\sin^{-1}(\sin 9)+\tan^{-1}(\tan 9)\ \forall\ x\in R$ तो n के सम्भव मान होंगे :

(A) 11 (B) 12 (C) 13 (D) 14

Ans. BCD

Sol. We have $x^2+2x+(n-10)>0\ \forall\ x\in R$ [$\because$ $\sin^{-1}(\sin 9)=3\pi-9$ and $\tan^{-1}(\tan 9)=9-3\pi$

$\therefore \quad \sin^{-1}(\sin 9)+\tan^{-1}(\tan 9)=3\pi-9+9-3\pi=0$]

$\therefore$ Discriminant <0

$\Rightarrow\ 4-4(n-10)<0 \ \Rightarrow\ 1-n+10<0 \ \Rightarrow\ n>11$

10. Let $f: R \to R$ defined by $f(x) = \cos^{-1}(-\{-x\})$ where $\{x\}$ is fractional part function. Then which of the following is/are correct?

(A) f is many one but not even function
(B) Range of f contains two prime numbers
(C) f is aperiodic
(D) Graph of f does not lie below x–axis

माना $f: R \to R$ पर परिभाषित फलन $f(x) = \cos^{-1}(-\{-x\})$ है तब निम्न में से सत्य कथन होंगे : (जहां $\{x\}$ भिन्नात्मक मान फलन को दर्शाता है ।)

(A) f बहुएकैकी लेकिन सम नहीं
(B) f के परिसर में दो अभाज्य संख्याएं हैं
(C) f आवर्ती फलन नहीं है
(D) f का आरेख x–अक्ष के नीचे नहीं होगा

Ans. ABD

Sol. We have $f(x) = \cos^{-1}(-\{-x\})$

D_f = all real

As $0 \le \{-x\} < 1 \ \forall x \in R$

$\Rightarrow \quad -1 < -\{-x\} \le 0$

So $\quad R_f = \left[\frac{\pi}{2}, \pi\right)$

Clearly, f is neither even nor odd.

But $\quad f(x+1) = f(x) \quad \Rightarrow \quad$ f is periodic with period 1.

11. The value of $\sum_{p=1}^{20}\sum_{q=1}^{20} 2\tan^{-1}\left(\frac{q}{p}\right)$ is :

$\sum_{p=1}^{20}\sum_{q=1}^{20} 2\tan^{-1}\left(\frac{q}{p}\right)$ का मान होगा :

(A) 200π
(B) 220π
(C) $\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\tan^{-1}\frac{p}{q} + \tan^{-1}\frac{q}{p}\right)$
(D) 400π

Ans. AC

Sol. $\sum_{p=1}^{20}\sum_{q=1}^{20} 2\tan^{-1}\left(\frac{q}{p}\right)$

$$2\left[\tan^{-1}\left(\frac{1}{1}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) + \ldots\ldots + \tan^{-1}\left(\frac{1}{20}\right) + \tan^{-1}\left(\frac{2}{1}\right)\right.$$

$$+\tan^{-1}\left(\frac{2}{2}\right)+\tan^{-1}\left(\frac{2}{3}\right)+.....+\tan^{-1}\left(\frac{2}{20}\right)+.....$$

$$+\tan^{-1}\left(\frac{20}{1}\right)+\tan^{-1}\left(\frac{20}{2}\right)+\tan^{-1}\left(\frac{20}{3}\right)+.....+\tan^{-1}\left(\frac{20}{20}\right)\Bigg]$$

$$=2\times\tan^{-1}(1)\times 20+2\times\frac{\pi}{2}[19+18+.....+1]$$

$$=2\times\left(\frac{\pi}{4}\right)\times 20+2\times\frac{\pi}{2}\left[\frac{19\times 20}{2}\right]$$

$\Rightarrow 10\pi + 190\pi$

$= 200\pi$

(C) $\displaystyle\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\tan^{-1}\left(\frac{p}{q}\right)+\tan^{-1}\frac{q}{p}\right)$

$$\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\tan^{-1}\frac{p}{q}+\cot^{-1}\frac{p}{q}\right)$$

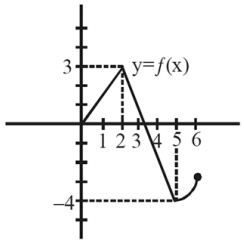
$$=\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\frac{\pi}{2}\right)$$

$$=\sum_{p=1}^{20}\left(\frac{\pi}{2}+\frac{\pi}{2}+.....+\frac{\pi}{2}\right)=\sum_{p=1}^{20}(10\pi)$$

$\Rightarrow 10\pi+10\pi+.....+10\pi$

$= 200\pi$

12. The graph shows a part of an even function f: $[-6,6] \to R$, then :



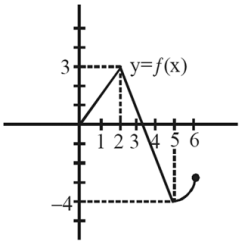
(A) Domain of $f(|x|+3)$ is $[-3,3]$

(B) Range of $f(|x|+3)$ is $\left[-4,\frac{2}{3}\right]$

(C) $\cos^{-1}\left(f\left(\frac{1}{3}\right)\right)+\cos^{-1}\left(f\left(\frac{43}{14}\right)\right)=\frac{2\pi}{3}$

(D) Number of solutions of the equation $f(f(x))=2$ is 4

दिया गया आरेख समफलन f: $[-6,6] \to R$ के एक भाग को दर्शाया हो, तो :



(A) $f(|x|+3)$ का प्रांत $[-3,3]$ है

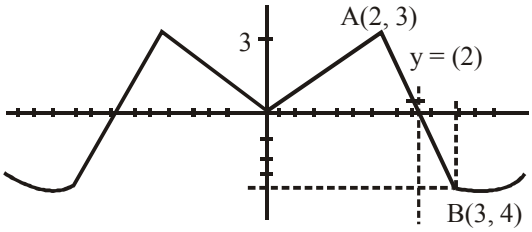
(B) $f(|x|+3)$ का परिसर $\left[-4,\frac{2}{3}\right]$ है

(C) $\cos^{-1}\left(f\left(\frac{1}{3}\right)\right)+\cos^{-1}\left(f\left(\frac{43}{14}\right)\right)=\frac{2\pi}{3}$

(D) समीकरण $f(f(x))=2$ के हलों की संख्या 4 है

Ans. ABC

Sol.



$f(x) \Rightarrow$ domain $x \in [6, 6]$

$f(x) \Rightarrow x \in [0, 6]$

$f(|x|+3) \Rightarrow x \in [-3, 3]$

$AB \Rightarrow y-3=-\frac{7}{3}(x-2)$

$7x + 3y = 23$

$x = 3$

$y = 3 - \frac{7}{3}(3-2) = 3 - \frac{7}{3}$

$y_{max} = \frac{2}{3}$

$y_{min} = -4$

Range of $f(|x|+3) \Rightarrow \left[-4,\frac{2}{3}\right]$

(C) $\Rightarrow y = \frac{3}{2}x$

$f\left(\frac{1}{3}\right) = \frac{1}{2} \qquad f\left(\frac{43}{14}\right) \Rightarrow y = 3 - \frac{7}{3}\left(\frac{43}{14} - 2\right) = 3 - \frac{7}{3}\left(\frac{15}{14}\right) = \frac{1}{2}$

$\cos^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$

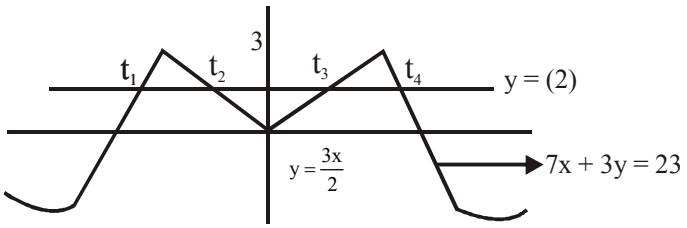
(D) $f(f(x)) = 2$

$f(x) = t$

$f(t) = 2$

$t_3 = 1.3 \qquad t_4 = 2.4$

$t_2 = -1.3 \qquad t_1 = -2.4$



$f(x) = t_1, f(x) = t_2, f(x) = t_3, f(x) = t_4$

$f(x) = -2.4 \qquad f(x) = -1.3 \qquad f(x) = 1.3 \qquad f(x) = 2.4$

Total no. of sol. = 12

13. Which of the following statement is/are true :

(A) $\cos^{-1}(\cos(1 + |\sin x|)) = 1 + |\sin x|$ for all real x

(B) $\sin^{-1}(\sin((\cos^2 x - \sin^2 x))) = \cos^2 x - \sin^2 x \ \forall\ x \in R$.

(C) $\tan^{-1}(\tan(x^2 + 2x + 2)) = x^2 + 2x + 2 \ \forall\ x \in R$

(D) $\tan^{-1}(\tan(3 \sin x)) = 3 \sin x \ \forall\ x \in R$

निम्न में से कौनसा/कौनसे कथन सत्य है/होंगे :

(A) सभी वास्तविक x के लिए $\cos^{-1}(\cos(1 + |\sin x|)) = 1 + |\sin x|$

(B) $\sin^{-1}(\sin((\cos^2 x - \sin^2 x))) = \cos^2 x - \sin^2 x \ \forall\ x \in R$

(C) $\tan^{-1}(\tan(x^2 + 2x + 2)) = x^2 + 2x + 2 \ \forall\ x \in R$

(D) $\tan^{-1}(\tan(3 \sin x)) = 3 \sin x \ \forall\ x \in R$

Ans. AB

Sol. (A) $\cos^{-1}(\cos(1 + |\sin x|)) = 1 + |\sin x|$

$\cos^{-1}(\cos[1, 2])$

$1 + |\sin x| \in [1,2]$

(B) $\sin^{-1}(\sin((\cos^2 x - \sin^2 x))) = \cos^2 x - \sin^2 x$

$\sin^{-1}\left[\sin\left(\cos 2x\right)\right]$

$\Rightarrow \cos 2x \Rightarrow \cos^2 x - \sin^2 x$

(C) $\tan^{-1}(\tan(x^2+2x+2)) = x^2+2x+2$

$\tan^{-1}\left(\tan\left((x+1)^2+1\right)\right) = x^2+2x+2$ for only $x \in \left[1, \frac{\pi}{2}\right]$

(D) $\tan^{-1}(\tan(3\sin x)) = 3\sin x$

$\tan^{-1}(\tan([-3, 3]))$

$\Rightarrow 3\sin x$; for only $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Exercise C - II

1. Solve the equation : $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

हल कीजिए : $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

Ans. $x = 1/3$

Sol. $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

$\sin^{-1}\left(\sqrt{1-6x^2}\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$

$\sqrt{1-6x}=3\sqrt{3}x^2$

$1 - 6x^2 = 27x^4$

$27x^4 + 6x^2 - 1 = 0$

$27x^4 + 9x^2 - 3x^2 - 1 = 0$

$9x^2\ (3x^2 + 1) - 1\ (3x^2 + 1) = 0$

$x^2=\frac{1}{9}\Rightarrow x=\pm\frac{1}{3}$ $\qquad$ $x^2=-\frac{1}{3}$ (not possible)

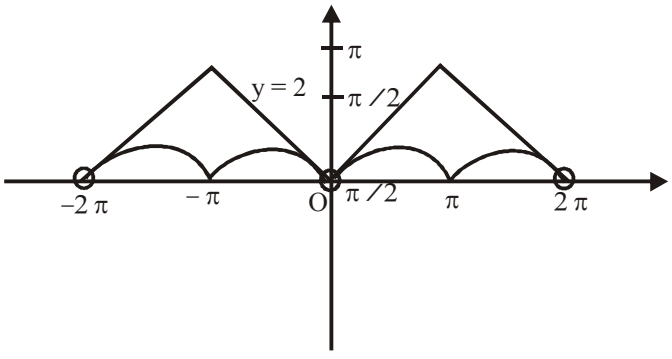
$x=\frac{1}{3}$, $x > 0$

2. Find the number of ordered pairs of (x, y), which satisfy $y = |\sin x|$ and $y = \cos^{-1}(\cos x)$, where $-2\pi \le x \le 2\pi$.

अन्तराल $-2\pi \le x \le 2\pi$ उन क्रमित युग्मों की संख्या ज्ञात करो जो $y = |\sin x|$ तथा $y = \cos^{-1}(\cos x)$ को सन्तुष्ट कर सके।

Ans. 3

Sol. $y = |\sin x|$; $y = \cos^{-1}(\cos x)$



3. Prove that $\cos \tan^{-1} \sin \cot^{-1} x = \sqrt{\frac{x^2+1}{x^2+2}}$.

सिद्ध कीजिए $\cos \tan^{-1} \sin \cot^{-1} x = \sqrt{\frac{x^2+1}{x^2+2}}$

Ans.

Sol. $\text{LHS} = \cos \tan^{-1} \sin \sin^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$

$= \cos \tan^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$

$= \cos \cos^{-1} \left(\frac{\sqrt{x^2+1}}{\sqrt{2+x^2}} \right)$

$= \sqrt{\frac{x^2+1}{x^2+2}}$

4. Prove that, $\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) = \begin{cases} \pi & , \text{ if } 0 < A < \frac{\pi}{4} \\ 0 & , \text{ if } \frac{\pi}{4} < A < \frac{\pi}{2} \end{cases}$.

सिद्ध कीजिए $\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) = \begin{cases} \pi & , \text{ if } 0 < A < \frac{\pi}{4} \\ 0 & , \text{ if } \frac{\pi}{4} < A < \frac{\pi}{2} \end{cases}$

Ans.

Sol. $\text{LHS} = \tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} \left(\cot A \right) + \tan^{-1} \left(\cot^3 A \right)$

Case – Ist : $0 < A < \frac{\pi}{4}$

$(\cot A)\,(\cot^3 A) > 1$

$\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \pi + \tan^{-1} \left(\frac{\cot A + \cot^3 A}{1 - \cot^4 A} \right)$

$$= \tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \pi + \tan^{-1}\left(\frac{\cot A\left(1+\cot^2 A\right)}{\left(1-\cot^2 A\right)\left(1+\cot^2 A\right)}\right)$$

$$= \pi + \tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}\left(\frac{\frac{1}{\tan A}}{\frac{\tan^2 A - 1}{\tan^2 A \tan A}}\right)$$

$$= \pi + \tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}\left(\frac{1}{2}\times\frac{-2\tan A}{1-\tan^2 A}\right)$$

$$= \pi + \tan^{-1}\left(\frac{1}{2}\tan 2A\right) - \tan^{-1}\left(\frac{1}{2}\tan 2A\right)$$

$= \pi$

Case – II[nd] : $\frac{\pi}{4} < A < \frac{\pi}{2}$

$$\Rightarrow \tan^{-1}\left(\frac{1}{2}\tan 2A\right) - \tan^{-1}\left(\frac{1}{2}\tan 2A\right)$$

$= 0$

5. Let $f(x) = \frac{\pi}{4} + \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) - \tan^{-1} x$ and $a_i \left(a_i < a_{i+1} \ \forall\ i = 1, 2, 3, ..., n\right)$ be the positive integral values of x for which $\operatorname{sgn}\left(f(x)\right) = 1$, where sgn(y) denotes signum function of y. Find $\sum_{i=1}^{n} a_i^2$.

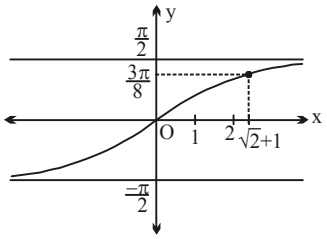
माना $f(x) = \frac{\pi}{4} + \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) - \tan^{-1} x$ तथा $a_i \left(a_i < a_{i+1} \ \forall\ i = 1, 2, 3, ..., n\right)$, x के सभी धनात्मक पूर्णांक मान है जिसके लिये $\operatorname{sgn}\left(f(x)\right) = 1$, जहाँ sgn(y), y का सिग्नम फलन है। $\sum_{i=1}^{n} a_i^2$ ज्ञात कीजिये।

Ans. 5

Sol. We have $f(x)=\frac{\pi}{4}+\cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)-\tan^{-1}x=\frac{\pi}{4}+\cot^{-1}x-\tan^{-1}x=\frac{3\pi}{4}-2\tan^{-1}x$

Now, $\text{sgn}(f(x))=1 \Rightarrow f(x)>0$

So, $\frac{3\pi}{4}-2\tan^{-1}x>0 \Rightarrow \tan^{-1}x<\frac{3\pi}{8} \Rightarrow x<\sqrt{2}+1.$



$\therefore$ Possible positive integral values of x are $1, 2$.

So, $a_1=1$ and $a_2=2$.

Hence $\sum_{i=1}^{n}a_i^2=\sum_{i=1}^{2}a_i^2=a_1^2+a_2^2=1+4=5$ Ans.

6. If $\tan\left(\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+.....+\tan^{-1}\left(\frac{1}{381}\right)\right)=\frac{m}{n}$. where $m, n \in N$, then find least value of $(m+n)$.

यदि $\tan\left(\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+.....+\tan^{-1}\left(\frac{1}{381}\right)\right)=\frac{m}{n}$. जहाँ $m, n \in N$, तो $(m+n)$ का न्यूनतम मान होगा।

Ans. 40

Sol. We have $\tan\left(\tan^{-1}\left(\frac{1}{1+1\cdot2}\right)+\tan^{-1}\left(\frac{1}{1+2\cdot3}\right)+.....+\tan^{-1}\left(\frac{1}{1+19\cdot20}\right)\right)$

$=\tan\left(\tan^{-1}\left(\frac{2-1}{1+2\cdot1}\right)+\tan^{-1}\left(\frac{3-2}{1+2\cdot3}\right)+.....+\tan^{-1}\left(\frac{20-19}{1+19\cdot20}\right)\right)$

$=\tan[\tan^{-1}2-\tan1+\tan^{-1}3-\tan^{-1}2+.....+\tan^{-1}20-\tan^{-1}19]=\tan(\tan^{-1}20-\tan^{-1}1)$

$=\tan\left(\tan^{-1}\left(\frac{20-1}{1+20\cdot1}\right)\right)=\frac{20-1}{1+20}=\frac{19}{21}=\frac{m}{n}$ (Given)

$\therefore$ $m=19, n=21$

Hence $(m+n)_{\text{Least}}=40$. Ans.

7. If $\sum_{n=0}^{\infty} 2 \operatorname{arc\,cot}\left(\frac{n^2+n+4}{2}\right) = k\pi$, then find the value of k.

यदि $\sum_{n=0}^{\infty} 2 \operatorname{arc\,cot}\left(\frac{n^2+n+4}{2}\right) = k\pi$, तो k का मान ज्ञात कीजिए।

Ans. 1

Sol. $\sum_{n=0}^{\infty} 2\cot^{-1}\left(\frac{n^2+n+4}{2}\right) = k\pi$

$$T_n = \tan^{-1}\left(\frac{2}{n^2+n+4}\right)$$

$$= \tan^{-1}\left(\frac{2}{4\left(1+\frac{n(n+1)}{4}\right)}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{1}{2}}{1+\frac{n}{2}\left(\frac{n+1}{2}\right)}\right)$$

$$T_n = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}\left(\frac{n}{2}\right)$$

$$T_0 = \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}(0)$$

$$T_1 = \tan^{-1}\left(\frac{2}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)$$

$$T_2 = \tan^{-1}\left(\frac{3}{2}\right) - \tan^{-1}\left(\frac{2}{2}\right)$$

$\vdots \qquad \vdots \qquad \vdots$

$$T_n = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}\left(\frac{n}{2}\right)$$

$$S_n = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}(0)$$

$$S_\infty = \frac{\pi}{2} - 0$$

$$2S_\infty = \pi = k\pi$$

$k = 1$

8. Let α, β and γ are the roots of the equation $x^3 + 6x + 3 = 0$ and

$$A = \cos^{-1}\left(\sin\left((\alpha+\beta)^{-1} + (\beta+\gamma)^{-1} + (\gamma+\alpha)^{-1}\right)\right),\ B = \cos\left(\tan^{-1}\left(\sin\left(\frac{\alpha+\beta+\gamma}{2}\right)\right)\right)\ C = \sec^{-1}$$

$$\left(\operatorname{cosec}\left((1-\alpha)(1-\beta)(1-\gamma)\right)\right).$$

(i) If the range of the quadratic trinomial $g(x) = x^2 - 2Bx + k$ is $[0, \infty)$, then find the range of k.

(ii) Find the value of $(5A + B - C)$.

(iii) Find the range of the function $f(x) = \frac{(5A-C)x^5 + 6Bx^2}{x^4 + (B-1)x^3 + 1}$.

यदि α, β तथा γ समीकरण $x^3 + 6x + 3 = 0$ के मूल है और

$$A = \cos^{-1}\left(\sin\left((\alpha+\beta)^{-1} + (\beta+\gamma)^{-1} + (\gamma+\alpha)^{-1}\right)\right),\ B = \cos\left(\tan^{-1}\left(\sin\left(\frac{\alpha+\beta+\gamma}{2}\right)\right)\right)\ C = \sec^{-1}$$

$$\left(\operatorname{cosec}\left((1-\alpha)(1-\beta)(1-\gamma)\right)\right).$$

(i) यदि द्विघात व्यंजक $g(x) = x^2 - 2Bx + k$ का परिसर $[0, \infty)$ हो, तो k का परिसर ज्ञात कीजिए।

(ii) $(5A + B - C)$ का मान ज्ञात कीजिए।

(iii) फलन $f(x) = \frac{(5A-C)x^5 + 6Bx^2}{x^4 + (B-1)x^3 + 1}$ का परिसर ज्ञात कीजिए।

Ans. (i) {1} (ii) 1 (iii) [0, 3]

Sol. $x^3 + 6x + 3 = 0$

$\alpha + \beta + \gamma = 0$ $\alpha\beta + \beta\gamma + \gamma\alpha = 6$ $\alpha\beta\gamma = -3$

$A = \cos^{-1}(\sin(\alpha+\beta)^{-1} + (\alpha+\beta)^{-1} + (\gamma+\alpha)^{-1})$

$$= \cos^{-1}\left[\sin\left(-\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)\right)\right]$$

$$= \cos^{-1}\left[\sin\left(\frac{-(\alpha\beta+\beta\gamma+\gamma\alpha)}{\alpha\beta\gamma}\right)\right]$$

$$= \cos^{-1}\left[\sin\left(\frac{6}{3}\right)\right]$$

$= \cos^{-1}[\sin(2)]$

$$= \cos^{-1}\left[\cos\left(\frac{\pi}{2}-2\right)\right]$$

$$A \Rightarrow 2-\frac{\pi}{2}$$

$$B = \cos\left[\tan^{-1}\left(\sin\left(\frac{\alpha+\beta+\gamma}{2}\right)\right)\right]$$

$= \cos[\tan^{-1}(\sin(0))]$

$= \cos[\tan^{-1}(0)]$

$= \cos(0)$

$B = 1$

$C = \sec^{-1}(\text{cosec}((1-\alpha)(1-\beta)(1-\gamma)))$

$x^3 + 6x + 3 = (x-\alpha)(x-\beta)(x-\gamma)$

$x = 1$

$1 + 6 + 3 = (1-\alpha)(1-\beta)(1-\gamma)$

$(1-\alpha)((1-\beta)(1-\gamma) = 10$

$C = \sec^{-1}(\text{cosec } 10)$

$$= \sec^{-1}\left(\sec\left(10-\frac{\pi}{2}\right)\right)$$

$$= 10-\frac{\pi}{2}-2\pi$$

$$C = 10-\frac{5\pi}{2}$$

(i) $g(x) = x^2 - 2Bx + k$

$B = 1$

$= x^2 - 2x + k$

$-\frac{D}{4a} = 0$

$\frac{-(4-4k)}{4} = 0$

$k - 1 = 0$

$k = 1$

(ii) $5A + B - C = 5\left(2 - \frac{\pi}{2}\right) + 1 - \left(10 - \frac{5\pi}{2}\right)$

$= 10 - \frac{5\pi}{2} + 1 - 10 + \frac{5\pi}{2}$

$= 1$

(iii) $f(x) = \frac{(5A - C)x^5 + 6Bx^2}{x^4 + (B-1)x^3 + 1}$

$= \frac{\left(10 - \frac{5\pi}{2} - 10 + \frac{5\pi}{2}\right)x^5 + 6x^2}{x^4 + (1-1)x^3 + 1}$

$y = \frac{6x^2}{x^4 + 1} \geq 0$ .....(1)

$yx^4 + y = 6x^2$

$yx^4 - 6x^2 + y = 0$

Case – I : $y \neq 0$

$D \geq 0$

$36 - 4y^2 \geq 0$

$+ 4(y^2 - 9) \leq 0$

$(y - 3)(y + 3) \leq 0$

$y \in [-3, 3] - \{0\}$ .....(2)

Case – II : $y = 0$

$0 \leq \frac{6x^2}{x^4 + 1}$

$x = 0$ .....(3)

$(2) \cup (3)$

$x = [-3, 3]$ .....(4)

$(1) \cap (4)$

$y = [0, 3]$

9. If $x \in \left[-1,-\frac{1}{2}\right]$ then express the function $f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x)$ in the form of $a\cos^{-1} x + b\pi$, where a and b are rational numbers.

यदि $x \in \left[-1,-\frac{1}{2}\right]$ तो $f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x)$ को $a\cos^{-1} x + b\pi$ के रूप में प्रदर्शित कीजिए, जहाँ a व b परिमेय संख्याएँ है।

Ans. $6\cos^{-1} x - \frac{9\pi}{2}$, so $a = 6, b = -\frac{9}{2}$

Sol. $f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x)$

$\Rightarrow -\sin^{-1}(4x^3 - 3x) + \cos^{-1}(4x^3 - 3x)$

$$\Rightarrow \left(-\frac{\pi}{2} + \cos^{-1}\left(4x^3 - 3x\right)\right) + \cos^{-1}\left(4x^3 - 3x\right)$$

$$= -\frac{\pi}{2} + 2\cos^{-1}\left(4x^3 - 3x\right)$$

$\theta = \cos^{-1} x \qquad \theta \in \left[\frac{2\pi}{3}, \pi\right]$

$$= -\frac{\pi}{2} + 2\cos^{-1}\left(\cos 3\theta\right) \qquad 3\theta \in [2\pi, 3\pi]$$

$$= -\frac{\pi}{2} + 2\left(3\theta - 2\pi\right)$$

$$= -\frac{\pi}{2} + 6\theta - 4\pi$$

$$\Rightarrow 6\theta - \frac{9\pi}{2}$$

$$\Rightarrow 6\cos^{-1} x - \frac{9\pi}{2}$$

10. If α and β are the roots of the equation $x^2 - 4x + 1 = 0$ $(\alpha > \beta)$ then find the value of

$$f(\alpha, \beta) = \frac{\beta^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right) + \frac{\alpha^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right).$$

यदि α व β समीकरण $x^2 - 4x + 1 = 0$ $(\alpha > \beta)$ के मूल है तो $f(\alpha, \beta) = \frac{\beta^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right) + \frac{\alpha^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right)$ का मान ज्ञात कीजिए।

Ans. 56

Sol. $x^2 - 4x + 1 = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

$\alpha + \beta = 4 \qquad \alpha\beta = 1$

$$x = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

$$\alpha = 2 + \sqrt{3}, \beta = 2 - \sqrt{3}$$

11. Find the integral values of K for which the system of equations; $\begin{bmatrix} \arccos x + (\arcsin y)^2 & = \frac{K\pi^2}{4} \\ (\arcsin y)^2 \cdot (\arccos x) & = \frac{\pi^4}{16} \end{bmatrix}$ possesses solutions and find those solutions.

K का पूर्णांक मान ज्ञात कीजिए जिसके लिये दिये गये समीकरणों का हल हो तथा वह हल भी ज्ञात कीजिए।

$$\begin{bmatrix} \arccos x + (\arcsin y)^2 & = \frac{K\pi^2}{4} \\ (\arcsin y)^2 \cdot (\arccos x) & = \frac{\pi^4}{16} \end{bmatrix}.$$

Ans. $k = 2, \left(\cos\frac{\pi^2}{4}, 1\right)$ and $\left(\cos\frac{\pi^2}{4}, -1\right)$

Sol. Let $\cos^{-1}x = p$, $\sin^{-1}y = q$

$$p + q^2 = \frac{k\pi^2}{4}$$

$$p \cdot q^2 = \frac{\pi^4}{16}$$

$$x^2 - \frac{k\pi^2}{4}x + \frac{\pi^4}{16} = 0 \begin{matrix} \nearrow p \\ \searrow q^2 \end{matrix}$$

$$x^2 - \frac{k\pi^2}{4}x + \left(\frac{\pi^2}{4}\right)^2 = 0$$

So $k = 2$

$$x = \frac{\pi^2}{4}$$

$$q^2 = \frac{\pi^2}{4}$$

$$\sin^{-1} y = \pm\frac{\pi}{2}$$

$$y = \sin\left(\pm\frac{\pi}{2}\right) = \pm 1$$

$$x = \cos\left(\frac{\pi^2}{4}\right)$$

12. Solve the following inequalities :

असमिकाओं को हल कीजिए :

(a) $\text{arc cot}^2 x - 5 \text{ arc cot} x + 6 > 0$ (b) $\text{arc sin} x > \text{arc cos} x$

(c) $\tan^2(\text{arc sin} x) > 1$

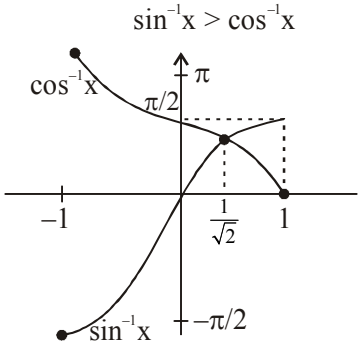
Ans. (a) $(-\infty, \cot 3) \cup (\cot 2, \infty)$ (b) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (c) $\left(-1, -\frac{\sqrt{2}}{2}\right) \cup \left(\frac{\sqrt{2}}{2}, 1\right)$

Sol. (a) $\left(\cot^{-1} x - 2\right)\left(\cot^{-1} x - 3\right) > 0$

$$\cot^{-1} x \in (-\infty, 2) \cup (3, \infty)$$

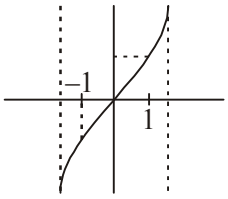
$$x \in (-\infty, \cot 3) \cup (\cot 2, \infty)$$

(b)



$$x \in \left(\frac{1}{\sqrt{2}}, 1\right]$$

(c)



$$\tan^2\left(\sin^{-1} x\right) > 1$$

Let θ

$$\tan^2 \theta > 1$$

$$\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$x \in \left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$$

13. (a) Solve the inequality : $(\text{arc sec } x)^2 - 6(\text{arc sec } x) + 8 > 0$

(b) If $\sin^2 x + \sin^2 y < 1$ for all $x, y \in R$ then prove that $\sin^{-1}(\tan x \cdot \tan y) \in (-\pi/2, \pi/2)$.

(c) Solve the following system of inequations :
$4 \text{ arc } \tan^2 x - 8 \text{ arc } \tan x + 3 < 0$ and $4 \text{ arc cot } x - \text{arc cot}^2 x - 3 \geq 0$.

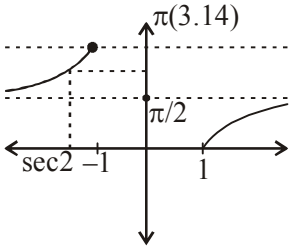
(a) दी गई असमिका को हल कीजिए : $(\text{arc sec } x)^2 - 6(\text{arc sec } x) + 8 > 0$

(b) यदि $\sin^2 x + \sin^2 y < 1 \ \forall \ x, y \in R$ तो सिद्ध कीजिये $\sin^{-1}(\tan x \cdot \tan y) \in \left(-\pi/2, \pi/2\right)$

(c) निम्न असमिकाओं के समूह का हल ज्ञात कीजिए :
$4 \text{ arc } \tan^2 x - 8 \text{ arc } \tan x + 3 < 0$ and $4 \text{ arc cot } x - \text{arc cot}^2 x - 3 \geq 0$.

Ans. (a) $(-\infty, \sec 2) \cup [1, \infty)$ (c) $\left(\tan\frac{1}{2}, \cot 1\right]$

Sol. (a) $\left(\sec^{-1} x - 2\right)\left(\sec^{-1} x - 4\right) > 0$



$\sec^{-1} x \in (-\infty, 2) \cup (4, \infty)$

$\sec^{-1} x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, 2\right)$

$x \in (-\infty, \sec 2) \cup [1, \infty)$

(b) $\sin^2 x < \cos^2 y$

$\Rightarrow \frac{\sin^2 x}{\cos^2 y} < 1$ _____(1)

Similarly,

$\frac{\sin^2 y}{\cos^2 x} < 1$ _____(2)

(1) × (2)

$\tan^2 x \cdot \tan^2 y < 1$

$-1 < \tan x \cdot \tan y < 1$

$-\frac{\pi}{2} < \sin^{-1}(\tan x \cdot \tan y) < \frac{\pi}{2}$

(c) $\left(2\tan^{-1} x - 1\right)\left(2\tan^{-1} x - 3\right) < 0$

$\tan^{-1} x \in \left(\frac{1}{2}, \frac{3}{2}\right)$

$x \in \left(\tan\frac{1}{2}, \tan\frac{3}{2}\right)$ _____(1)

$\left(\cot^{-1} x\right)^2 - 4\cot^{-1} x + 3 \le 0$

$\left(\cot^{-1} x - 1\right)\left(\cot^{-1} x - 3\right) \le 0$

$\cot^{-1} x \in [1, 3]$

$x \in [\cot 3, \cot 1]$ _____(2)

$(1) \cap (2)$

$\left(\tan\frac{1}{2}, \cot 1\right)$

Paragraph for Question Nos. 14 to 16

For $x \in \left(0, \frac{\pi}{4}\right)$, Let $S_n = \sum_{r=1}^{2n} \sin\left(\sin^{-1} x^{3r-2}\right)$, $C_n = \sum_{r=1}^{2n} \cos\left(\cos^{-1} x^{3r-1}\right)$ and $T_n = \sum_{r=1}^{2n} \tan\left(\tan^{-1} x^{3r}\right)$ where $n \in N$ and $n \geq 3$.

$x \in \left(0, \frac{\pi}{4}\right)$ के लिए, माना $S_n = \sum_{r=1}^{2n} \sin\left(\sin^{-1} x^{3r-2}\right)$, $C_n = \sum_{r=1}^{2n} \cos\left(\cos^{-1} x^{3r-1}\right)$ तथा $T_n = \sum_{r=1}^{2n} \tan\left(\tan^{-1} x^{3r}\right)$ जहाँ $n \in N$ एवं $n \geq 3$.

14. The correct order of S_n, C_n and T_n is given by :

S_n, C_n एवं T_n का सही क्रम होगा :

(A) $S_n > T_n > C_n$ (B) $S_n < C_n < T_n$ (C) $S_n < T_n < C_n$ (D) $S_n > C_n > T_n$

Ans. D

Sol. We have $S_n = x + x^4 + x^7 + x^{10} + \ldots\ldots$ (2n terms)

$C_n = x^2 + x^5 + x^8 + x^{11} + \ldots\ldots$ (2n terms)

$T_n = x^3 + x^6 + x^9 + x^{12} + \ldots\ldots$ (2n terms)

Clearly $S_n > C_n > T_n$ as x is a proper fraction $\therefore\ x > x^2 > x^3$ & so on

15. The value of $\lim_{n\to\infty} (S_n + C_n + T_n)$ is equal to :

$\lim_{n\to\infty} (S_n + C_n + T_n)$ का मान बराबर होगा :

(A) $\frac{1}{1-x}$ (B) $\frac{x}{1-x}$ (C) $\frac{1}{1-x}$ (D) $\frac{x}{1+x}$

Ans. B

Sol. We have $S_n = x + x^4 + x^7 + x^{10} + \ldots\ldots$ (2n terms)

$C_n = x^2 + x^5 + x^8 + x^{11} + \ldots\ldots$ (2n terms)

$T_n = x^3 + x^6 + x^9 + x^{12} + \ldots\ldots$ (2n terms)

$\lim\limits_{n\to\infty} (S_n + C_n + T_n) = x + x^2 + x^3 + \ldots\ldots \infty = \dfrac{x}{1-x}$

16. The value of 'x' for which $S_n = C_n + T_n$, is :

x का वह मान जिसके लिए $S_n = C_n + T_n$ हो, होगा :

(A) $\sin\dfrac{\pi}{5}$ (B) $2\sin\dfrac{\pi}{5}$ (C) $2\sin\dfrac{\pi}{10}$ (D) $\sin\dfrac{\pi}{10}$

Ans. C

Sol. We have $S_n = x + x^4 + x^7 + x^{10} + \ldots\ldots$ (2n terms)

$C_n = x^2 + x^5 + x^8 + x^{11} + \ldots\ldots$ (2n terms)

$T_n = x^3 + x^6 + x^9 + x^{12} + \ldots\ldots$ (2n terms)

We have $S_n = C_n + T_n \Rightarrow x\dfrac{\left((x^3)^{2n}-1\right)}{(x^3-1)} = x^2\dfrac{\left((x^3)^{2n}-1\right)}{(x^3-1)} + x^3\dfrac{\left((x^3)^{2n}-1\right)}{(x^3-1)}$

But $x \neq 1$, as $x \in \left(0, \dfrac{\pi}{4}\right)$, so we get $x = x^2 + x^3$

$\Rightarrow \quad x^2 + x - 1 = 0 \ (x \neq 0) \quad \Rightarrow \quad x = \dfrac{\sqrt{5}-1}{2} = 2\sin\dfrac{\pi}{10} \in \left(0, \dfrac{\pi}{4}\right)$

MATCH THE COLUMN

17. Match the following columns :

Column-I	**Column-II**
(A) $f(x) = \begin{cases} x\tan^{-1}\dfrac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$	(P) even
(B) $g(x) = \begin{cases} x\cot^{-1} x, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$	(Q) odd
(C) $h(x) = \begin{cases} x^2\tan^{-1}\dfrac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$	(R) neither even nor odd

(D) $k(x) = \begin{cases} x\cot^{-1}x^2, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ (S) even as well as odd

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $f(x) = \begin{cases} x\tan^{-1}\frac{1}{x}, & \text{यदि } x \neq 0 \\ 0, & \text{यदि } x = 0 \end{cases}$	(P) सम
(B) $g(x) = \begin{cases} x\cot^{-1}x, & \text{यदि } x \neq 0 \\ 0, & \text{यदि } x = 0 \end{cases}$	(Q) विषम
(C) $h(x) = \begin{cases} x^2\tan^{-1}\frac{1}{x}, & \text{यदि } x \neq 0 \\ 0, & \text{यदि } x = 0 \end{cases}$	(R) न तो सम न ही विषम
(D) $k(x) = \begin{cases} x\cot^{-1}x^2, & \text{यदि } x \neq 0 \\ 0, & \text{यदि } x = 0 \end{cases}$	(S) सम तथा विषम दोनों

Ans. (A) P (B) R (C) Q (D) Q

Sol. (A) $f(-x) = f(x) \Rightarrow f$ is even

(B) g(x) is neither even nor odd.

(C) $h(-x) = -h(x) \Rightarrow h(x)$ is odd.

(D) $k(-x) = -k(x) \Rightarrow k(x)$ is odd

18. Match the following column :

Column -I	**Column -II**
(A) Let $f: R \to R$ be defined as $f(x) = \sqrt[3]{x} + \tan^{-1}x$, then f(x) is	(P) one-one
(B) Let $f: (-\infty, \infty) \to \{-1, 0, 1\}$ be defined as $f(x) = \sin^3\pi\left(\text{sgn}(x^2+3x+5)\right)$ then f(x) is (where sgn x denotes signum function of x)	(Q) into
(C) Let $f: [-2, 2] \to (0, e^2]$ be defined as $f(x) = e^{\sqrt{x^2}}$, then f(x) is	(R) odd
(D) Let $f: (-1, 5) \to [0, 3]$ be defined as $f(x) = \sqrt{5+4x-x^2}$, then f(x) is	(S) non-invertible
	(T) aperiodic

निम्न कॉलम को सुमेलित कीजिए :

Column -I	**Column -II**
(A) माना फलन $f: R \to R$ इस प्रकार की $f(x) = \sqrt[3]{x} + \tan^{-1} x$, तो f(x) होगा	(P) एकैकी
(B) माना फलन $f: (-\infty, \infty) \to \{-1, 0, 1\}$ इस प्रकार की $f(x) = \sin^3 \pi\left(\text{sgn}(x^2 + 3x + 5)\right)$ तो f(x) होगा (जहाँ sgn x, x का सिग्नम फलन है.)	(Q) अन्तःक्षेपी
(C) माना फलन $f:[-2, 2] \to (0, e^2]$ इस प्रकार की $f(x) = e^{\sqrt{x^2}}$, तो f(x) होगा	(R) विषम
(D) माना फलन $f: (-1, 5) \to [0, 3]$ इस प्रकार की $f(x) = \sqrt{5 + 4x - x^2}$, तो f(x) होगा	(S) अव्युत्क्रमणीय
	(T) आवर्तकालीय नहीं है

Ans. (A) PRT (B) QRS (C) QST (D) QST

Sol. (A) We have $f(x) = \sqrt[3]{x} + \tan^{-1} x$

Clearly range of f(x) is R. (As f(x) is increasing in its domain.)

Hence f(x) is one-one and onto function, so f(x) is invertible.

Also f(x) is odd and aperiodic function.

(B) We have $f(x) = \sin^3 \pi\left(\text{sgn}(x^2 + 3x + 5)\right)$

(As $x^2 + 3x + 5 > 0 \ \forall \ x \in R$)

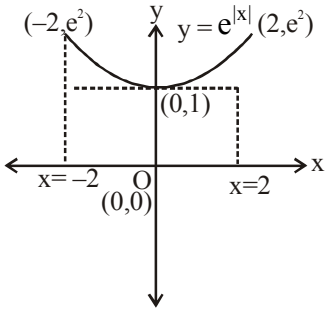
$\therefore$ $\text{sgn}(x^2 + 3x + 5) = 1$

Hence $f(x) = \sin^3 \pi = 0 \ \forall \ x \in R$

Clearly f(x) is into, odd and non-invertible.

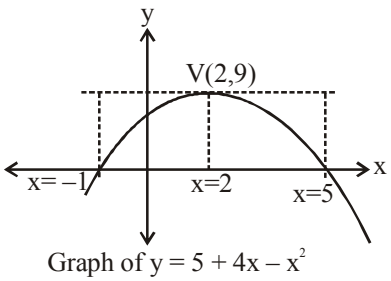
(C) We have $f(x) = e^{\sqrt{x^2}} = e^{|x|}$

Clearly f(x) is non-invertible and aperiodic and into function.



(D) We have $f(x) = \sqrt{5 + 4x - x^2}$

Clearly f(x) is into, non-invertible and aperiodic function.



Graph of $y = 5 + 4x - x^2$

19. Match the following column :

Column -I	**Column -II**
(A) $f(x)=\sin^{-1}\left(\frac{2}{\lvert\sin x-1\rvert+\lvert\sin x+1\rvert}\right)$	(P) f(x) is periodic
(B) $f(x)=\cos^{-1}\left(\lvert x-1\rvert-\lvert x-2\rvert\right)$	(Q) Domain of f(x) is R
(C) $f(x)=\sin^{-1}\left(\frac{\pi}{\lvert\sin^{-1}x-\frac{\pi}{2}\rvert+\lvert\sin^{-1}x+\frac{\pi}{2}\rvert}\right)$	(R) Range consists only irrational number
(D) $f(x)=\cos\left(\cos^{-1}\lvert x\rvert\right)+\sin^{-1}(\sin x)-\text{cosec}^{-1}(\text{cosec } x)+\text{cosec}^{-1}\lvert x\rvert$	(S) f(x) is constant function

निम्न कॉलम को सुमेलित कीजिए :

Column -I	**Column -II**
(A) $f(x)=\sin^{-1}\left(\frac{2}{\lvert\sin x-1\rvert+\lvert\sin x+1\rvert}\right)$	(P) f(x) आवर्ती है
(B) $f(x)=\cos^{-1}\left(\lvert x-1\rvert-\lvert x-2\rvert\right)$	(Q) f(x) का प्रान्त R है
(C) $f(x)=\sin^{-1}\left(\frac{\pi}{\lvert\sin^{-1}x-\frac{\pi}{2}\rvert+\lvert\sin^{-1}x+\frac{\pi}{2}\rvert}\right)$	(R) परिसर में अपरिमेय संख्याएं है
(D) $f(x)=\cos\left(\cos^{-1}\lvert x\rvert\right)+\sin^{-1}(\sin x)-\text{cosec}^{-1}(\text{cosec } x)+\text{cosec}^{-1}\lvert x\rvert$	(S) f(x) अचर फलन है

Ans. (A) PQRS (B) Q (C) RS (D) RS

Sol. (A) $f(x)=\sin^{-1}\left(\frac{2}{|\sin x-1|+|\sin x+1|}\right)$

$\because \sin x-1\le 0,\ \sin x+1\ge 0$

$f(x)=\sin^{-1}\left(\frac{2}{1-\sin x+1+\sin x}\right)$

$f(x)=\frac{\pi}{2}$

(B) $f(x)=\cos^{-1}\left(|x-1|-|x-2|\right)$

Case (I) $x\in(-\infty,1)$

$f(x) = \cos^{-1}(-x+1+x-2)$

$= \pi$

Case (II) $x \in (1,2)$

$f(x) = \cos^{-1}(-x+1-x+2)$

$= \cos^{-1}(3-2x)$

Case (III) $x \in (2,\infty)$

$f(x) = \cos^{-1}(x-1-x+2)$

$f(x) = 0$

(C) $f(x) = \sin^{-1}\left(\dfrac{\pi}{-\sin^{-1}x + \dfrac{\pi}{2} + \sin^{-1}x + \dfrac{\pi}{2}}\right)$

$= \dfrac{\pi}{2}$

(D) Domain = {–1, 1}

$f(\pm 1) = 1 + (\pm 1) - (\pm 1) + \dfrac{\pi}{2}$

Range $= 1 + \dfrac{\pi}{2}$